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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 2

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\401-E-31001-AK.pvdb

Date of Analysis : Jul 4, 2026 1:26pm

PV Elite 28, January 2026

Vessel Design Summary:

ASME Code, Section VIII Division 2, 2025

Diameter Spec : 1300.000 mm ID
 Vessel Design Length, Tangent to Tangent 2005.90 cm
 Specified Datum Line Distance 0.00 cm
 Shell Side Design Temperature 250 °C
 Channel Side Design Temperature 260 °C
 Shell Side Design Pressure 163.158 kgf/cm²
 Channel Side Design Pressure 163.158 kgf/cm²
 Shop Shell Side Test Pressure 256.419 kgf/cm²
 Shop Channel Side Test Pressure 255.658 kgf/cm²
 Wind Design Code IS 875 (PART 3) Amd. 1&2 (2020)
 Earthquake Design Code IS 1893 RSM Amd. 1&2 (2018)

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-387 11	2	...	K11789	+IT
Head	SA-387 11	2	...	K11789	+IT
Flange	SA-336 F11	2	...	K11572	+IT
Nozzle	SA-336 F11	2	...	K11572	+IT
Nozzle Flg	SA-336 F11	2	...	K11572	
Tubes	SA-213 T11	...	...	K11597	
Tubesheet	SA-336 F11	2	...	K11572	+IT
Flg Bolting	SA-193 B7	...	64 < t <= 100	G41400	
Hrz Bolting	SA-193 B7	...	<= 64	G41400	
Saddles	SA-387 11	2	...	K11789	+IT

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (kgf/cm² & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
LHS Ch Cover	163.169	0.00	170.000	1.5000	N/A	No
LHS Ch flg	163.165	0.00	166.000	1.5000	N/A	No
LHS Ch Shell	163.165	0.00	166.000	1.5000	N/A	No
Main Shell	163.165	0.00	167.000	1.5000	N/A	No
RHS Ch Shell	163.165	0.00	167.000	1.5000	N/A	No
RHS Dish End	163.158	1.05	168.000	1.5000	No	No

Liquid Level: 214.00 cm Dens.: 49.977 kg/m³ Sp. Gr.: 0.050

Element Types and Properties:

Element Type	"To" Elev cm	Element Length cm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Body Flg	37.8	37.5	383.0	375.0	367.7	...	1.00	1.00
Body Flg	92.8	55.0	418.0	410.0	399.6	...	1.00	1.00
Cylinder	272.8	180.0	68.0	68.0	66.7	...	1.00	1.00
Cylinder	1820.8	1520.0	68.0	68.0	66.2	...	1.00	1.00
Cylinder	2000.8	180.0	68.0	68.0	66.2	...	1.00	1.00
Ellipse	2005.9	5.1	72.0	60.0	58.3	...	1.00	1.00



External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure kgf/cm ²	External M.A.W.P. kgf/cm ²
10	20	375	257.623	...	No Calc
20	30	410	385.47	...	No Calc
30	Tsht	68	No Calc	...	117
Tsht	40	68	No Calc	...	No Calc
40	70	68	No Calc	...	84
70	80	68	No Calc	...	84
80	90	60	4.68551	1.05461	109

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners cm	Allowable Length Bet. Stiffeners cm	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	No Calc	No Calc	No Calc	No Calc
30	Tsht	234.997	No Calc	No Calc	No Calc
Tsht	40	No Calc	No Calc	No Calc	No Calc
40	70	1715.92	No Calc	No Calc	No Calc
70	80	1715.92	No Calc	No Calc	No Calc
80	90	No Calc	No Calc	No Calc	No Calc

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N1-Tube Inlet	WRC-297	0.984	Shell	Passed
N3-Shell Inlet	WRC-297	0.977	Shell	Passed
N4-Shell outle	WRC-297	0.977	Shell	Passed
N2-Tube Outlet	WRC-297	0.977	Shell	Passed

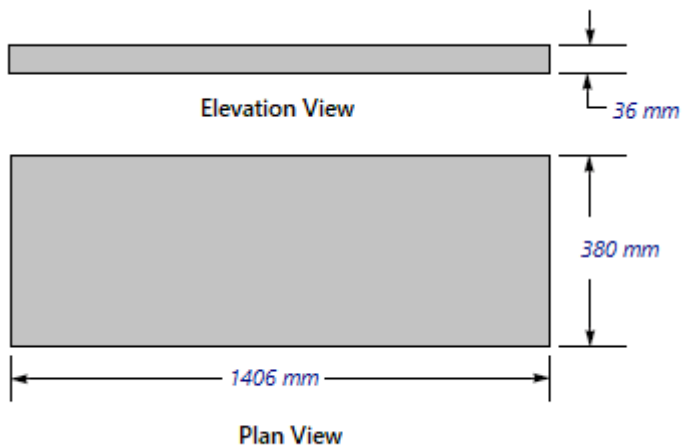
Foundation/Support Summary

Saddle Parameters:

Saddle Width	200.000 mm
Saddle Bearing Angle	120.000 deg.
Centerline Dimension	1300.000 mm
Wear Pad Width	560.000 mm
Wear Pad Thickness	12.000 mm
Wear Pad Bearing Angle	132.000 deg.
Distance from Saddle to Tangent	3000.000 mm

Baseplate Length	1406.000 mm
Baseplate Thickness	36.000 mm
Baseplate Width	380.000 mm
Number of Ribs (including outside ribs)	5
Rib Thickness	28.000 mm
Web Thickness	28.000 mm
Height of Center Web	600.000 mm
Number of Bolts in Baseplate	4

Baseplate Sketch



Maximum Tensile Bolt Load 6. kgf

Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	101226.20	kgf
Maximum Transverse Saddle Shear Load	6861.31	kgf
Maximum Longitudinal Saddle Shear Load	13722.61	kgf

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	91230.40	kgf
Maximum Transverse Saddle Shear Load	1170.75	kgf
Maximum Longitudinal Saddle Shear Load	181.91	kgf

Weights:

Fabricated - Bare w/o removable internals	104218.5	kg
Shop Test - Fabricated + water (full)	127263.1	kg
Shipping - Fab. + rem. intls.+ shipping app.	105700.0	kg
Erected - Fab. + rem. intls.+ insul. (etc)	105700.0	kg
Empty - Fab. + intls. + details + wghts.	105700.0	kg
Operating - Empty + operating liquid (no ca)	106834.5	kg

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Nozzle Calculation Summary (kgf/cm² & mm):

Description	MAWP	Ext	MAPNC	Tbl 4.5.2	Weld	Pass/Fail
N1-Tube Inlet	166	...	...	OK	OK	Passed
N3-Shell Inlet	167	...	...	OK	OK	Passed
N4-Shell outlet	167	...	...	OK	OK	Passed
N2-Tube Outlet	167	...	...	OK	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 167 Nozzle : N3-Shell Inlet [Shellside]
 Minimum MAWP Nozzles : 166 Nozzle : N1-Tube Inlet [Tubeside]

Note: MAWPs (Internal Case) shown above are at the High Point.

VIII-2 Nozzle to Gross Structural Discontinuity Check:

No Nozzles were too close to any gross structural discontinuities per 4.5.13.1

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Inside Radius mm
30	N1-Tube Inlet	1828.175	90.000	222.963
40	N3-Shell Inlet	4508.175	270.000	200.738
40	N4-Shell outlet	16008.176	90.000	222.963
70	N2-Tube Outlet	19108.174	90.000	200.738

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia mm	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
N3-Shell Inlet	450 mm	100	None	457.2	29.362	...	...	555	...
N2-Tube Outlet	450 mm	100	None	457.2	29.362	...	...	555	...
N1-Tube Inlet	500 mm	100	None	508.0	32.537	...	...	573	...
N4-Shell outlet	500 mm	100	None	508.0	32.537	...	...	573	...

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
N3-Shell In	SA-336 F11	72.000	10.000	...	...	...
N2-Tube Out	SA-336 F11	72.000	10.000	...	...	...
N1-Tube Inl	SA-336 F11	72.000	10.000	...	...	...
N4-Shell ou	SA-336 F11	72.000	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum cm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
N3-Shell Inlet	450.818	270.0	350.00	0.00	Main Shell
N2-Tube Outlet	1910.817	90.0	350.00	0.00	RHS Ch Shell
N1-Tube Inlet	182.818	90.0	350.00	0.00	LHS Ch Shell
N4-Shell outlet	1600.818	90.0	350.00	0.00	Main Shell

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Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	BLIND FLANGE: 383.0mm THK X 2140.0mm OD	SA-336 F11
1	WELD NECK FLANGE: 418.0mm THK X 2140.0mm OD	SA-336 F11
1	CYLINDER: 68.0mm THK X 1300.0mm ID X 180.0cm	SA-387 11
1	CYLINDER: 68.0mm THK X 1300.0mm ID X 1520.0cm	SA-387 11
1	ELLIPTICAL HEAD: 2.0 X 1, 72.0mm THK X 1300.0mm ID X 5.1cm	SA-387 11
2	INSULATION: 375mm X 75mm THK	Hot Insulation
1	INSULATION: 550mm X 75mm THK	Hot Insulation
2	INSULATION: 1800mm X 75mm THK	Hot Insulation
1	TUBESHEET: 1444.0mm X 280.0mm THK	SA-336 F11
2615	TUBES: 15200.0mm X 14.0mm DIA X 2.0mm THK	SA-213 T11
1	FLOATING TUBESHEET: 1386.0mm X 280.0mm THK	SA-336 F11
2	SADDLE: 200.0mm X 120 DEG	SA-387 11
1	INSULATION: 15199mm X 105mm THK	Hot Insulation
1	GASKET: 1364.0mm OD X 1332.0mm ID	...
36	BODY FLANGE BOLTS: 82.6mm DIA	SA-193 B7
72	NUTS FOR BODY FLANGE BOLTS: 82.6mm DIA	...
1	CYLINDRICAL SEGMENT 68.0mm THK X 3048.0mm X 180.0cm	SA-387 11
1	CYLINDRICAL SEGMENT 68.0mm THK X 1036.1mm X 180.0cm	SA-387 11
1	NAMEPLATE	...

Minimum Design Metal Temperature Results Summary (°C):

Description	Curve	Basic MDMT	Reduced MDMT	PWHT	Thickness ratio	Gov Thk mm	E*
Notes							
Main Shell [8]	!	-29		No			
RHS Ch Shell [8]	!	-29		No			
RHS Dish End [10]	!	-29		No			
RHS Dish End [7]	!	-29		No			
N3-Shell Inlet[1]	!	-29		No			
N4-Shell outle[1]	!	-29		No			
N2-Tube Outlet[1]	!	-29		No			
Tubesheet: SS[13]	!	-29		No			
Warmest MDMT:		-29					
LHS Ch Cover [11]	!	-29		No			
LHS Ch flg [11]	!	-29		No			
LHS Ch Shell [8]	!	-29		No			
N1-Tube Inlet [1]	!	-29		No			
Tubesheet: CS[14]	!	-29		No			
Bolting [21]		-40					
Warmest MDMT:		-29					
Exchanger Side	Computed MDMT		Required MDMT		Pass/Fail		
Shell	-29.0		-28.9		Pass		
Channel/Tube	-29.0		-28.9		Pass		

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per equation 3.1.
- [5] - ASME Flange MDMT Calcs; Thickness ratio per equation 3.3.
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per 3.11.2.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld



Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.

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Units used in this Analysis (Kgfc^{m2}):

Name	System Unit	Constant	User Unit
Length	Feet	30.4800	cm
Force	Pounds	0.4536	kgf
Mass	Pounds	0.4536	kg
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	kgf-m.
Stress	lbs./sq.in.	0.0703	kgf/cm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0703	kgf/cm ²
Elast. Modulus	lbs./sq.in.	0.0703	kgf/cm ²
Pipe Density	lbs./cu.in.	0.0277	kg/cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Fluid Density	lbs./cu.ft.	16.0183	kg/m ³
Wind Speed	miles/hr	1.6093	km/hr
Tray Weight	lbs./sq.ft.	0.0005	kgf/cm ²
Inertia	in.4	41.6231	cm ⁴
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8824	kgf/m ²
Elevation	Feet	0.3048	m
Volume	in.3	16.3871	cm ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	163.16	kgf/cm ²
Channel Side Design Pressure	163.16	kgf/cm ²
Shell Side Design Temperature	250.0	°C
Channel Side Design Temperature	260.0	°C
Radiography, Shell Side	None	
Radiography, Channel Side	None	
Service Type, Shell Side	None	
Service Type, Channel Side	None	
MDMT (CET), Shell Side	-28.9	°C
MDMT (CET), Tube Side	-28.9	°C
User defined MAWP, Shell Side	0	kgf/cm ²
User defined MAWP, Channel Side	0	kgf/cm ²
User defined MAPnc, Shell Side	0	kgf/cm ²
User defined MAPnc, Channel Side	0	kgf/cm ²
User defined Test Pres., Shell Side	0	kgf/cm ²
User defined Test Pres., Channel Side	0	kgf/cm ²

Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	N	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175

0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
2.800	0.025
0.000	0.000

Consider Vortex Shedding N

Shop Pressure Test:

Type of Pressure Test	Hydrostatic
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	IS 875 (PART 3):2015-2020	
Basic Wind Speed for IS-875	180	km/hr
Wind Zone Number	5	
Base Elevation	179	cm
Percent Wind for Hydrotest	33.0	
Risk Factor	[k1] 1.0	
Terrain Category	2	
Equipment Class	Class A	
Topography Factor	[k3] 1.0	
Importance Factor	[k4] 1.15	
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	

Seismic Design Code	IS 1893-RSM
Importance Factor for IS-1893	2.000
Soil Factor	1.000
Zone Number	2.000
Percent Seismic for Hydrotest	0.000
Period of Vibration (User defined) Sec	0.000
Damping Factor	2.000

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
-------------	------

0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
2.800	0.025
0.000	0.000

Consider Vertical Acceleration	Yes
Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N
Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Using Metric Material Databases, ASME II D	Yes
Using ASME VIII-1 Appendix 46	No
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/6
Element From Node	10
Element To Node	20
Element Type	Flange
Description	LHS Ch Cover
Distance "FROM" to "TO"	37.5 cm
Flange Outside Diameter	2140 mm
Element Thickness	375 mm
Internal Corrosion Allowance	1.5 mm
Nominal Thickness	383 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	163.16 kgf/cm ²
Design Temperature Internal Pressure	260 °C
Design External Pressure	0 kgf/cm ²
Design Temperature External Pressure	260 °C
Effective Diameter Multiplier	1.2
Material Name	SA-336 F11 +IT
Allowable Stress, Ambient	1876.2 kgf/cm ²
Allowable Stress, Operating	1521.4 kgf/cm ²
Allowable Stress, Hydrotest	2663.9 kgf/cm ²
Material Density	0.00775 kg/cm ³
P Number Thickness	0 mm
Yield Stress, Operating	2282 kgf/cm ²
MDMT Curve	Impact Tested



External Pressure Chart Name	CS-2
UNS Number	K11572
Class / Thickness / Grade	2 ::
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No
Element From Node	10
Detail Type	Liquid
Detail ID	Liquid: 10
Dist. from "FROM" Node / Offset dist	0 cm
Height/Length of Liquid	214 cm
Liquid Density	49.977 kg/m ³
Element From Node	10
Detail Type	Insulation
Detail ID	Hot Insulation
Dist. from "FROM" Node / Offset dist	0 cm
Height/Length of Insulation	37.5 cm
Thickness of Insulation	75 mm
Density	0.00015 kg/cm ³

Element#	2/6
Element From Node	20
Element To Node	30
Element Type	Flange
Description	LHS Ch flg
Distance "FROM" to "TO"	55 cm
Flange Inside Diameter	1300 mm
Element Thickness	410 mm
Internal Corrosion Allowance	1.5 mm
Nominal Thickness	418 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	163.16 kgf/cm ²
Design Temperature Internal Pressure	260 °C
Design External Pressure	0 kgf/cm ²
Design Temperature External Pressure	260 °C
Effective Diameter Multiplier	1.2
Material Name	SA-336 F11 +IT
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element From Node	20
Detail Type	Liquid
Detail ID	Liquid: 20
Dist. from "FROM" Node / Offset dist	0 cm
Height/Length of Liquid	130 cm
Liquid Density	49.977 kg/m ³
Element From Node	20
Detail Type	Insulation
Detail ID	Hot Insulation
Dist. from "FROM" Node / Offset dist	0 cm
Height/Length of Insulation	55 cm
Thickness of Insulation	75 mm
Density	0.00015 kg/cm ³

Element#	3/6
Element From Node	30
Element To Node	40
Element Type	Cylinder
Description	LHS Ch Shell
Distance "FROM" to "TO"	180 cm
Inside Diameter	1300 mm
Element Thickness	68 mm
Internal Corrosion Allowance	1.5 mm



Nominal Thickness	68	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	163.16	kgf/cm ²
Design Temperature Internal Pressure	260	°C
Design External Pressure	0	kgf/cm ²
Design Temperature External Pressure	260	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-387 11	+IT
Allowable Stress, Ambient	2110.7	kgf/cm ²
Allowable Stress, Operating	1711	kgf/cm ²
Allowable Stress, Hydrotest	3002.9	kgf/cm ²
Material Density	0.00775	kg/cm ³
P Number Thickness	0	mm
Yield Stress, Operating	2563.5	kgf/cm ²
MDMT Curve	Impact	Tested
External Pressure Chart Name	CS-3	
UNS Number	K11789	
Class / Thickness / Grade	2 ::	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	30	
Detail Type	Liquid	
Detail ID	Liquid: 30	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Liquid	130	cm
Liquid Density	49.977	kg/m ³

Element From Node	30	
Detail Type	Insulation	
Detail ID	Hot Insulation	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Insulation	180	cm
Thickness of Insulation	75	mm
Density	0.00015	kg/cm ³

Element From Node	30	
Detail Type	Nozzle	
Detail ID	N1-Tube Inlet	
Dist. from "FROM" Node / Offset dist	90	cm
Nozzle Diameter	500	mm
Nozzle Schedule	100	
Nozzle Class	None	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	724.64	kgf
Grade of Attached Flange	None	
Nozzle Matl	SA-336 F11	+IT

Element From Node	30	
Detail Type	Weight	
Detail ID	bAFFLE	
Dist. from "FROM" Node / Offset dist	90	cm
Miscellaneous Weight	200	kgf
Offset from Element Centerline	0	mm

Element#	4/6	
Element From Node	40	
Element To Node	70	
Element Type	Cylinder	
Description	Main Shell	
Distance "FROM" to "TO"	1520	cm
Inside Diameter	1300	mm
Element Thickness	68	mm
Internal Corrosion Allowance	1.5	mm
Nominal Thickness	68	mm



External Corrosion Allowance	0	mm
Design Internal Pressure	163.16	kgf/cm ²
Design Temperature Internal Pressure	250	°C
Design External Pressure	0	kgf/cm ²
Design Temperature External Pressure	250	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-387 11	+IT
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	40	
Detail Type	Saddle	
Detail ID	Left Saddle	
Dist. from "FROM" Node / Offset dist	300	cm
Width of Saddle	200	mm
Height of Saddle at Bottom	1300	mm
Saddle Contact Angle	120.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	560	mm
Thickness of Wear Plate	12	mm
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.0	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	cm

Element From Node	40	
Detail Type	Saddle	
Detail ID	New Saddle	
Dist. from "FROM" Node / Offset dist	1220	cm
Width of Saddle	200	mm
Height of Saddle at Bottom	1300	mm
Saddle Contact Angle	120.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	560	mm
Thickness of Wear Plate	12	mm
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.30000001	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	cm

Element From Node	40	
Detail Type	Liquid	
Detail ID	Liquid: 40	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Liquid	130	cm
Liquid Density	49.977	kg/m ³

Element From Node	40	
Detail Type	Insulation	
Detail ID	Hot Insulation	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Insulation	1520	cm
Thickness of Insulation	105	mm
Density	0.00015	kg/cm ³

Element From Node	40	
Detail Type	Nozzle	
Detail ID	N3-Shell Inlet	
Dist. from "FROM" Node / Offset dist	150	cm
Nozzle Diameter	450	mm
Nozzle Schedule	100	
Nozzle Class	None	
Layout Angle	270.0	



Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 663.18 kgf
 Grade of Attached Flange None
 Nozzle Matl SA-336 F11 +IT

Element From Node 40
 Detail Type Nozzle
 Detail ID N4-Shell outlet
 Dist. from "FROM" Node / Offset dist 1300 cm
 Nozzle Diameter 500 mm
 Nozzle Schedule 100
 Nozzle Class None
 Layout Angle 90.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 663.18 kgf
 Grade of Attached Flange None
 Nozzle Matl SA-336 F11 +IT

Element# 5/6
 Element From Node 70
 Element To Node 80
 Element Type Cylinder
 Description RHS Ch Shell
 Distance "FROM" to "TO" 180 cm
 Inside Diameter 1300 mm
 Element Thickness 68 mm
 Internal Corrosion Allowance 1.5 mm
 Nominal Thickness 68 mm
 External Corrosion Allowance 0 mm
 Design Internal Pressure 163.16 kgf/cm²
 Design Temperature Internal Pressure 250 °C
 Design External Pressure 0 kgf/cm²
 Design Temperature External Pressure 250 °C
 Effective Diameter Multiplier 1.2
 Material Name SA-387 11 +IT
 Efficiency, Longitudinal Seam 1.0
 Efficiency, Circumferential Seam 1.0
 Weld is pre-Heated No

Element From Node 70
 Detail Type Liquid
 Detail ID Liquid: 70
 Dist. from "FROM" Node / Offset dist 0 cm
 Height/Length of Liquid 130 cm
 Liquid Density 49.977 kg/m³

Element From Node 70
 Detail Type Insulation
 Detail ID Hot Insulation
 Dist. from "FROM" Node / Offset dist 0 cm
 Height/Length of Insulation 180 cm
 Thickness of Insulation 75 mm
 Density 0.00015 kg/cm³

Element From Node 70
 Detail Type Nozzle
 Detail ID N2-Tube Outlet
 Dist. from "FROM" Node / Offset dist 90 cm
 Nozzle Diameter 450 mm
 Nozzle Schedule 100
 Nozzle Class None
 Layout Angle 90.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 663.18 kgf
 Grade of Attached Flange None
 Nozzle Matl SA-336 F11 +IT



Element#	6/6
Element From Node	80
Element To Node	90
Element Type	Elliptical
Description	RHS Dish End
Distance "FROM" to "TO"	5.08 cm
Inside Diameter	1300 mm
Element Thickness	60 mm
Internal Corrosion Allowance	1.5 mm
Nominal Thickness	72 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	163.16 kgf/cm ²
Design Temperature Internal Pressure	250 °C
Design External Pressure	1.0546 kgf/cm ²
Design Temperature External Pressure	250 °C
Effective Diameter Multiplier	1.2
Material Name	SA-387 11 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	No
Element From Node	80
Detail Type	Insulation
Detail ID	Hot Insulation
Dist. from "FROM" Node / Offset dist	0 cm
Height/Length of Insulation	37.58 cm
Thickness of Insulation	75 mm
Density	0.00015 kg/cm ³



XY Coordinate Calculations:

From	To	X (Horiz.) cm	Y (Vert.) cm	DX (Horiz.) cm	DY (Vert.) cm
LHS Ch Cover		37.8175	...	37.5	...
LHS Ch flg		92.8175	...	55	...
LHS Ch Shell		272.818	...	180	...
Main Shell		1820.82	...	1520	...
RHS Ch Shell		2000.82	...	180	...
RHS Dish End		2005.9	...	5.08	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Flange Input Data Values

Description: New Flange :

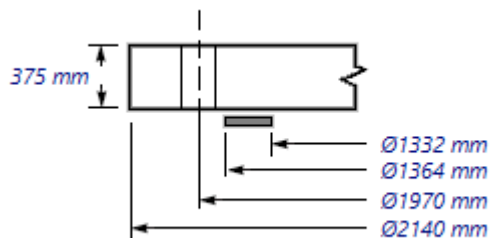
LHS Ch Cover

Flange Type		Blind	
Design Pressure	P	163.17	kgf/cm ²
Design Temperature		260	°C
Internal Corrosion Allowance	ci	1.5000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Outside Diameter	A	2140.000	mm
Flange Thickness	t	375.0000	mm
Flange Material		SA-336 F11 +IT	
Flange Material UNS number		K11572	
Flange Allowable Stress At Temperature	Sfo	1521.35	kgf/cm ²
Flange Allowable Stress At Ambient	Sfa	1876.18	kgf/cm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	1621.26	kgf/cm ²
Bolt Allowable Stress At Ambient	Sa	1621.26	kgf/cm ²

Warning:

It seems as if the selected bolt material does not match the bolt diameter range allowed per the material specification. Please check the material selection for the given nominal bolt diameter.

Perimeter along the Center of the Bolts	L	6188.938	mm
Diameter of Bolt Circle	C	1970.000	mm
Nominal Bolt Diameter	a	82.5500	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		36	
Flange Face Outside Diameter	Fod	1375.000	mm
Flange Face Inside Diameter	Fid	1300.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	1364.000	mm
Gasket Inside Diameter	Gi	1332.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	703.07	kgf/cm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm



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Gasket Contact Width,	$N = (G_o - G_i)/2$	16.000	mm
Basic Gasket Width,	$b_o = N/2$	8.000	mm
Effective Gasket Width,	$b = C_b * \sqrt{b_o}$	7.127	mm
Gasket Reaction Diameter,	$G = G_o - 2 * b$	1349.745	mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 1350^2 \cdot 163$$

$$= 2334698.500 \text{ kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 7.13 \cdot \pi \cdot 1350 \cdot 3 \cdot 163$$

$$= 295884.344 \text{ kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(2334698 + 295884 + 0, 0)$$

$$= 2630583.000 \text{ kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 703 \cdot 7.13 \cdot \pi \cdot 1350 + 0 \cdot 0 \cdot 0$$

$$= 212486.484 \text{ kgf}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G)/Sb, Wg/Sa)$$

$$= \max((2630583 + 0 + 4 \cdot 0/1350)/1621, 212486/1621)$$

$$= 1622.554 \text{ cm}^2$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	1622.554	1738.913	
Radial Distance between Bolts and Edge:	76.200	85.000	
Circ. Spacing between the Bolts:	171.450	171.697	...

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi (Go + Gi))$$

$$= 1739 \cdot 1621 / (703 \cdot \pi (1364 + 1332))$$

$$= 47.344 \text{ mm [Note: Exceeds actual gasket width, 16.000]}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 1621 \cdot 1739$$

$$= 2819231.50 \text{ kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo$$

$$= 2630583.00 \text{ kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1970 - 1350) / 2$$

$$= 310.1275 \text{ mm}$$

Flange Stresses

Tangential Flange Stress, Flat Head (4.6.2.3), Operating [STo]:

$$= 1.9 \cdot Wm1 \cdot hG \cdot Bsc / (t^2 \cdot G) + C \cdot Z \cdot Peq \cdot G^2 / t^2$$

$$= 1.9 \cdot 2630583 \cdot 310 \cdot 1 / (374^2 \cdot 1350) + 0.3 \cdot 1 \cdot 163 \cdot 1350^2 / 374^2$$

$$= 1462.48 \text{ kgf/cm}^2$$

Tangential Flange Stress, Flat Head (4.6.2.4), Seating [STa]:

$$= 1.9 \cdot W \cdot hG \cdot Bsc / (t^2 \cdot G)$$

$$= 1.9 \cdot 2819231 \cdot 310 \cdot 1 / (374^2 \cdot 1350)$$

$$= 882.25 \text{ kgf/cm}^2$$

Bolt Stress, Operating [BSo]:

$$= Wm1 / Ab$$

$$= 2630583 / 1739$$



= 1512.78 kgf/cm²

Bolt Stress, Seating [BSa]:

= (Wm2 / Ab)
 = (212486/1739)
 = 122.20 kgf/cm²

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Tangential Flange	1462.48	1521.35	882.25 1876.18
Bolting	1512.78	1621.26	122.20 1621.26

Reqd. Blind Flange Thickness at Center 367.702 mm
 Reqd. Blind Flange Thickness at Gasket 276.247 mm

Estimated M.A.W.P. (Operating) 170 kgf/cm²
 Estimated Finished Weight of Flange at given Thk. 10453.7 kg
 Estimated Unfinished Weight of Forging at given Thk 10453.7 kg

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Tangential Flange	1450.81	1876.18	875.21 1876.18
Bolting	1512.78	1621.26	122.20 1621.26

Minimum Req'd Thickness, ambient, no Corrosion: 329.768 mm

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Flange Input Data Values

Description: Lhs Ch Flg :

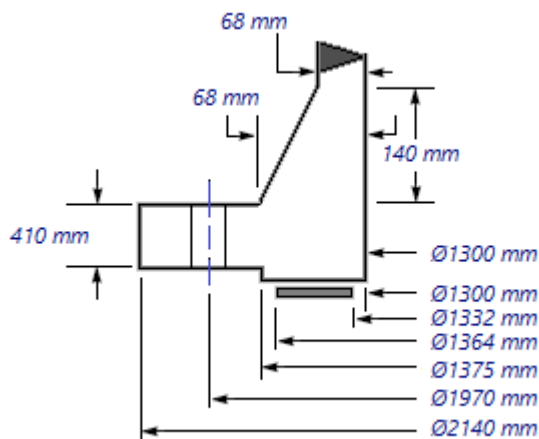
LHS Ch flg

Flange Type	Integral Weld Neck		
Design Pressure	P	163.16	kgf/cm ²
Design Temperature		260	°C
Internal Corrosion Allowance	ci	1.5000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	1300.000	mm
Flange Outside Diameter	A	2140.000	mm
Flange Thickness	t	410.0000	mm
Thickness of Hub at Small End	go	68.0000	mm
Thickness of Hub at Large End	gl	68.0000	mm
Length of Hub	h	140.0000	mm
Flange Material	SA-336 F11 +IT		
Flange Material UNS number	K11572		
Flange Allowable Stress At Temperature	Sfo	1521.35	kgf/cm ²
Flange Allowable Stress At Ambient	Sfa	1876.18	kgf/cm ²
Bolt Material	SA-193 B7		
Bolt Allowable Stress At Temperature	Sb	1621.26	kgf/cm ²
Bolt Allowable Stress At Ambient	Sa	1621.26	kgf/cm ²

Warning:

It seems as if the selected bolt material does not match the bolt diameter range allowed per the material specification. Please check the material selection for the given nominal bolt diameter.

Diameter of Bolt Circle	C	1970.000	mm
Nominal Bolt Diameter	a	82.5500	mm
Type of Threads	TEMA Thread Series		
Number of Bolts	36		
Flange Face Outside Diameter	Fod	1375.000	mm
Flange Face Inside Diameter	Fid	1300.000	mm
Flange Facing Sketch	Sketch 1a		
Gasket Outside Diameter	Go	1364.000	mm
Gasket Inside Diameter	Gi	1332.000	mm
Gasket Factor	m	2.0000	
Gasket Design Seating Stress	y	175.77	kgf/cm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm



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Hub Small End Required Thickness due to Internal Pressure:

$$= (P(R + Ca) / (S - 0.5 \cdot P)) + (Ca + Ca_{ext})$$

$$= (163(1300 + 1.5) / (1711 - 0.5 \cdot 163)) + (1.5)$$

$$= 66.7383 \text{ mm}$$

Small End Hub MAWP:

$$= S \cdot \log((2 \cdot T + R) / R)$$

$$= 1711 \cdot \log((2 \cdot 66.5 + 652) / 652)$$

$$= 166.298 \text{ kgf/cm}^2$$

Small End Hub MAPnc:

$$= S_a \cdot \log((2.0 \cdot T + R) / R)$$

$$= 2111 \cdot \log((2.0 \cdot 68 + 650) / 650)$$

$$= 210.009 \text{ kgf/cm}^2$$

Corroded Flange Thickness, $t_c = T - c_i$	408.500 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	1303.000 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	66.500 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	66.500 mm
Code R Dimension, $R = ((C - B_{cor}) / 2) - g1_{cor}$	267.000 mm

Gasket Contact Width, $N = (G_o - G_i) / 2$	16.000 mm
Basic Gasket Width, $b_o = N / 2$	8.000 mm
Effective Gasket Width, $b = C_b \cdot \sqrt{b_o}$	7.127 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	1349.745 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi / 4 \cdot G^2 \cdot P$$

$$= \pi / 4 \cdot 1350^2 \cdot 163$$

$$= 2334638.750 \text{ kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 7.13 \cdot \pi \cdot 1350 \cdot 2 \cdot 163$$

$$= 197251.172 \text{ kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi / 4 \cdot B_{cor}^2 \cdot P$$

$$= \pi / 4 \cdot 1303^2 \cdot 163$$

$$= 2175730.000 \text{ kgf}$$

Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 2334638 - 2175730$$

$$= 158908.703 \text{ kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_o, 1 + W_o, 2 + H'p, 0)$$

$$= \max(2334638 + 197251 + 0, 0)$$

$$= 2531889.750 \text{ kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 176 \cdot 7.13 \cdot \pi \cdot 1350 + 0 \cdot 0 \cdot 0$$

$$= 53121.621 \text{ kgf}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a)$$

$$= \max((2531889 + 0 + 4 \cdot 0 / 1350) / 1621, 53122 / 1621)$$

$$= 1561.679 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 82.5 + 6 \cdot 408 / (2 + 0.5)$$

$$= 1145.500 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 1970 \cdot \sin(\pi / 36)$$

$$= 171.697 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{Bs / (2a + t)}, 1)$$

$$= \max(\sqrt{172 / (2 \cdot 82.5 + 408)}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	1561.679	1738.913	
Radial Distance between Hub and Bolts:	95.250	267.000	
Radial Distance between Bolts and Edge:	76.200	85.000	
Circ. Spacing between the Bolts:	171.450	171.697	1145.500

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 1739 \cdot 1621 / (176 \cdot \pi(1364 + 1332))$$

$$= 189.375 \text{ mm [Note: Exceeds actual gasket width, 16.000]}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 1621 \cdot 1739$$

$$= 2819231.50 \text{ kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 2531889 - 2334638$$

$$= 197251.06 \text{ kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1970 - 1350) / 2$$

$$= 310.1275 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((1970 - 1303) / 2 + 310)$$

$$= 321.8138 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (1970 - 1303 - 66.5) / 2$$

$$= 300.2500 \text{ mm}$$

Summary of Moments for Internal Pressure: (kgf-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	2175730.	300.2500	1.0000	653264.
Face Pressure, Mt	158909.	321.8138	1.0000	51139.
Gasket Load, Mg	197251.	310.1275	1.0000	61173.
Gasket Seating, Matm	2819232.	310.1275	1.0000	874322.

Total Moment for Operation, Mo 765576. kgf-m.
 Total Moment for Gasket seating, Mg 874322. kgf-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 294.363 mm
 Hub Ratio, h/h0 = Defined as 2 2.000



Thickness Ratio, $g1/g0 = (g1Cor/goCor)$ 1.000

Note: Because $g1 = g0$, the ratio $h/h0$ has been set to 2.

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.926
 Factor V per 4.16.5 0.549
 Factor f per 4.16.5 1.000

Stress factor from Table 4.16.4 [K]:

= A / B
 = 2140/1303
 = 1.642

Stress factor from Table 4.16.4 [Y]:

= $(0.66845 + (5.7169(K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$
 = $(0.66845 + (5.7169(1.64^2 \cdot \log_{10}(1.64))) / (1.64^2 - 1)) / (1.64 - 1)$
 = 4.088

Stress factor from Table 4.16.4 [T]:

= $(K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$
 = $(1.64^2(1 + 8.55246 \cdot \log_{10}(1.64)) - 1) / ((1.0472 + 1.9448 \cdot 1.64^2) \cdot (1.64 - 1))$
 = 1.650

Stress factor from Table 4.16.4 [U]:

= $(K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$
 = $(1.64^2(1 + 8.55246 \cdot \log_{10}(1.64)) - 1) / (1.36136(1.64^2 - 1) \cdot (1.64 - 1))$
 = 4.492

Stress factor from Table 4.16.4 [Z]:

= $(K^2 + 1) / (K^2 - 1)$
 = $(1.64^2 + 1) / (1.64^2 - 1)$
 = 2.178

Stress factor from Table 4.16.4 for integral flanges [d]:

= $U \cdot g0^2 \cdot h0 / V$
 = $4.49 \cdot 66.5^2 \cdot 294 / 0.55$
 = 10649082.000 mm³

Stress factor from Table 4.16.4 for integral flanges [e]:

= F / h0
 = 0.93/294
 = 0.003 1/mm

Stress factor from Table 4.16.4 [L]:

= $(t \cdot e + 1) / T + t^3/d$
 = $(408 \cdot 0.0031 + 1) / 1.65 + 408^3/10649082$
 = 7.786

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

= $(f \cdot Mop / Bcor) / (L \cdot g1^2)$
 = $(1 \cdot 765576 / 1303) / (7.79 \cdot 66.5^2)$
 = 1706.34 kgf/cm²

Longitudinal Hub Stress, Seating [SHa]:

= $(f \cdot Matm / Bcor) / (L \cdot g1^2)$
 = $(1 \cdot 874322 / 1303) / (7.79 \cdot 66.5^2)$
 = 1948.71 kgf/cm²

Radial Flange Stress, Operating [SRo]:

= $(\beta \cdot Mop / Bcor) / (L \cdot t^2)$
 = $(2.71 \cdot 765576 / 1303) / (7.79 \cdot 408^2)$
 = 122.68 kgf/cm²

Radial Flange Stress, Seating [SRa]:

= $(\beta \cdot Matm / Bcor) / (L \cdot t^2)$
 = $(2.71 \cdot 874322 / 1303) / (7.79 \cdot 408^2)$
 = 140.11 kgf/cm²

Tangential Flange Stress, Operating [STo]:
 $= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRO$
 $= (4.09 \cdot 765576 / (408^2 \cdot 1303)) - 2.18 \cdot 123$
 $= 1172.13 \text{ kgf/cm}^2$

Tangential Flange Stress, Seating [STa]:
 $= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRA$
 $= (4.09 \cdot 874322 / (408^2 \cdot 1303)) - 2.18 \cdot 140$
 $= 1338.62 \text{ kgf/cm}^2$

Average Flange Stress, Operating [SAo]:
 $= (SHo + \max(SRO, STo)) / 2$
 $= (1706 + \max(123, 1172)) / 2$
 $= 1439.23 \text{ kgf/cm}^2$

Average Flange Stress, Seating [SAa]:
 $= (SHA + \max(SRA, STa)) / 2$
 $= (1949 + \max(140, 1339)) / 2$
 $= 1643.67 \text{ kgf/cm}^2$

Bolt Stress, Operating [BSo]:
 $= Wm1 / Ab$
 $= 2531889 / 1739$
 $= 1456.02 \text{ kgf/cm}^2$

Bolt Stress, Seating [BSa]:
 $= (Wm2 / Ab)$
 $= (53122 / 1739)$
 $= 30.55 \text{ kgf/cm}^2$

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Longitudinal Hub	1706.34	2282.03	1948.71 2814.27
Radial Flange	122.68	1521.35	140.11 1876.18
Tangential Flange	1172.13	1521.35	1338.62 1876.18
Maximum Average	1439.23	1521.35	1643.67 1876.18
Bolting	1456.02	1621.26	30.55 1621.26

Minimum Required Flange Thickness 399.593 mm

Estimated M.A.W.P. (Operating) 166 kgf/cm²
 Estimated Finished Weight of Flange at given Thk. 7528.7 kg
 Estimated Unfinished Weight of Forging at given Thk 9674.1 kg

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 874322 / 1 \cdot 100000 \cdot 0.55 / (7.29 \cdot 2081087 \cdot 66.5^2 \cdot 294 \cdot 0.3)$
 $= 0.422 \text{ (should be } \leq 1)$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 765576 / 1 \cdot 100000 \cdot 0.55 / (7.29 \cdot 1926428 \cdot 66.5^2 \cdot 294 \cdot 0.3)$
 $= 0.399 \text{ (should be } \leq 1)$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 874322 / 1 \cdot 100000 \cdot 0.55 / (7.79 \cdot 2081087 \cdot 66.5^2 \cdot 294 \cdot 0.3)$
 $= 0.395 \text{ (should be } \leq 1)$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 765576 / 1 \cdot 100000 \cdot 0.55 / (7.79 \cdot 1926428 \cdot 66.5^2 \cdot 294 \cdot 0.3)$



= 0.374 (should be ≤ 1)

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: kgf/cm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	1694.14	2814.27	1929.82	2814.27
Radial Flange	125.93	1876.18	143.45	1876.18
Tangential Flange	1156.23	1876.18	1317.09	1876.18
Maximum Average	1425.19	1876.18	1623.45	1876.18
Bolting	1456.02	1621.26	30.55	1621.26

Minimum Req'd Thickness, ambient, no Corrosion: 382.956 mm

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd kgf/cm ²	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) kgf/cm ²	Element#
LHS Ch Cover		163.17	383	1.5	2140	1521.4	1
LHS Ch flg		163.16	418	1.5	1300	1521.4	2
LHS Ch Shell		163.16	68	1.5	1300	1711	3
Main Shell		163.16	68	1.5	1300	1723.3	4
RHS Ch Shell		163.16	68	1.5	1300	1723.3	5
RHS Dish End		163.16	72	1.5	1300	1723.3	6

Element Required Thickness and MAWP :

From	To	Design Pressure kgf/cm ²	M.A.W.P. Corroded kgf/cm ²	M.A.P. New & Cold kgf/cm ²	Minimum Thickness mm	Required Thickness mm	Element#
LHS Ch Cover		163.158	170	175	375	367.702	1
LHS Ch flg		163.158	166	182	410	399.593	2
LHS Ch Shell		163.158	166	210	68	66.6864	3
Main Shell		163.158	167	210	68	66.2013	4
RHS Ch Shell		163.158	167	210	68	66.2013	5
RHS Dish End		163.158	168	211	60	58.3034	6

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:

For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

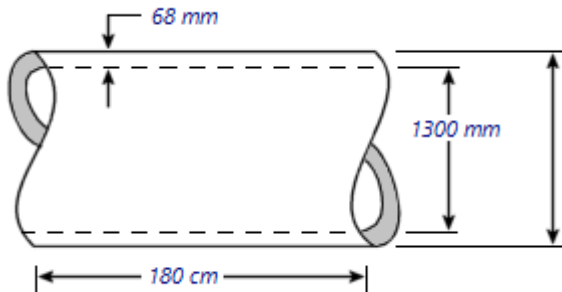
Shell Side MAWP = 167 kgf/cm²
 Shell Side MAPnc = 210 kgf/cm²
 Channel Side MAWP = 166 kgf/cm²
 Channel Side MAPnc = 175 kgf/cm²

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: LHS Ch Shell

Cylindrical Shell Design Information:					
Design Pressure	P	163.165 kgf/cm²	Int. Design Temperature	260 °C	
Material		SA-387 11	Ext. Design Temperature	260 °C	
Length		180.000 cm	External Pressure Chart	CS-3	
			UNS Number	K11789	
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	1711.0 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	2110.7 kgf/cm²
Inside Diameter	Di	1300.000 mm	Actual Stress	1598.5 kgf/cm²	
Spec. Min. Thk t _e , or sa		68.0000 mm	Spec. Nominal Thk.	68.000 mm	
Surface Area		81203.9 cm²	SG of Contents	0.050	
Element Volume		2389181.5 cm³	Weight of Contents	119.4 kgf	
Empty Weight		4077.0 kgf	Operating Weight	4196.4 kgf	
			Comp. All. Ope. Stress	1064.2 kgf/cm²	

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 1303 \cdot (\exp(163/(1711 \cdot 1)) - 1) + 1.5 + 0$
 $= 65.2 + 1.5 + 0 = 66.7 \text{ mm}$ Equation 4.3.1

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 1711 \cdot 1 \cdot \ln((2 \cdot 66.5 + 1303) / 1303) - 0.0065$
 $= 166 - 0.0065 = 166 \text{ kgf/cm}^2$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D)$
 $= 2111 \cdot 1 \cdot \ln((2 \cdot 68 + 1300) / 1300)$
 $= 210 \text{ kgf/cm}^2$

Computed Stress Intensity at Design Pressure [Si]:
 $= P \cdot D / (E (D_o - D))$
 $= 163 \cdot 1303 / (1 (1436 - 1303))$
 $= 1598.5255 \text{ kgf/cm}^2$

Computed Membrane Stress at Design Pressure [Sm]:
 $= P / (E \cdot \ln((2 \cdot t + D) / D))$



$$= 163 / (1 \cdot \ln((2 \cdot 66.5 + 1303) / 1303))$$
$$= 1678.7871 \text{ kgf/cm}^2$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 4.971 %

Minimum Design Metal Temperature Results:

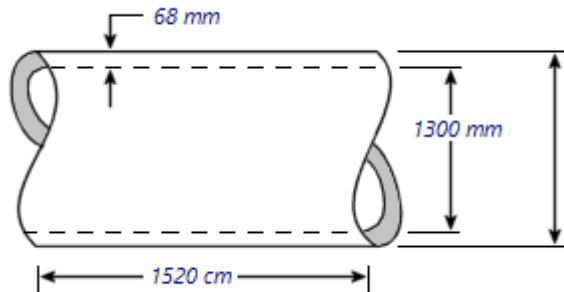
Note: This Element/Detail was specified as being Impact Tested.

Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Main Shell

Cylindrical Shell Design Information:					
Design Pressure	P	163.165 kgf/cm²	Int. Design Temperature	250 °C	
Material		SA-387 11	Ext. Design Temperature	250 °C	
Length		1520.000 cm	External Pressure Chart	CS-3	
			UNS Number	K11789	
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	1723.3 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	2110.7 kgf/cm²
Inside Diameter	Di	1300.000 mm	Actual Stress	1598.5 kgf/cm²	
Spec. Min. Thk t _e , or sa		68.0000 mm	Spec. Nominal Thk.	68.000 mm	
Surface Area		685721.8 cm²	SG of Contents	0.050	
Element Volume		14169292.0 cm³	Weight of Contents	708.1 kgf	
Empty Weight		34427.9 kgf	Operating Weight	35136.1 kgf	
			Comp. All. Ope. Stress	1083.1 kgf/cm²	

Radiography Information:			
...			...
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 1303 \cdot (\exp(163/(1723 \cdot 1)) - 1) + 1.5 + 0$
 $= 64.7 + 1.5 + 0 = 66.2 \text{ mm}$ Equation 4.3.1

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 1723 \cdot 1 \cdot \ln((2 \cdot 66.5 + 1303) / 1303) - 0.0065$
 $= 167 - 0.0065 = 167 \text{ kgf/cm}^2$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D)$
 $= 2111 \cdot 1 \cdot \ln((2 \cdot 68 + 1300) / 1300)$
 $= 210 \text{ kgf/cm}^2$

Computed Stress Intensity at Design Pressure [Si]:
 $= P \cdot D / (E (D_o - D))$
 $= 163 \cdot 1303 / (1 (1436 - 1303))$
 $= 1598.5255 \text{ kgf/cm}^2$

Computed Membrane Stress at Design Pressure [Sm]:
 $= P / (E \cdot \ln((2 \cdot t + D) / D))$



$$= 163 / (1 \cdot \ln((2 \cdot 66.5 + 1303) / 1303))$$
$$= 1678.7871 \text{ kgf/cm}^2$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 4.971 %

Minimum Design Metal Temperature Results:

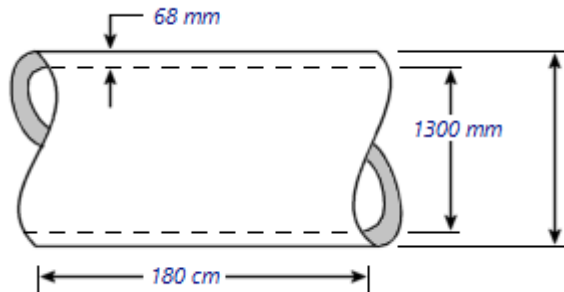
Note: This Element/Detail was specified as being Impact Tested.

Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: RHS Ch Shell

Cylindrical Shell Design Information:					
Design Pressure	P	163.165 kgf/cm²	Int. Design Temperature	250 °C	
Material		SA-387 11	Ext. Design Temperature	250 °C	
Length		180.000 cm	External Pressure Chart	CS-3	
			UNS Number	K11789	
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope) S	1723.3 kgf/cm²	
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb) Sa	2110.7 kgf/cm²	
Inside Diameter	Di	1300.000 mm	Actual Stress	1598.5 kgf/cm²	
Spec. Min. Thk t _e , or sa		68.0000 mm	Spec. Nominal Thk.	68.000 mm	
Surface Area		81203.9 cm²	SG of Contents	0.050	
Element Volume		2276469.0 cm³	Weight of Contents	113.8 kgf	
Empty Weight		4077.0 kgf	Operating Weight	4190.8 kgf	
			Comp. All. Ope. Stress	1083.1 kgf/cm²	

Radiography Information:			
...			...
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 1303 \cdot (\exp(163/(1723 \cdot 1)) - 1) + 1.5 + 0$
 $= 64.7 + 1.5 + 0 = 66.2 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 1723 \cdot 1 \cdot \ln((2 \cdot 66.5 + 1303) / 1303) - 0.0065$
 $= 167 - 0.0065 = 167 \text{ kgf/cm}^2$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D)$
 $= 2111 \cdot 1 \cdot \ln((2 \cdot 68 + 1300) / 1300)$
 $= 210 \text{ kgf/cm}^2$

Computed Stress Intensity at Design Pressure [Si]:
 $= P \cdot D / (E (D_o - D))$
 $= 163 \cdot 1303 / (1 (1436 - 1303))$
 $= 1598.5255 \text{ kgf/cm}^2$

Computed Membrane Stress at Design Pressure [Sm]:
 $= P / (E \cdot \ln((2 \cdot t + D) / D))$



$$= 163 / (1 \cdot \ln(2 \cdot 66.5 + 1303) / 1303)$$
$$= 1678.7871 \text{ kgf/cm}^2$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 4.971 %

Minimum Design Metal Temperature Results:

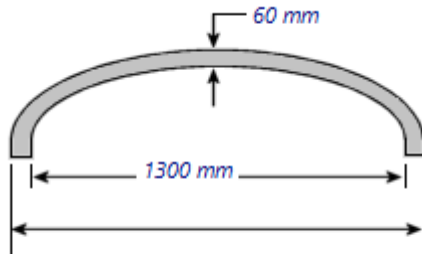
Note: This Element/Detail was specified as being Impact Tested.

Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: RHS Dish End

Elliptical Head Design Information:							
Design Pressure	P	163.158	kgf/cm ²	Int. Design Temperature		250	°C
Material		SA-387	11	Ext. Design Temperature		250	°C
Length		5.080	cm	External Pressure Chart		CS-3	
				UNS Number		K11789	
Int. Corr. All.	c	1.5000	mm	Allowable Stress (ope)	S	1723.3	kgf/cm ²
Ext. Corr. All.	ce	0.0000	mm	Allowable Stress (amb)	Sa	2110.7	kgf/cm ²
Inside Diameter	Di	1300.000	mm	Actual Stress		1674.5	kgf/cm ²
Spec. Min. Thk t,e, or sa		60.0000	mm	Spec. Nominal Thk.		72.000	mm
Surface Area		24907.1	cm ²	SG of Contents		0.000	
Element Volume		355014.7	cm ³	Weight of Contents		0.0	kgf
Empty Weight		1379.4	kgf	Operating Weight		1379.4	kgf
				Comp. All. Ope. Stress		1083.1	kgf/cm ²
				Aspect Ratio		2.00	

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 16.250 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 8.125 mm

Material treatment options: +IT

Head Calculation - Section 4.3.6 (and Section 4.3.7 for Ellipsoidal Head)

Computed Minimum Required Thickness [t]:
 tr = 56.8 mm

$$\begin{aligned}
 t &= t_r + c_i + c_o \\
 &= 56.8 + 1.5 + 0 \\
 &= 58.3 \text{ mm - see below for the derivation}
 \end{aligned}$$

For Elliptical heads: Compute [h, k, r,] per 4.3.7.1

$$\begin{aligned}
 h &= D / (2 \cdot \text{Aspect Ratio}) \\
 &= 1300 / (2 \cdot 2) \\
 &= 325.000 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 k &= D / (2 \cdot h) \\
 &= 1300 / (2 \cdot 325) \\
 &= 2.000
 \end{aligned}$$

$$r = D (0.5/k - 0.08) + c_a$$

$$= 1300 (0.5/2 - 0.08) + 1.5$$

$$= 222.500 \text{ mm}$$

$$L = D(0.44 \cdot k + 0.02) + ca$$

$$= 1300 (0.44 \cdot 2 + 0.02) + 1.5$$

$$= 1171.500 \text{ mm}$$

Step 3: Geometric Constants [βTh , ϕTh , Rth]:
 Equations 4.3.8 to 4.3.11 as appropriate
 t (corroded) and P for the calc.: $t = 56.8 \text{ mm}$, $P = 163 \text{ kgf/cm}^2$

Angle [βTh]:

$$= \arccos[(0.5 \cdot D - r)/(L - r)]$$

$$= \arccos[(0.5 \cdot 1303 - 222)/(1172 - 222)]$$

$$= 1.1 \text{ radians - Equation 4.3.8}$$

$$\phi Th = \sqrt{(L \cdot t) / r}$$

$$= \sqrt{(1172 \cdot 56.8) / 222}$$

$$= 1.16 \text{ radians - Equation 4.3.9}$$

$\phi Th > \beta Th$, Thus:

$$Rth = 0.5 \cdot D = 0.5 \cdot 1303 = 652 \text{ mm - Equation 4.3.11}$$

Step 4: Coefficients: Equations 4.3.12 to 4.3.15 [$C1$ and $C2$]:

$r/D > 0.08$, Thus:

$$C1 = 0.692(r/D) + 0.605$$

$$= 0.692(222/1303) + 0.605$$

$$= 0.72 \text{ - Equation 4.3.13}$$

$$C2 = 1.46 - 2.6(r/D) = 1.46 - 2.6(222/1303)$$

$$= 1.02, \text{ Equation 4.3.15}$$

Step 5: Pressure to produce knuckle elastic buckling [$Peth$]:

$$= C1 \cdot Et \cdot t^2 / (C2 \cdot Rth(0.5 \cdot Rth - r))$$

$$= 0.72 \cdot 1934021 \cdot 56.8^2 / (1.02 \cdot 652(0.5 \cdot 652 - 222))$$

$$= 66030 \text{ kgf/cm}^2, \text{ Equation 4.3.16}$$

Step 6: Internal Pressure for max stress in knuckle equals yield [Py]:

$$C3 = Sy = 2580 \text{ kgf/cm}^2, \text{ Not in the creep range - 4.3.6.1(f)}$$

$$Py = C3 \cdot t / (C2 \cdot Rth(0.5 \cdot Rth / r - 1))$$

$$= 2580 \cdot 56.8 / (1.02 \cdot 652(0.5 \cdot 652 / 222 - 1))$$

$$= 477 \text{ kgf/cm}^2, \text{ Equation 4.3.17}$$

Step 7: Value of Internal Press. resulting in buckling of knuckle [G and Pck]:

$$G = Peth/Py = 66030/477 = 138$$

$G > 1$, Thus:

Buckling Pressure [Pck]:

$$= ((0.77508 \cdot G - 0.20354 \cdot G^2 + 0.019274 \cdot G^3) /$$

$$(1 + 0.19014 \cdot G - 0.089534 \cdot G^2 + 0.0093965 \cdot G^3)) Py$$

$$= ((0.77508 \cdot 138 - 0.20354 \cdot 138^2 + 0.019274 \cdot 138^3) /$$

$$(1 + 0.19014 \cdot 138 - 0.089534 \cdot 138^2 + 0.0093965 \cdot 138^3)) \cdot 477$$

$$= 972 \text{ kgf/cm}^2, \text{ Equation 4.3.19}$$

Step 8: Compute Allowable Pressure base on buckling of knuckle [Pak]:

$$= Pck/1.5 = 972/1.5 = 648 \text{ kgf/cm}^2 \text{ - Equation 4.3.21}$$

Step 9: Compute allowable pressure based on rupture of crown [Pac]:

$$= 2 \cdot S \cdot E / (L/t + 0.5)$$

$$= 2 \cdot 1723 \cdot 1 / (1172/56.8 + 0.5)$$

$$= 163 \text{ kgf/cm}^2, \text{ Equation 4.3.22}$$

Step 10: Compute final pressure [Pa]:

$$= \min(Pak, Pac) = \min(648, 163)$$

$$= 163 \text{ kgf/cm}^2, \text{ Equation 4.3.23: (same as original design pressure)}$$

Allowable pressure in Crown at given thickness [Pac]:

$$= 2 \cdot S \cdot E / (L/t + 0.5)$$



$$= 2 \cdot 1723 \cdot 1 / (1172 / 58.5 + 0.5) \\ = 167.913 \text{ kgf/cm}^2$$

Computed Maximum Allowable Pressure [MAWP]:
= MaxPressure - Pliq = 168 - 0 = 168 kgf/cm²

Computed Maximum Allowable Pressure New and Cold [MAPnc]:
= 211 kgf/cm²

Computed Membrane Stress Intensity at Design Pressure [Sm]:
= 1674.5 kgf/cm²

Required Thickness of Straight Flange = 66.199 mm

Percent Fiber Elongation per Table 6.1

For Heads formed from Segments $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 21.012 %
For one Piece doubly Curved Heads $100 \cdot \ln(D_b / (D_f - 2t))$ 31.470 %

Where D_b is the blank diameter for one piece doubly curved surfaces [D_b]:
= square root($4 \cdot \text{Head Surface Area} / 3.14159$)
= square root($4 \cdot 24907 / 3.14159$)
= 1780.806 mm

MDMT Calculations in the Knuckle Portion:

Note: This Element/Detail was specified as being Impact Tested.

MDMT Calculations in the Head Straight Flange:

Note: This Element/Detail was specified as being Impact Tested.

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 16.250 mm
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 8.125 mm

Material treatment options: +IT

EXTERNAL PRESSURE RESULTS for: RHS Dish End

Elliptical Head From 80 to 90 Ext. Chart: CS-2 at 250 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:
= $0.075 \cdot E_y(t / R_o)$
= $0.075 \cdot 1934021 (58.5 / 1181)$
= 7183.965 kgf/cm²

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:
= Fhe / Sy
= 7184.0 / 2579.8
= 2.785

Engineering Yield to Tensile Stress ratio [R]:
= $\sigma_{ys} / \sigma_{uts}$
= 2580 / 5273
= 0.489

Ratio [AE]:
= Fhe / E
= 7184 / 1934021
= 0.003715

Factor [m1]:
= $(\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$
= $(\ln(0.49) + (0.20000E-04 - 0.002)) / \ln(\ln(1 + 0.20000E-04) / \ln(1 + 0.002))$
= 0.156

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys}) m_1$$

$$= 2580 (1 + 0.0020) / \ln(1 + 0.0020) 0.15570091$$

$$= 6803.810 \text{ kgf/cm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m_2 f(m_2 f_2 - R)$$

$$= 0.6 (1 - 0.49)$$

$$= 0.306$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m_2)) / (m_2 m_2)$$

$$= (5273 \cdot \exp(0.31)) / (0.31 \cdot 0.31)$$

$$= 10293.345 \text{ kgf/cm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R_{1.5} - 0.5 \cdot R_{2.5} - R_{3.5}$$

$$= 1.5 \cdot 0.491.5 - 0.5 \cdot 0.492.5 - 0.493.5$$

$$= 0.348$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	2636	-1.879	0.276E-05	0.242E-05	0.727E-05	-0.582E-05	0.140E+06	0.151E-01
2	1318	-4.694	0.644E-07	0.644E-07	0.151E-05	-0.151E-05	0.155E+07	0.286E-02
3	1977	-3.287	0.581E-06	0.577E-06	0.379E-05	-0.375E-05	0.584E+06	0.329E-03
4	2306	-2.583	0.134E-05	0.130E-05	0.537E-05	-0.513E-05	0.294E+06	0.413E-02
5	2142	-2.935	0.896E-06	0.884E-06	0.454E-05	-0.445E-05	0.418E+06	0.141E-02
6	2059	-3.111	0.724E-06	0.718E-06	0.416E-05	-0.410E-05	0.495E+06	0.446E-03
7	2018	-3.199	0.649E-06	0.644E-06	0.397E-05	-0.392E-05	0.538E+06	0.371E-04
8	1997	-3.243	0.614E-06	0.610E-06	0.388E-05	-0.384E-05	0.561E+06	0.151E-03
9	2008	-3.221	0.631E-06	0.627E-06	0.392E-05	-0.388E-05	0.549E+06	0.582E-04
10	2013	-3.210	0.640E-06	0.635E-06	0.395E-05	-0.390E-05	0.544E+06	0.109E-04
11	2016	-3.204	0.645E-06	0.640E-06	0.396E-05	-0.391E-05	0.541E+06	0.130E-04
12	2014	-3.207	0.643E-06	0.638E-06	0.395E-05	-0.391E-05	0.542E+06	0.102E-05
13	2014	-3.208	0.641E-06	0.637E-06	0.395E-05	-0.390E-05	0.543E+06	0.495E-05
14	2014	-3.208	0.642E-06	0.637E-06	0.395E-05	-0.391E-05	0.543E+06	0.197E-05
15	2014	-3.207	0.642E-06	0.637E-06	0.395E-05	-0.391E-05	0.542E+06	0.476E-06
16	2014	-3.207	0.642E-06	0.637E-06	0.395E-05	-0.391E-05	0.542E+06	0.271E-06
17	2014	-3.207	0.642E-06	0.637E-06	0.395E-05	-0.391E-05	0.542E+06	0.102E-06

Factor of Safety [FS]:

$$\text{Since } F_{ic} > 0.55 F_y \text{ and } F_{ic} < F_y$$

$$= 2.407 - 0.741 \cdot F_{ic} / F_y$$

$$= 2.407 - 0.741 \cdot 2015/2580$$

$$= 1.828$$

Allowable Circumferential Compressive Stress [Fha]:

$$= F_{ic} / FS$$

$$= 2015/1.83$$

$$= 1101.871 \text{ kgf/cm}^2$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot F_{ha} [t/R_o]$$

$$= 2 \cdot 1102 [58.5/1181]$$

$$= 109.145 \text{ kgf/cm}^2$$

Hydrostatic Test Pressure Results:

Hydrotest Pressure Calculation [Shell Side]:

= Test Factor • ShellSide MAWP(St/S)

= 1.25 • 167 (1.22)

= 256.419 kgf/cm²

Hydrotest Pressure Calculation [Channel Side]:

= Test Factor • Channel Side MAWP(St/S)

= 1.25 • 166 (1.23)

= 255.658 kgf/cm²

Horizontal Test performed per: per VIII-2, Section 8.2

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (kgf/cm²):

From To	Stress	Allowable	Ratio	Pressure
LHS Ch Shell	2570.8	3002.9	0.856	255.79
Main Shell	2578.5	3002.9	0.859	256.55
RHS Ch Shell	2578.5	3002.9	0.859	256.55
RHS Dish End	2565.5	3002.9	0.854	256.55

Theoretical Maximum Allowable test pressure Shellside is : 298.651 kgf/cm²

Theoretical Maximum Allowable test pressure Tubeside : 298.651 kgf/cm²

Stress ratios for Nozzle and Pad Materials (kgf/cm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N1-Tube Inlet	Nozzle	1876.18	1521.35	1.233
N3-Shell Inlet	Nozzle	1876.18	1529.52	1.227
N4-Shell outlet	Nozzle	1876.18	1529.52	1.227
N2-Tube Outlet	Nozzle	1876.18	1529.52	1.227

Minimum 1.227

Stress ratios for Pressurized Vessel Elements (kgf/cm²):

Description	Ambient	Operating	Ratio
LHS Ch Cover	1876.18	1521.35	1.233
LHS Ch flg	1876.18	1521.35	1.233
LHS Ch Shell	2110.70	1711.02	1.234
Main Shell	2110.70	1723.26	1.225
RHS Ch Shell	2110.70	1723.26	1.225
RHS Dish End	2110.70	1723.26	1.225

Minimum 1.225

Stress ratios for Exchanger Materials (kgf/cm²):

Description	Ambient	Operating	Ratio
Tube Material	1407.13	1144.08	1.230
Tubesheet Material	1876.18	1521.35	1.233

Minimum 1.230

Hoop Stress in Nozzle Wall during Pressure Test (kgf/cm²):

Description	Ambient	Operating	Ratio
-------------	---------	-----------	-------



N1-Tube Inlet	1865.97	2663.87	0.700
N3-Shell Inlet	1866.14	2663.87	0.701
N4-Shell outlet	1871.52	2663.87	0.703
N2-Tube Outlet	1866.14	2663.87	0.701

Elements Suitable for Test Pressure.

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Input Echo, Tubesheet Number 1, Description:

Shell Data:

Main Shell Description: Main Shell

Shell Test Pressure		256.42 kgf/cm ²
Shell Thickness	ts	68.0000 mm
Shell Internal Corrosion Allowance	cas	0.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	1300.000 mm
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-387 11
Shell Material UNS Number		K11789
Shell Allowable Stress at Temperature	Ss	2110.70 kgf/cm ²
Shell Allowable Stress at Ambient		2110.70 kgf/cm ²

Channel Data:

Channel Description: LHS Ch Shell

Channel Type:		Cylinder
Channel Test Pressure		255.66 kgf/cm ²
Channel Thickness	tc	68.0000 mm
Channel Internal Corrosion Allowance	cac	0.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	1300.000 mm
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-387 11
Channel Material UNS Number		K11789
Channel Allowable Stress at Temperature	Sc	2110.70 kgf/cm ²
Channel Allowable Stress at Ambient		2110.70 kgf/cm ²

Tube Data:

Number of Tube Holes	Nt	2615
Tube Wall Thickness	et	2.0000 mm
Tube Outside Diameter	D	14.0000 mm
Total Straight Tube Length	Lt	15200.00 mm
Straight Tube Length (bet. inner tubsht faces) L		14640.00 mm
Design Temperature of the Tubes		21.11 °C
Tube Material		SA-213 T11
Tube Material UNS Number		K11597
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		1407.13 kgf/cm ²
Tube Allowable Stress At Ambient		1407.13 kgf/cm ²
Tube Yield Stress At design Temperature	Syt	2090.31 kgf/cm ²
Tube Pitch (Center to Center Spacing)	P	19.0000 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.0000 mm
Groove Weld Leg	ag	2.0000 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		4.21.2
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	577.862 mm
Largest Center-to-Center Tube Distance	Ul	0.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	6.0000 mm

Tubesheet Data:

Tubesheet TYPE: Stationary, Integral both Sides, Conf. a
Exchanger TYPE: With an Immersed Floating Head, Conf. a

Tubesheet Design Metal Temperature	T	21.11 °C
Tubesheet Material		SA-336 F11
Tubesheet Material UNS Number		K11572
Tubesheet Allowable Stress at Temperature	S	1876.18 kgf/cm ²
Tubesheet Allowable Stress at Ambient	Tt	1876.18 kgf/cm ²
Thickness of Tubesheet	h	280.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	0.0000 mm

Tubesheet Corr. Allowance (Channel side) Catc 0.0000 mm
 Tubesheet Outside Diameter A 1444.000 mm

Area of the Untubed Lanes AL 0.0 cm²

Additional Data for Fixed/Floating Tubesheet Exchangers:

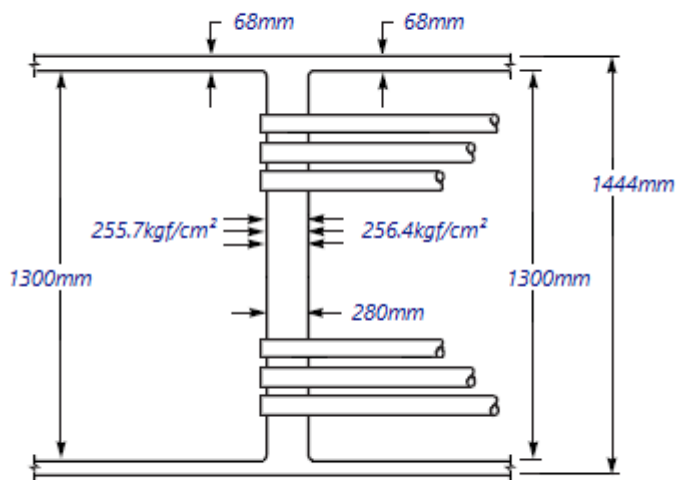
Unsupported Tube Span under consideration l 0.000 mm
 Tube End condition corresponding to Span (l) k 0.00

Ignore Radial Thermal Exp. effects (4.18.8/4.18.9) Yes
 Perform Differential Pressure Design No
 Run Multiple Load Cases Yes
 Junction Stress Reduction option Perform Plastic Calculation

Tubesheet Integral with Both
 Tubesheet Extended as Flange No

Is Exchanger in Creep range (skip EP, Use 3S for Sps) No

Tubesheet Integral w/Shell and Channel



Configuration a:

Messages regarding the Tubesheet & Exchanger Input Data:

Warning - Tube end condition k should be bet 0 - 1 (.00).

ASME Division 2 Tubesheet Results per Part 4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
 Shell - TM-1 Material Group C

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20811E+07 kgf/cm ²
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.20811E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20811E+07 kgf/cm ²

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
 Channel - TM-1 Material Group C

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20811E+07 kgf/cm ²



Elastic Mod. at Ambient Temperature 21.1 °C 0.20811E+07 kgf/cm²

Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
 Tubes - TM-1 Material Group C

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20811E+07 kgf/cm ²
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.20811E+07 kgf/cm ²
Elastic Mod. at Tubsht. Design Temp.	21.1 °C	0.20811E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20811E+07 kgf/cm ²

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
 TubeSheet - TM-1 Material Group C

Elastic Mod. at Design Temperature	21.1 °C	0.20811E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20811E+07 kgf/cm ²

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.
 Please insure these properties are consistent with the
 type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i$ Equation 4.3.1
 $= 0.5 \cdot 10 \cdot (\exp(256/(1407 \cdot 1)) - 1) + 0$
 $= 1 + 0$
 $= 1 \text{ mm}$

Tube Required Thickness under External Pressure (Shellside pressure) :

Determine Parameter [Mx]:

$= L / \sqrt{(R_o \cdot t)}$
 $= 14640 / \sqrt{(7 \cdot 2)}$
 $= 3912.705$

Compute the Factor [Ch]:

Since $M_x > 2 \cdot (D_o/t)^{0.94}$, $Ch = 0.55 / (D_o/t)$
 $= 0.55 \cdot (14/2)$
 $= 0.0786$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$= 1.6 \cdot Ch \cdot E_y (t / D_o)$
 $= 1.6 \cdot 0.079 \cdot 2081087 (2/14)$
 $= 37374.625 \text{ kgf/cm}^2$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$= Fhe / F_y$
 $= 37374.6/2090.3$
 $= 17.880$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$= \sigma_{ys} / \sigma_{uts}$
 $= 2090/4218$
 $= 0.496$

Ratio [AE]:

$= Fhe / E$
 $= 37375/2081087$
 $= 0.017959$

Factor [m1]:

$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$
 $= (\ln(0.5) + (0.20000E-04 - 0.002)) / \ln(\ln(1 + 0.20000E-04) / \ln(1 + 0.002))$
 $= 0.153$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys}) m_1$$

$$= 2090 (1 + 0.0020) / \ln(1 + 0.0020) 0.15293255$$

$$= 5418.821 \text{ kgf/cm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m_2 f(m_2^2 - R)$$

$$= 0.6 (1 - 0.5)$$

$$= 0.303$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m_2)) / (m_2^2)$$

$$= (4218 \cdot \exp(0.3)) / (0.3 \cdot 0.3)$$

$$= 8197.895 \text{ kgf/cm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R^{1.5} - 0.5 \cdot R^{2.5} - R^{3.5}$$

$$= 1.5 \cdot 0.515 - 0.5 \cdot 0.525 - 0.535$$

$$= 0.351$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	2109	-1.949	0.324E-05	0.290E-05	0.883E-05	-0.731E-05	0.123E+06	0.782E-03
2	3163	0.873	0.306E-04	-0.416E-04	0.225E-04	0.449E-04	0.176E+05	0.162E+00
3	2636	-0.538	0.112E-04	-0.365E-05	0.148E-04	0.167E-04	0.254E+05	0.860E-01
4	2372	-1.244	0.623E-05	0.356E-05	0.116E-04	-0.350E-05	0.545E+05	0.256E-01
5	2241	-1.597	0.454E-05	0.355E-05	0.102E-04	-0.656E-05	0.823E+05	0.929E-02
6	2175	-1.773	0.384E-05	0.326E-05	0.948E-05	-0.713E-05	0.101E+06	0.365E-02
7	2142	-1.861	0.353E-05	0.308E-05	0.915E-05	-0.726E-05	0.111E+06	0.130E-02
8	2125	-1.905	0.338E-05	0.299E-05	0.899E-05	-0.729E-05	0.117E+06	0.228E-03
9	2117	-1.927	0.331E-05	0.294E-05	0.891E-05	-0.730E-05	0.120E+06	0.284E-03
10	2121	-1.916	0.335E-05	0.297E-05	0.895E-05	-0.730E-05	0.118E+06	0.302E-04
11	2123	-1.911	0.337E-05	0.298E-05	0.897E-05	-0.729E-05	0.118E+06	0.983E-04
12	2122	-1.914	0.336E-05	0.297E-05	0.896E-05	-0.729E-05	0.118E+06	0.339E-04
13	2122	-1.915	0.335E-05	0.297E-05	0.896E-05	-0.729E-05	0.118E+06	0.182E-05
14	2121	-1.916	0.335E-05	0.297E-05	0.895E-05	-0.729E-05	0.118E+06	0.142E-04
15	2121	-1.915	0.335E-05	0.297E-05	0.895E-05	-0.729E-05	0.118E+06	0.620E-05
16	2122	-1.915	0.335E-05	0.297E-05	0.896E-05	-0.729E-05	0.118E+06	0.219E-05

Factor of Safety [FS]:

Since Fic [2122] >= Fy[2090], FS = 1.667

Allowable Circumferential Compressive Stress [Fha]:

$$= F_{ic} / FS$$

$$= 2122 / 1.67$$

$$= 1272.957 \text{ kgf/cm}^2$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot F_{ha} (t / D_o)$$

$$= 2 \cdot 1273 (2 / 14)$$

$$= 363.702 \text{ kgf/cm}^2$$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	1.5521 mm
Allowable Internal Pressure at Corroded thickness	473.46 kgf/cm ²
Required Internal Design Pressure	255.66 kgf/cm ²
Allowable External Pressure at Corroded thickness	363.70 kgf/cm ²
Required External Design Pressure	256.42 kgf/cm ²
Required Thickness due to Shell Side pressure	1.5521 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Results for ASME Stationary Tubesheet Calculations for Configuration a:

Results for Tubesheet Calculations Original Thickness:



ASME Section VIII-2 Part 4.18.9.4 Step 1:

Tube expansion depth ratio [ρ]:

$$\begin{aligned} &= l_{tx} / h \text{ (modified for corrosion if present)} \\ &= 0/280 \\ &= 0.0000 \text{ (must be } 0 \leq \rho \leq 1 \text{)} \end{aligned}$$

Effective tube hole diameter [d^*]:

$$\begin{aligned} &= \max(dt - 2 \cdot tt(EtT/E) \cdot (StT/S) \cdot (\rho); dt - 2 \cdot tt) \\ &= \max(14 - 2 \cdot 2 (2081087/2081087) \cdot \\ &\quad (1407/1876) \cdot (0); 14 - 2 \cdot 2) \\ &= 14.0000 \text{ mm} \end{aligned}$$

Equivalent outer tube limit circle diameter [Do]:

$$\begin{aligned} &= 2 \cdot r_o + dt \\ &= 2 \cdot 578 + 14 \\ &= 1169.724 \text{ mm} \end{aligned}$$

Determine the basic ligament efficiency for shear [μ]:

$$\begin{aligned} &= (p - dt) / p \\ &= (19 - 14) / 19 \\ &= 0.263 \end{aligned}$$

Equivalent outer tube limit radius [ao]:

$$\begin{aligned} &= Do / 2 \\ &= 1170/2 \\ &= 584.8620 \text{ mm} \end{aligned}$$

Effective tube pitch [p^*]:

$$\begin{aligned} &= p / \sqrt{ (1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot Do \cdot p) / (\pi \cdot Do^2)) } \\ &= 19 / \sqrt{ (1 - 4 \cdot \min(0 \cdot 100 ; 4 \cdot 1170 \cdot 19) / (\pi \cdot 1170^2)) } \\ &= 19.0000 \text{ mm} \end{aligned}$$

Effective ligament efficiency for bending [μ^*]:

$$\begin{aligned} &= (p^* - d^*) / p^* \\ &= (19 - 14) / 19 \\ &= 0.2632 \end{aligned}$$

Ratio [ps]:

$$\begin{aligned} &= as / ao \\ &= 650/585 \\ &= 1.1114 \end{aligned}$$

Ratio [pc]:

$$\begin{aligned} &= ac / ao \\ &= 650/585 \\ &= 1.1114 \end{aligned}$$

Compute parameter [xt]:

$$\begin{aligned} &= 1 - Nt((dt - 2 \cdot tt) / (2 \cdot ao))^2 \\ &= 1 - 2615 ((14 - 2 \cdot 2) / (2 \cdot 585))^2 \\ &= 0.8089 \end{aligned}$$

Determine parameter [xs]:

$$\begin{aligned} &= 1 - Nt(dt / (2 \cdot ao))^2 \\ &= 1 - 2615 (14 / (2 \cdot 585))^2 \\ &= 0.6254 \end{aligned}$$

Determine the value [$h'g$]:

$$\begin{aligned} &= \max((hg - CATC); 0) \text{ (For pressure only cases)} \\ &= \max((6 - 0); 0) \\ &= 6.000 \text{ mm} \end{aligned}$$

ASME Section VIII-2 Part 4.18.9.4 Step 2:

Shell coefficient [β_s]:

$$\begin{aligned} &= ((12(1 - vs^2))^{1/4}) / ((Ds + ts)ts) \\ &= ((12(1 - 0.3^2))^{1/4}) / ((1300 + 68)68) \\ &= 0.0060 \text{ 1/mm} \end{aligned}$$

Shell coefficient [ks]:

$$= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - \nu_s^2))$$

$$= 0.006 \cdot 2081087 \cdot 68^3 / (6(1 - 0.3^2))$$

$$= 714303168.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Shell coefficient [λs]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(1300 + 68) \cdot 714303168 / 280^3 \cdot (1 + 280 \cdot 0.006 + (280 \cdot 0.006)^2 / 2)$$

$$= 1084721.1250 \text{ kgf/cm}^2$$

Shell coefficient [δs]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot \nu_s / (2(D_s + t_s)))$$

$$= 1300^2 / (4 \cdot 2081087 \cdot 68) \cdot (1 - 1300 \cdot 0.3 / (2(1300 + 68)))$$

$$= 0.25600\text{E-}02 \text{ mm/kgf/cm}^2$$

Channel coefficient [βc]:

$$= ((12(1 - \nu_c^2))^{1/4}) / ((D_c + t_c)t_c)$$

$$= ((12(1 - 0.3^2))^{1/4}) / ((1300 + 68)68)$$

$$= 0.0060 \text{ 1/mm}$$

Channel coefficient [kc]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - \nu_c^2))$$

$$= 0.006 \cdot 2081087 \cdot 68^3 / (6(1 - 0.3^2))$$

$$= 714303168.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λc]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (Cm\Delta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot Cq\Delta)$$

$$= 6(1300 + 68 / \cos(0)) \cdot 714303168 / 280^3 \cdot (1 + 280 \cdot 0.006 \cdot (1 + 1) / 2 + (280 \cdot 0.006)^2 / 2 \cdot 1)$$

$$= 1084721.1 \text{ kgf/cm}^2$$

Channel coefficient [δc]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot \nu_c / (2(D_c + t_c)))$$

$$= 1300^2 / (4 \cdot 2081087 \cdot 68) \cdot (1 - 1300 \cdot 0.3 / (2(1300 + 68)))$$

$$= 0.25600\text{E-}02 \text{ mm/kgf/cm}^2$$

Channel coefficient [zetac]:

$$zetac = 0$$

ASME Section VIII-2 Part 4.18.9.4 Step 3:

$$h/p = 14.736844 ; \mu^* = 0.263158 ; E^* = 485641 \text{ kgf/cm}^2$$

$$E^*/E = 0.233360 ; \nu^* = 0.383525$$

Note: As h/p (14.737) is > 2, data values for h/p = 2 were used.

Tube Bundle Stiffness Factor [Xa]:

$$= ((24(1 - \nu^*)Nt \cdot Et \cdot t_t(d_t - t_t)ao^2) / (E^* \cdot L \cdot H^3))^{1/4}$$

$$= ((24(1 - 0.38^2)2615 \cdot 2081087 \cdot 2(14 - 2)585^2) / (485642 \cdot 14640 \cdot 280^3))^{1/4}$$

$$= 1.5558$$

Values from Table 4.18.3

$$Z_d = 0.393525 ; Z_v = 0.170662 ; Z_m = 0.692948$$

$$Z_a = 0.578496\text{E}+00 ; Z_w = 0.170662$$

ASME Section VIII-2 Part 4.18.9.4 Step 4:

Diameter ratio [K]:

$$= A / D_o$$

$$= 1444/1170$$

$$= 1.2345$$

Compute coefficient [F]:

$$= (1 - \nu^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$



$$= (1 - 0.38) / 485642 \cdot (1084721 + 1084721 + 2081087 \cdot \ln(1.23))$$

$$= 3.3104$$

Compute parameter [φ]:

$$= (1 + \nu) F$$

$$= (1 + 0.38) 3.31$$

$$= 4.5800$$

Compute parameter [Q1]:

$$= (\rho_s - 1 - \phi \cdot Z_v) / (1 + \phi \cdot Z_m)$$

$$= (1.11 - 1 - 4.58 \cdot 0.17) / (1 + 4.58 \cdot 0.69)$$

$$= -0.160590$$

ASME Section VIII-2 Part 4.18.9.4 Step 5:

Determine factor [γ_b]:

$$= 0 \text{ (config a)}$$

Calculate parameter [ω_s]:

$$= \rho_s \cdot k_s \cdot \beta_s \cdot \delta S (1 + h \cdot \beta_s)$$

$$= 1.11 \cdot 714303168 \cdot 0.006 \cdot 0.002559997 (1 + 280 \cdot 0.006)$$

$$= 32326.8672 \text{ mm}^2$$

Calculate parameter [ω*_s]:

$$= a o^2 (\rho_s^2 - 1) \cdot (\rho_s - 1) / 4 - \omega_s$$

$$= 585^2 (1.11^2 - 1) \cdot (1.11 - 1) / 4 - 32327$$

$$= -30087.2539 \text{ mm}^2$$

Parameter [ω_c]:

$$= \rho_c \cdot k_c [\delta C (C_m \theta + \beta C + C_q \Delta \cdot h \cdot \beta C^2) - \zeta c (C_m \theta + h \cdot C_q \theta + h \cdot \beta C / 2)]$$

$$= 1.11 \cdot 714303168 [0.002559997 (1 + 0.006 \cdot 1 \cdot 280) - 0.000 (1/280 + 1 \cdot 0.006/2)]$$

$$= 32326.8672 \text{ mm}^2$$

Parameter [ω*_c]:

$$= a o^2 [(\rho_c^2 + 1) \cdot (\rho_c - 1) / 4 - (\rho_s - 1) / 2] - \omega_c$$

$$= 585^2 [(1.11^2 + 1) \cdot (1.11 - 1) / 4 - (1.11 - 1) / 2] - 32327$$

$$= -30087.2539 \text{ mm}^2$$

Pressure [P*_S]:

$$= 0 \text{ For Pressure only cases or Configurations d,e,f,A,B,C,D}$$

Pressure [P*_C]:

$$= 0 \text{ For Pressure only cases or Configurations b,c,d,B,C,D}$$

ASME Section VIII-2 Part 4.18.9.4 Step 6:

$$P_e = P_s - P_t$$

$$= 256 - 256 = 0.76 \text{ kgf/cm}^2$$

ASME Section VIII-2 Part 4.18.9.4 Step 7:

Determine factor [Q2]:

$$= [(\omega^* \cdot P_s - \omega^* \cdot P_t) - (\omega_s \cdot P^* \cdot s - \omega_c \cdot P^* \cdot c)] + W^* \cdot \gamma_b / (2 \cdot \pi) / (1 + \phi \cdot Z_m)$$

$$= [(-30087 \cdot 256 - -30087 \cdot 256) - (32327 \cdot 0 - 32327 \cdot 0)] \cdot 0.01 + 0 \cdot 0 / (2 \cdot \pi) / (1 + 4.58 \cdot 0.69)$$

$$= -54.8573 \text{ kgf}$$

Calculate factor [Q3]:

$$= Q1 + 2 \cdot Q2 / (P_e \cdot a o^2)$$

$$= -0.16 + 2 \cdot -54.9 / (0.76 \cdot 585^2)$$

$$= -0.20274$$

F_m Value from Table 4.18.3 = 0.103104

Tubesheet Bending Stress - original thickness [σ]:

$$\begin{aligned} &= (1.5 \cdot F_m / \mu \cdot) \cdot (2 \cdot a_o / (H - h'g))^2 \cdot P_e \\ &= (1.5 \cdot 0.1 / 0.26) \cdot (2 \cdot 585 / (280 - 6))^2 \cdot 0.76 \\ &= 8.1506 \text{ kgf/cm}^2 \end{aligned}$$

Allowable Tubesheet Bending Stress [σ allowed]:
 $= 1.5 \cdot \beta T \cdot S_y = 1.5 \cdot 0.95 \cdot 2804 = 3996 \text{ kgf/cm}^2, 4.18.233$

Tubesheet Bending Stress - final thickness [σ_f]:
 $= (1.5 \cdot F_m / \mu \cdot) \cdot (2 \cdot a_o / (h - h'g))^2 \cdot P_e$
 $= (1.5 \cdot 0.022 / 0.26) \cdot (2 \cdot 585 / (11.7 - 6))^2 \cdot 0.76$
 $= 3995.6655 \text{ kgf/cm}^2$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:
 $= h + \text{Cats} + \text{Catc}$
 $= 11.7 + 0 + 0$
 $= 11.6773 \text{ mm}$

ASME Section VIII-2 Part 4.18.9.4 Step 8:

Shear Stress check [τ_{limit}]:
 $= (2 \cdot \mu \cdot h / a_o) \cdot (0.8 \cdot \beta T \cdot S_y)$
 $= (2 \cdot 0.26 \cdot 280 / 585) \cdot 2131$
 $= 536.9745 \text{ kgf/cm}^2$

Shear Stress is not required to be computed; [P_e] $\leq$ τ_{limit}

Tubesheet Shear Stress is probably low, use the following req. thk:

Tubesheet thickness (Incl. Corr.) = 3.8100 mm
Tubesheet Shear Stress = 221.9513 kgf/cm²

Reqd Tubesheet Thickness for Given Loadings (including CA) [Hreqd]:
 $= \max(\text{HreqB}; \text{HreqS})$
 $= \max(11.7; 3.81)$
 $= 11.6773 \text{ mm}$

ASME Section VIII-2 Part 4.18.9.4 Step 9:

The F_{tmin} and F_{tmax} Coefficients from Table 4.18.4:
 $F_{\text{tmin}} = 0.92, F_{\text{tmax}} = 1.05$

First Extreme Tube Axial Stress from among all the tubes [σ_{t1}]:
 $= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{\text{tmin}}) / (X_t - X_s)$
 $= ((256 \cdot 0.63 - 256 \cdot 0.81) - (0.76) \cdot 0.92) /$
 $(0.81 - 0.63)$
 $= -256.8777 \text{ kgf/cm}^2$

Second Extreme value of Tube Axial Stress from among all the tubes [σ_{t2}]:
 $= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{\text{tmax}}) / (X_t - X_s)$
 $= ((256 \cdot 0.63 - 256 \cdot 0.81) - (0.76) \cdot 1.05) /$
 $(0.81 - 0.63)$
 $= -257.4258 \text{ kgf/cm}^2$

Maximum Tube Axial Stress [$\sigma_{t,\text{max}}$]:
 $= \max(|\sigma_{t,1}|; |\sigma_{t,2}|) = 257 \text{ kgf/cm}^2$

Allowable Tube Stress [$\sigma_{t,A}$]:
 $= \beta T \cdot S_{y,t} = 1985.7927 \text{ kgf/cm}^2$

Check for Buckling as some of the tubes are in compression.

Determine the Factor of Safety [Fs]:
 $= \max((3.25 - 0.25(Z_d + Q3 \cdot Z_w)X_a^4); 1.25)$
 $= \max((3.25 - 0.25(0.39 + -0.2 \cdot 0.17)1.56^4); 1.25)$
 $= 1.2500 \text{ (Should be } \leq 2)$

Determine the factor [rt]:
 $= (\sqrt{dt^2 + (dt - 2 \cdot tt)^2}) / 4$
 $= (\sqrt{14^2 + (14 - 2 \cdot 2)^2}) / 4$
 $= 4.3012 \text{ mm}$

Determine the factor [Ct]:

$$\begin{aligned} &= \sqrt{(2 \cdot \pi^2 \cdot Et / Syt)} \\ &= \sqrt{(2 \cdot \pi^2 \cdot 2081087 / 2090)} \\ &= 149.0780 \end{aligned}$$

Determine the factor [Ft]:

$$\begin{aligned} &= k \cdot L / r \\ &= 0 \cdot 0 / 4.3 \\ &= 0.0000 \end{aligned}$$

Buckling Allowable Stress [Stb]:

$$\begin{aligned} &= (Sy / t / Fs) \cdot (1 - Ft / (2 \cdot Ct)) \\ &= (2090 / 1.25) \cdot (1 - 0 / (2 \cdot 149)) \\ &= 1985.793 \text{ kgf/cm}^2 \quad (\text{Never greater than } Sot) \end{aligned}$$

Note: The Axial Compressive stress in tubes is within limits.

Largest tube-to-tubesheet Joint Load [Wt]:

$$\begin{aligned} &= \sigma_{t,max} \cdot \text{Tube Area} \\ &= 257 \cdot 0.75 \\ &= 194.09 \text{ kgf} \end{aligned}$$

Tube Metal Area [At]:

$$\begin{aligned} &= \pi (do - t) t \\ &= \pi (14 - 2) 2 \\ &= 0.754 \text{ cm}^2 \end{aligned}$$

Tube Weld Size Results per VIII-2 4.21.2:

Tube Strength [Ft]:

$$\begin{aligned} &= \pi \cdot t (do - t) S \\ &= \pi \cdot 2 (14 - 2) 1986 \\ &= 1497.250 \text{ kgf} \end{aligned}$$

Groove Weld Strength [Fg]:

$$\begin{aligned} &= 0.85 \cdot \pi \cdot ag (do + 0.67 \cdot ag) Sw \quad (\text{but not } > Ft) \\ &= 0.85 \cdot \pi \cdot 2 (14 + 0.67 \cdot 2) 1986 \\ &= 1497.2505 \text{ kgf} \end{aligned}$$

Weld Strength Factor [fw]:

$$\begin{aligned} &= S / Sw \quad \text{where } Sw = \min(S \text{ and } St) \\ &= 1986 / 1986 \\ &= 1.000 \end{aligned}$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$\begin{aligned} &= 0.55 \cdot \pi \cdot af (do + 0.67 \cdot af) Sw \quad (\text{but not } > Ft) \\ &= 0.55 \cdot \pi \cdot 2 (14 + 0.67 \cdot 2) 1986 \\ &= 1052.6918 \text{ kgf} \end{aligned}$$

Minimum Length of Weld [ar]:

$$\begin{aligned} &= 2 \left(\sqrt{(0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t) fw \cdot fd} - 0.75 \cdot do \right) \\ &= 2 \left(\sqrt{(0.75 \cdot 14)^2 + 1.07 \cdot 2 \cdot (14 - 2) 1 \cdot 1} - 0.75 \cdot 14 \right) \\ &= 2.318 \text{ mm} \end{aligned}$$

Minimum Required Fillet Weld Leg [afr]: 1.2000 mm

Minimum Required Groove Weld Leg [agr]: 1.2000 mm

Tube-Tubesheet Jt allowable, 1497 is $\geq$ tube strength 1497 kgf

Note: This tube-tubesheet joint is a Full Strength joint

ASME Section VIII-2 Part 4.18.9.4 Step 10:

Note:



For a given Shell thickness of 68 mm, the minimum Shell length adjacent to the tubesheet should be 535.178 mm

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	8.2 <=	3995.8 kgf/cm ²	Pass
Tubesheet Shear Stress	3.0 <=	2131.1 kgf/cm ²	Pass
Maximum Tube Stress	257.4 <=	1985.8 kgf/cm ²	Pass
Minimum Tube Stress (Buckling)	-257.4 <=	-1985.8 kgf/cm ²	Pass
Maximum Force on any one Tube	194.1 <=	1497.3 kgf	Pass

Thickness Results for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	11.677	280.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.200	2.000	Ok
Tube-Tubesheet Groove Weld Leg :	1.200	2.000	Ok

Stationary Tubesheet results per ASME Div. 2 Part 4.18.9 2025

Results for 3 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend	Stresses Allwd	Shear	Allwd	Case Type	Pass/ Fail
D1uc	232.401 ...	2738	3996	1015	2131	Ps+Pt-Th	D1	Ok
D2uc	232.726 ...	2746	3996	1018	2131	Ps+Pt-Th	D2	Ok
D3uc	11.677 ...	8	3996	3	2131	Ps+Pt-Th	D3	Ok
Max:	232.7258 ... mm	0.687		0.478	(Str. Ratio)			

Load Case Definitions:

[Ps & Pt]:
 Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[(+/-)Th]:
 With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:
 With or Without Corrosion Allowance

[D1, D2, D3]:
 Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:
 Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:
 Operating Load Cases using the Maximum and Minimum Operating Pressures and
 Operating Temperatures

Tube, Shell and Channel Stress Summary:

Case#	Ten	---- Tube Stresses Allwd	Tube Loads Cmp Allwd	Tube Loads Ld Allwd	Shell Stress Allwd	Channel Stress Allwd	Stress	Pass
D1uc	338	1985	...	255	1497	...	4504	Ok
D2uc	596	1985	-596	449	1497	...	4504	Ok
D3uc	257	1985	-257	194	1497	...	4504	Ok
Max	RATIO	0.300	0.300	0.300	...	...		

Summary of Thickness Comparisons for 3 Load Cases:



Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	232.726	280.000	Ok
Tube thickness :	1.552	2.000	Ok
Tube-tubesheet fillet weld leg :	1.200	2.000	Ok
Tube-tubesheet groove weld leg :	1.200	2.000	Ok

Min Shell length of thk, (68.000) adj. to tubesheet: 535.178 mm

Note: This is a full strength Tube to Tubesheet Joint.

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Input Echo, Tubesheet Number 1, Description:

Shell Data:

Main Shell Description: Main Shell

Shell Test Pressure		256.42	kgf/cm ²
Shell Thickness	ts	68.0000	mm
Shell Internal Corrosion Allowance	cas	0.0000	mm
Shell External Corrosion Allowance	casext	0.0000	mm
Inside Diameter of Shell	Ds	1300.000	mm
Shell Temperature for Internal Pressure	Ts	21.11	°C
Shell Material		SA-387 11	
Shell Material UNS Number		K11789	
Shell Allowable Stress at Temperature	Ss	2110.70	kgf/cm ²
Shell Allowable Stress at Ambient		2110.70	kgf/cm ²

Channel Data:

Channel Description: LHS Ch Shell

Channel Type:		Cylinder	
Channel Test Pressure		255.66	kgf/cm ²
Channel Thickness	tc	80.0000	mm
Channel Internal Corrosion Allowance	cac	0.0000	mm
Channel External Corrosion Allowance	cacext	0.0000	mm
Inside Diameter of Channel	Dc	1226.000	mm
Channel Design Temperature	TEMPC	21.11	°C
Channel Flange Matl		SA-387 11	
Channel Material UNS Number		K11789	
Channel Allowable Stress at Temperature	Sc	2110.70	kgf/cm ²
Channel Allowable Stress at Ambient		2110.70	kgf/cm ²

Tube Data:

Number of Tube Holes	Nt	2615	
Tube Wall Thickness	et	2.0000	mm
Tube Outside Diameter	D	14.0000	mm
Total Straight Tube Length	Lt	15200.00	mm
Straight Tube Length (bet. inner tubsht faces) L		14640.00	mm
Design Temperature of the Tubes		21.11	°C
Tube Material		SA-213 T11	
Tube Material UNS Number		K11597	
Is this a Welded Tube		No	
Tube Material Specification used		Smls. tube	
Tube Allowable Stress at Temperature		1407.13	kgf/cm ²
Tube Allowable Stress At Ambient		1407.13	kgf/cm ²
Tube Yield Stress At design Temperature	Syt	2090.31	kgf/cm ²
Tube Pitch (Center to Center Spacing)	P	19.0000	mm
Tube Layout Pattern		Triangular	
Fillet Weld Leg	af	2.0000	mm
Groove Weld Leg	ag	2.0000	mm
Tube-Tubesheet Joint Weld Type		Full Strength	
Method for Tube-Tubesheet Jt. Allowable		4.21.2	
Is the Tube Inset		No	
Radius to Outermost Tube Hole Center	ro	577.862	mm
Largest Center-to-Center Tube Distance	Ul	0.0000	mm
Length of Expanded Portion of Tube	ltx	0.0000	mm
Tube-side pass partition groove depth	hg	5.0000	mm

Tubesheet Data:

Tubesheet TYPE: Floating, Tubesheet Integral, Conf A
Exchanger TYPE: With an Immersed Floating Head, Conf. a

Tubesheet Design Metal Temperature	T	21.11	°C
Tubesheet Material		SA-336 F11	
Tubesheet Material UNS Number		K11572	
Tubesheet Allowable Stress at Temperature	S	1876.18	kgf/cm ²
Tubesheet Allowable Stress at Ambient	Tt	1876.18	kgf/cm ²
Thickness of Tubesheet	h	280.0000	mm
Tubesheet Corr. Allowance (Shell side)	Cats	0.0000	mm



Tubesheet Corr. Allowance (Channel side) Catc 0.0000 mm
 Tubesheet Outside Diameter A 1386.000 mm

Area of the Untubed Lanes AL 0.0 cm²

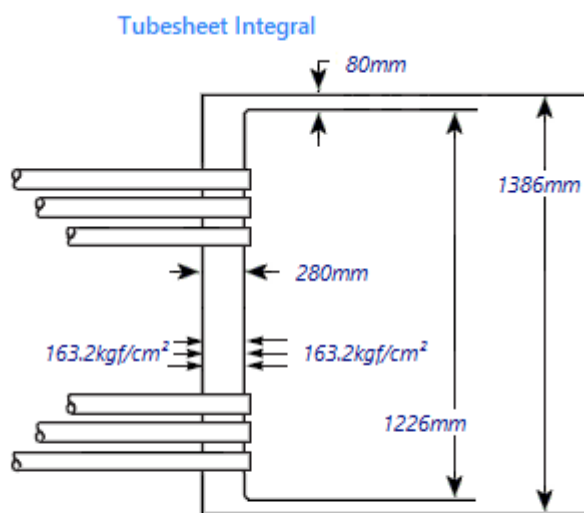
Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration l 0.000 mm
 Tube End condition corresponding to Span (l) k 0.00

Ignore Radial Thermal Exp. effects (4.18.8/4.18.9) Yes
 Perform Differential Pressure Design No
 Run Multiple Load Cases Yes
 Junction Stress Reduction option Perform Plastic Calculation

Tubesheet Integral with Both
 Tubesheet Extended as Flange No

Is Exchanger in Creep range (skip EP, Use 3S for Sps) No



Messages regarding the Tubesheet & Exchanger Input Data:

Warning - Tube end condition k should be bet 0 - 1 (.00).

ASME Division 2 Tubesheet Results per Part 4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
 Shell - TM-1 Material Group C

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20811E+07 kgf/cm ²
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.20811E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20811E+07 kgf/cm ²

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
 Channel - TM-1 Material Group C

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20811E+07 kgf/cm ²



Elastic Mod. at Ambient Temperature 21.1 °C 0.20811E+07 kgf/cm²

Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
 Tubes - TM-1 Material Group C

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20811E+07 kgf/cm ²
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.20811E+07 kgf/cm ²
Elastic Mod. at Tubsht. Design Temp.	21.1 °C	0.20811E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20811E+07 kgf/cm ²

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
 TubeSheet - TM-1 Material Group C

Elastic Mod. at Design Temperature	21.1 °C	0.20811E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20811E+07 kgf/cm ²

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.
 Please insure these properties are consistent with the
 type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i$ Equation 4.3.1
 $= 0.5 \cdot 10 \cdot (\exp(256/(1407 \cdot 1)) - 1) + 0$
 $= 1 + 0$
 $= 1 \text{ mm}$

Tube Required Thickness under External Pressure (Shellside pressure) :

Determine Parameter [Mx]:

$= L / \sqrt{(R_o \cdot t)}$
 $= 14640 / \sqrt{(7 \cdot 2)}$
 $= 3912.705$

Compute the Factor [Ch]:

Since $M_x > 2 \cdot (D_o/t)^{0.94}$, $Ch = 0.55 / (D_o/t)$
 $= 0.55 \cdot (14/2)$
 $= 0.0786$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$= 1.6 \cdot Ch \cdot E_y (t / D_o)$
 $= 1.6 \cdot 0.079 \cdot 2081087 (2/14)$
 $= 37374.625 \text{ kgf/cm}^2$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$= Fhe / F_y$
 $= 37374.6/2090.3$
 $= 17.880$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$= \sigma_{ys} / \sigma_{uts}$
 $= 2090/4218$
 $= 0.496$

Ratio [AE]:

$= Fhe / E$
 $= 37375/2081087$
 $= 0.017959$

Factor [m1]:

$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$
 $= (\ln(0.5) + (0.20000E-04 - 0.002)) / \ln(\ln(1 + 0.20000E-04) / \ln(1 + 0.002))$
 $= 0.153$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys}) m_1$
 $= 2090 (1 + 0.0020) / \ln(1 + 0.0020) 0.15293255$
 $= 5418.821 \text{ kgf/cm}^2$

Curve Fitting Exponent for, Ferritic Steel [m2]:
 $= m_2 f(m_2^2 - R)$
 $= 0.6 (1 - 0.5)$
 $= 0.303$

Curve Fitting Constant [A2]:
 $= (\sigma_{uts} \cdot \exp(m_2)) / (m_2^2 m_2)$
 $= (4218 \cdot \exp(0.3)) / (0.3 \cdot 0.3)$
 $= 8197.895 \text{ kgf/cm}^2$

Material Parameter for Stress-Strain Curve Model [K]:
 $= 1.5 \cdot R^{1.5} - 0.5 \cdot R^{2.5} - R^{3.5}$
 $= 1.5 \cdot 0.515 - 0.5 \cdot 0.525 - 0.535$
 $= 0.351$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	2109	-1.949	0.324E-05	0.290E-05	0.883E-05	-0.731E-05	0.123E+06	0.782E-03
2	3163	0.873	0.306E-04	-0.416E-04	0.225E-04	0.449E-04	0.176E+05	0.162E+00
3	2636	-0.538	0.112E-04	-0.365E-05	0.148E-04	0.167E-04	0.254E+05	0.860E-01
4	2372	-1.244	0.623E-05	0.356E-05	0.116E-04	-0.350E-05	0.545E+05	0.256E-01
5	2241	-1.597	0.454E-05	0.355E-05	0.102E-04	-0.656E-05	0.823E+05	0.929E-02
6	2175	-1.773	0.384E-05	0.326E-05	0.948E-05	-0.713E-05	0.101E+06	0.365E-02
7	2142	-1.861	0.353E-05	0.308E-05	0.915E-05	-0.726E-05	0.111E+06	0.130E-02
8	2125	-1.905	0.338E-05	0.299E-05	0.899E-05	-0.729E-05	0.117E+06	0.228E-03
9	2117	-1.927	0.331E-05	0.294E-05	0.891E-05	-0.730E-05	0.120E+06	0.284E-03
10	2121	-1.916	0.335E-05	0.297E-05	0.895E-05	-0.730E-05	0.118E+06	0.302E-04
11	2123	-1.911	0.337E-05	0.298E-05	0.897E-05	-0.729E-05	0.118E+06	0.983E-04
12	2122	-1.914	0.336E-05	0.297E-05	0.896E-05	-0.729E-05	0.118E+06	0.339E-04
13	2122	-1.915	0.335E-05	0.297E-05	0.896E-05	-0.729E-05	0.118E+06	0.182E-05
14	2121	-1.916	0.335E-05	0.297E-05	0.895E-05	-0.729E-05	0.118E+06	0.142E-04
15	2121	-1.915	0.335E-05	0.297E-05	0.895E-05	-0.729E-05	0.118E+06	0.620E-05
16	2122	-1.915	0.335E-05	0.297E-05	0.896E-05	-0.729E-05	0.118E+06	0.219E-05

Factor of Safety [FS]:
 Since Fic [2122] $\geq$ Fy[2090], FS = 1.667

Allowable Circumferential Compressive Stress [Fha]:
 $= \text{Fic} / \text{FS}$
 $= 2122 / 1.67$
 $= 1272.957 \text{ kgf/cm}^2$

Allowable External Pressure at the given Thickness [Pa]:
 $= 2 \cdot \text{Fha} (t / D_o)$
 $= 2 \cdot 1273 (2 / 14)$
 $= 363.702 \text{ kgf/cm}^2$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	1.5521 mm
Allowable Internal Pressure at Corroded thickness	473.46 kgf/cm ²
Required Internal Design Pressure	255.66 kgf/cm ²
Allowable External Pressure at Corroded thickness	363.70 kgf/cm ²
Required External Design Pressure	256.42 kgf/cm ²
Required Thickness due to Shell Side pressure	1.5521 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Results for ASME Floating Tubesheet Calculations for Configuration A:

Results for Tubesheet Calculations Original Thickness:



ASME Section VIII-2 Part 4.18.9.4 Step 1:

Tube expansion depth ratio [ρ]:

$$\begin{aligned} &= l_{tx} / h \text{ (modified for corrosion if present)} \\ &= 0/280 \\ &= 0.0000 \text{ (must be } 0 \leq \rho \leq 1 \text{)} \end{aligned}$$

Effective tube hole diameter [d^*]:

$$\begin{aligned} &= \max(dt - 2 \cdot tt(Et/E) \cdot (St/S) \cdot (\rho); dt - 2 \cdot tt) \\ &= \max(14 - 2 \cdot 2 (2081087/2081087) \cdot \\ &\quad (1407/1876) \cdot (0); 14 - 2 \cdot 2) \\ &= 14.0000 \text{ mm} \end{aligned}$$

Equivalent outer tube limit circle diameter [Do]:

$$\begin{aligned} &= 2 \cdot r_o + dt \\ &= 2 \cdot 578 + 14 \\ &= 1169.724 \text{ mm} \end{aligned}$$

Determine the basic ligament efficiency for shear [μ]:

$$\begin{aligned} &= (p - dt) / p \\ &= (19 - 14) / 19 \\ &= 0.263 \end{aligned}$$

Equivalent outer tube limit radius [ao]:

$$\begin{aligned} &= Do / 2 \\ &= 1170/2 \\ &= 584.8620 \text{ mm} \end{aligned}$$

Effective tube pitch [p^*]:

$$\begin{aligned} &= p / \sqrt{ (1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot Do \cdot p) / (\pi \cdot Do^2)) } \\ &= 19 / \sqrt{ (1 - 4 \cdot \min(0 \cdot 100 ; 4 \cdot 1170 \cdot 19) / (\pi \cdot 1170^2)) } \\ &= 19.0000 \text{ mm} \end{aligned}$$

Effective ligament efficiency for bending [μ^*]:

$$\begin{aligned} &= (p^* - d^*) / p^* \\ &= (19 - 14) / 19 \\ &= 0.2632 \end{aligned}$$

Ratio [ps]:

$$\begin{aligned} &= as / ao \\ &= 613/585 \\ &= 1.0481 \end{aligned}$$

Ratio [pc]:

$$\begin{aligned} &= ac / ao \\ &= 613/585 \\ &= 1.0481 \end{aligned}$$

Compute parameter [xt]:

$$\begin{aligned} &= 1 - Nt((dt - 2 \cdot tt) / (2 \cdot ao))^2 \\ &= 1 - 2615 ((14 - 2 \cdot 2) / (2 \cdot 585))^2 \\ &= 0.8089 \end{aligned}$$

Determine parameter [xs]:

$$\begin{aligned} &= 1 - Nt(dt / (2 \cdot ao))^2 \\ &= 1 - 2615 (14 / (2 \cdot 585))^2 \\ &= 0.6254 \end{aligned}$$

Determine the value [$h'g$]:

$$\begin{aligned} &= \max((hg - CATC); 0) \text{ (For pressure only cases)} \\ &= \max((5 - 0); 0) \\ &= 5.000 \text{ mm} \end{aligned}$$

ASME Section VIII-2 Part 4.18.9.4 Step 2:

Intermediate parameters for Tubesheet Gasketed on the Shell Side:

β_s , k_s , λ_s , δ_s are all = 0

Channel coefficient [β_c]:

$$= ((12(1 - \nu_c^2))^{1/4}) / \sqrt{((D_c + t_c)t_c)} \\ = ((12(1 - 0.3^2))^{1/4}) / \sqrt{(1226 + 80)80} \\ = 0.0056 \text{ 1/mm}$$

Channel coefficient [kc]:

$$= \beta C \cdot E_c \cdot t_c^3 / (6(1 - \nu_c^2)) \\ = 0.0056 \cdot 2081087 \cdot 80^3 / (6(1 - 0.3^2)) \\ = 1097505280.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λc]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot kc / h^3 \cdot (CmTheta + \\ h \cdot \beta c \cdot (Cm\Delta + CqTheta) / 2 + (h \cdot \beta c)^2 / 2 \cdot Cq\Delta) \\ = 6(1226 + 80 / \cos(0)) \cdot 1097505280 / 280^3 \cdot (1 + \\ 280 \cdot 0.0056 \cdot (1 + 1) / 2 + (280 \cdot 0.0056)^2 / 2 \cdot 1) \\ = 1494404.8 \text{ kgf/cm}^2$$

Channel coefficient [δc]:

$$= D_c^2 / (4 \cdot E_c \cdot T_c) \cdot (1 - D_c \cdot \nu_c / (2(D_c + t_c))) \\ = 1226^2 / (4 \cdot 2081087 \cdot 80) \cdot (1 - 1226 \cdot 0.3 / (2(1226 + 80))) \\ = 0.19392E-02 \text{ mm/kgf/cm}^2$$

Channel coefficient [zetac]:

$$zetac = 0$$

ASME Section VIII-2 Part 4.18.9.4 Step 3:

$$h/p = 14.736844 ; \mu^* = 0.263158 ; E^* = 485641 \text{ kgf/cm}^2 \\ E^*/E = 0.233360 ; \nu^* = 0.383525$$

Note: As h/p (14.737) is > 2 , data values for $h/p = 2$ were used.

Tube Bundle Stiffness Factor [Xa]:

$$= ((24(1 - \nu^*)Nt \cdot Et \cdot tt(dt - tt)ao^2) / (E^* \cdot L \cdot H^3))^{1/4} \\ = ((24(1 - 0.38^2)2615 \cdot 2081087 \cdot 2 \\ (14 - 2)585^2) / (485642 \cdot 14640 \cdot 280^3))^{1/4} \\ = 1.5558$$

Values from Table 4.18.3

$$Z_d = 0.393525 ; Z_v = 0.170662 ; Z_m = 0.692948 \\ Z_a = 0.578496E+00 ; Z_w = 0.170662$$

ASME Section VIII-2 Part 4.18.9.4 Step 4:

Diameter ratio [K]:

$$= A / D_o \\ = 1386 / 1170 \\ = 1.1849$$

Compute coefficient [F]:

$$= (1 - \nu^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K)) \\ = (1 - 0.38) / 485642 \cdot (0 + 1494404 + \\ 2081087 \cdot \ln(1.18)) \\ = 2.3452$$

Compute parameter [φ]:

$$= (1 + \nu^*)^F \\ = (1 + 0.38)^{2.35} \\ = 3.2446$$

Compute parameter [Q1]:

$$= (ps - 1 - \phi \cdot Z_v) / (1 + \phi \cdot Z_m) \\ = (1.05 - 1 - 3.24 \cdot 0.17) / (1 + 3.24 \cdot 0.69) \\ = -0.155655$$

ASME Section VIII-2 Part 4.18.9.4 Step 5:

Determine factor [yb]:

= 0 (config a)

Calculate parameter [ws]:

$$= \rho_s \cdot k_s \cdot \beta_s \cdot \delta S (1 + h \cdot \beta_s)$$

$$= 1.05 \cdot 0 \cdot 0 \cdot 0.000 (1 + 280 \cdot 0)$$

$$= 0.0000 \text{ mm}^2$$

Calculate parameter [ω*s]:

$$= a_o^2 (\rho_s^2 - 1) \cdot (\rho_s - 1) / 4 - \omega_s$$

$$= 585^2 (1.05^2 - 1) \cdot (1.05 - 1) / 4 - 0$$

$$= 405.3960 \text{ mm}^2$$

Parameter [wc]:

$$= \rho_c \cdot k_c [\delta C (C_m \Theta + \beta C + C_q \Delta \cdot h \cdot \beta C^2) - \zeta c (C_m \Theta + h \cdot C_q \Theta \cdot h \cdot \beta C / 2)]$$

$$= 1.05 \cdot 1097505280 [0.001939229 (1 + 0.0056 \cdot 1 \cdot 280) - 0.000 (1/280 + 1 \cdot 0.0056/2)]$$

$$= 32300.4102 \text{ mm}^2$$

Parameter [ω*c]:

$$= a_o^2 [(\rho_c^2 + 1) \cdot (\rho_c - 1) / 4 - (\rho_s - 1) / 2] - \omega_c$$

$$= 585^2 [(1.05^2 + 1) \cdot (1.05 - 1) / 4 - (1.05 - 1) / 2] - 32300$$

$$= -31895.0156 \text{ mm}^2$$

Pressure [P*S]:

= 0 For Pressure only cases or Configurations d,e,f,A,B,C,D

Pressure [P*C]:

= 0 For Pressure only cases or Configurations b,c,d,B,C,D

ASME Section VIII-2 Part 4.18.9.4 Step 6:

$$P_e = P_s - P_t$$

$$= 256 - 256 = 0.76 \text{ kgf/cm}^2$$

ASME Section VIII-2 Part 4.18.9.4 Step 7:

Determine factor [Q2]:

$$= [((\omega_s \cdot P_s - \omega_c \cdot P_t) - (\omega_s \cdot P_s - \omega_c \cdot P_c)) + W \cdot \gamma_b / (2 \cdot \pi)] / (1 + \phi \cdot Z_m)$$

$$= [(405 \cdot 256 - (-31895 \cdot 256) - (0 \cdot 0 - 32300 \cdot 0)) \cdot 0.01 + 0 \cdot 0 / (2 \cdot \pi)] / (1 + 3.24 \cdot 0.69)$$

$$= 25422.6230 \text{ kgf}$$

Calculate factor [Q3]:

$$= Q_1 + 2 \cdot Q_2 / (P_e \cdot a_o^2)$$

$$= -0.16 + 2 \cdot 25423 / (0.76 \cdot 585^2)$$

$$= 19.37733$$

Fm Value from Table 4.18.3 = 9.714251

Tube sheet Bending Stress - original thickness [σ]:

$$= (1.5 \cdot F_m / \mu) \cdot (2 \cdot a_o / (H - h'g))^2 \cdot P_e$$

$$= (1.5 \cdot 9.71 / 0.26) \cdot (2 \cdot 585 / (280 - 5))^2 \cdot 0.76$$

$$= 762.3618 \text{ kgf/cm}^2$$

Allowable Tube sheet Bending Stress [σ allowed]:

$$= 1.5 \cdot \beta_T \cdot S_y = 1.5 \cdot 0.95 \cdot 2804 = 3996 \text{ kgf/cm}^2, 4.18.233$$

Tube sheet Bending Stress - final thickness [σf]:

$$= (1.5 \cdot F_m / \mu) \cdot (2 \cdot a_o / (h - h'g))^2 \cdot P_e$$

$$= (1.5 \cdot 0.0089 / 0.26) \cdot (2 \cdot 585 / (8.64 - 5))^2 \cdot 0.76$$

$$= 3995.5798 \text{ kgf/cm}^2$$

Reqd Tube sheet Thickness, for Bending Stress (including CA) [HReqB]:

$$= h + C_{at} + C_{atc}$$

$$= 8.64 + 0 + 0$$

$$= 8.6446 \text{ mm}$$



H reqd. = HReqB = 8.6446 mm

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	762.4 <=	3995.8 kgf/cm²	Pass

Thickness Results for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	8.645	280.000	Ok
Tube-Tubesheet Fillet Weld Leg :	0.000	2.000	Ok
Tube-Tubesheet Groove Weld Leg :	0.000	2.000	Ok

Floating Tubesheet results per ASME Div. 2 Part 4.18.9 2025

Results for 3 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Stresses Bend Allwd Shear Allwd	Case Type	Pass/Fail
D1uc	235.142 ...	2809	3996 ...	Ps+Pt-Th D1	Ok
D2uc	246.192 ...	3297	3996 ...	Ps+Pt-Th D2	Ok
D3uc	8.645 ...	762	3996 ...	Ps+Pt-Th D3	Ok
Max:	246.1921 ... mm	0.825	...	(Str. Ratio)	

Load Case Definitions:

[Ps & Pt]:
 Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[(+)-Th]:
 With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:
 With or Without Corrosion Allowance

[D1, D2, D3]:
 Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:
 Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:
 Operating Load Cases using the Maximum and Minimum Operating Pressures and
 Operating Temperatures

Tube, Shell and Channel Stress Summary:

Case#	---- Tube Stresses Ten Allwd Cmp Allwd	Tube Loads Ld Allwd	Shell Stress Allwd	Channel Stress Allwd	Stress Allwd	Pass Fail
D1uc			... 4504	... 4504	... 4504	Ok
D2uc			... 4504	... 4504	... 4504	Ok
D3uc			... 4504	... 4504	... 4504	Ok
Max RATIO	...	...	...	...	...	

Summary of Thickness Comparisons for 3 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	246.192	280.000	Ok
Tubesheet thickness flanged extension :	0.000	284.000	Ok
Tube thickness :	1.552	2.000	Ok

PV Elite 28 Licensee: SL User
FileName : 401-E-31001-AK
ASME Fl-TS Calc: Pressure Test

Case: 1 1:26p Jul 4, 2026

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HEXAGON

Tube-tubesheet fillet weld leg :	0.000	2.000	Ok
Tube-tubesheet groove weld leg :	0.000	2.000	Ok

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 2, 2025

External Pressure Calculations:

From	To	Section Length cm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B kgf/cm ²
10	20	No Calc	...	373.5	No Calc	No Calc
20	30	No Calc	...	408.5	No Calc	No Calc
30	Tsht	234.997	1436	66.5	No Calc	No Calc
Tsht	40	No Calc	...	66.5	No Calc	No Calc
40	70	1715.92	1436	66.5	No Calc	No Calc
70	80	1715.92	1436	66.5	No Calc	No Calc
80	90	No Calc	1420	58.5	No Calc	No Calc

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure kgf/cm ²	External M.A.W.P. kgf/cm ²
10	20	375	257.623	...	No Calc
20	30	410	385.47	...	No Calc
30	Tsht	68	No Calc	...	117
Tsht	40	68	No Calc	...	No Calc
40	70	68	No Calc	...	84
70	80	68	No Calc	...	84
80	90	60	4.68551	1.05461	109

Minimum

84

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners cm	Allowable Length Bet. Stiffeners cm	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	No Calc	No Calc	No Calc	No Calc
30	Tsht	234.997	No Calc	No Calc	No Calc
Tsht	40	No Calc	No Calc	No Calc	No Calc
40	70	1715.92	No Calc	No Calc	No Calc
70	80	1715.92	No Calc	No Calc	No Calc
80	90	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kg	Element ID Volume cm ³	Corroded Metal Wgt. kg	Corroded ID Volume cm ³	Extra due Misc % kg
10	20	10676.7	...	10676.7	...	1067.67
20	30	7669.45	743210	7642.93	744069	766.945
30	40	4076.99	2389181	3991.43	2400221	407.699
40	70	34427.9	14169292	33705.4	14262518	3442.79
70	80	4076.99	2276469	3991.43	2287508	407.699
80	90	1379.38	355015	1350.65	357322	137.938
Total		62307	19933168.00	61358	20051640.00	6230



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kg	X Offset, Dtl. Cent. cm	Y Offset, Dtl. Cent. cm	Z Offset, Dtl. Cent. cm	Description
10	Liqd	...	18.75	...	...	Liquid: 10
10	Insl	53.6559	18.75	...	...	Hot Insulation
20	Liqd	37.1438	27.5	...	...	Liquid: 20
20	Insl	38.3833	27.5	...	...	Hot Insulation
30	Liqd	119.405	90	...	...	Liquid: 30
30	Insl	96.2365	90	...	...	Hot Insulation
30	Nozl	797.099	90	87.1463	...	N1-Tube Inlet
30	Wght	200	90	...	...	bAFFLE
40	Sadl	677.822	300	93.5	...	Left Saddle
40	Sadl	677.822	1220	93.5	...	New Saddle
40	Liqd	708.147	760	...	...	Liquid: 40
40	Insl	1160.32	760	...	...	Hot Insulation
40	Nozl	729.497	150	-84.9238	...	N3-Shell Inlet
40	Nozl	729.497	1300	87.1463	...	N4-Shell outlet
70	Liqd	113.772	90	...	...	Liquid: 70
70	Insl	96.2365	90	...	...	Hot Insulation
70	Nozl	729.497	90	84.9238	...	N2-Tube Outlet
80	Insl	36.7254	18.79	...	...	Hot Insulation
30	FTsh	2948.36	194	...	...	
30	Tube	25550	940	...	...	Tubes
30	FlTs	2640.62	1700	...	...	

Total Weight of Each Detail Type:

Saddles	1355.6
Liquid	978.5
Insulation	1481.6
Nozzles	2985.6
Weights	200.0
Exchanger Components	31139.0
Liquid in Tubes	156.0

Sum of the Detail Weights 38296.3 kg

Weight Summation Results: (kg)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	68538.2	68538.2	68538.2	68538.2	68538.2	68538.2
Saddles	1355.6	1355.6	1355.6	1355.6	1355.6	1355.6
Nozzles	2985.6	2985.6	2985.6	2985.6	2985.6	2985.6
Wld Weights	200.0	200.0	200.0	200.0	200.0	200.0



Exchanger	31139.0	31139.0	31139.0	31139.0	31139.0	31139.0
Insulation	...	...	1481.6	1481.6	1481.6	1481.6
Ope. Liquid	...	...	...	...	...	978.5
Tube Ope Lqd	...	...	...	...	...	156.0
Test Liquid	...	19924.2	...	...	...	...
Tube Tst Lqd	...	3120.4	...	...	...	...
Totals	104218.5	127263.1	105700.0	105700.0	105700.0	106834.5

Field Installation Options:

Miscellaneous Weight Percent: 10.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	104218.5 kg
Shop Test Wt.	- Fabricated weight + water (full)	127263.1 kg
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	105700.0 kg
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	105700.0 kg
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	105700.0 kg
Operating Wt.	- Empty weight + operating liq. uncorroded	106834.5 kg
Oper. Wt. + CA	- Corr wt. + operating liquid	105790.7 kg

Exchanger Tube Data

Volume of Exchanger tubes :	3121801.8 cm ³
Weight of Ope Liq in tubes :	156.0 kg
Weight of Water in tubes :	3120.4 kg

The Corroded Weight and thickness are used in the Horizontal Vessel Analysis (Ope Case) and Earthquake Load Calculations.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	61717.2	...
20	30	57112.9	22789.1
30	40	81203.9	73513.3
40	70	685722	620779
70	80	81203.9	73513.3
80	90	24907.1	20394.1

Total	991866.7	810988.4 cm ²
Total	1067.6	872.9 ft ²



Nozzle Flange MAWP Results: (kgf/cm² & °C)

Nozzle	Flange Rating		Des.			Equiv.	Max Pressure		
Description	Ope.	Ambient	Temp	Class	Grade/Group	Press	UG-44	50%	DNV

Shellside Flange Rating

Channelside Flange Rating

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Wind Load Calculation Results:

Wind Load Calculations per India Std. IS 875 (PART 3):2015, Amd. 1&2 (2020):

Input Values:

Wind Design Code	IS 875 (PART 3):2015-2020	
Basic Wind Speed for IS-875	180	km/hr
Wind Zone Number	5	
Base Elevation	179	cm
Percent Wind for Hydrotest	33.0	
Risk Factor	[k1]	1.0
Terrain Category		2
Equipment Class	Class A	
Topography Factor	[k3]	1.0
Importance Factor	[k4]	1.15
Damping Factor (Beta) for Wind (Ope)		0.0100
Damping Factor (Beta) for Wind (Empty)		0.0000
Damping Factor (Beta) for Wind (Filled)		0.0000
Use Gust Response Factor (Dynamic Analysis)		No

Calculated Values:

Vessel Length to Diameter ratio: 12.678

Design Wind Speed [Vz]:

= Basic Wind Speed • k1 • k2,1 • k3 • k4 (for the 1st element)

= 50 • 1 • 1 • 1 • 1.15

= 57.496 meters/s

Design Wind Pressure per 4.5.2 [pd]:

= max(0.7 • pz, Kd • Ka • Kc • 0.6 • Vz²)

= max(0.7 • 1984, 1 • 0.87 • 1 • 0.6 • 57.5²)

= 1726.8 N/m²

Element Wind Load (operating condition):

= Wind Area • Cf • pd

= 1.03 • 0.93 • 1727

= 1649.2 N

From	Height m	k1	k2	k3	Vz m/sec	Cf
10	3.09	1.0000	1.0000	1.0000	57.50	0.9268
20	3.09	1.0000	1.0000	1.0000	57.50	0.9268
30	3.09	1.0000	1.0000	1.0000	57.50	0.9268
40	3.09	1.0000	1.0000	1.0000	57.50	0.9268
70	3.09	1.0000	1.0000	1.0000	57.50	0.9268
80	3.09	1.0000	1.0000	1.0000	57.50	0.9268

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation cm	Pressure kgf/m ²	Force kgf
bAFFLE	0.00	309.00	176.16	0.00

Wind Load Calculation:

From	To	Wind Height cm	Wind Diameter cm	Wind Area cm ²	Wind Pressure kgf/m ²	Element Wind Load kgf
10	20	309	274.8	10305	176.156	184.994
20	30	309	174	9570	176.156	171.799
30	40	309	190.32	34257.6	176.156	614.987
40	70	309	197.52	300230	176.156	5389.69
70	80	309	190.32	34257.6	176.156	614.987



80	90	309	188.4	6629.22	176.156	119.007
----	----	-----	-------	---------	---------	---------

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Seismic Analysis Results per IS 1893 (Part 4): 2024

Input Data:

Importance Factor as Entered by User I 2.0000
Zone Number 2

Soil Type Rocky or Hard Soil
Response reduction factor (Force factor) R 2.0000
Time Period of Vibration (program or user) 0.0303 Seconds
Value of Sa/g from user Spectrum Data 0.0939
Percent Damping 2.0 %
Considering MCE for load determination No

Calculated results:

Horizontal Seismic Coefficient per 7.3.1 [Ah]:
 $= (Sa/g)_{ss} \cdot DampVal \cdot (I/R)$
 $= (0.094 \cdot 1.4) \cdot 1$
 $= 0.131$

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
2.800	0.025
0.000	0.000

Actual Time Period = 0.03 secs
Computed Sa/g = 0.094

Note:

It was indicated to apply the damping factor, this assumes that the Site Spectrum specified is for 5% damping. If your Sa/g values are already per the actual damping value of the vessel then do not apply the damping factor.

Earthquake Element Load, for the first Element:
 $= \text{Earthquake Weight} \cdot Ah \cdot \text{Scalar}$
 $= 13054 \cdot 0.13 \cdot 1$
 $= 1715.3 \text{ kgf}$

Vertical load per 8.4:
 $= 2/3 \cdot Ah \cdot W \cdot \text{Scalar}$
 $= 2/3 \cdot 0.13 \cdot 104435 \cdot 1$
 $= 9148.4 \text{ kgf}$

Earthquake Load Calculation:



From	To	Earthquake Height cm	Earthquake Weight kgf	Element Ope Load kgf
10	20	107	13054.4	1715.33
20	30	65	13054.4	1715.33
30	40	65	13054.4	1715.33
40	Sad1	65	13054.4	1715.33
Sad1	70	65	13054.4	1715.33
40	70	65	13054.4	1715.33
70	80	65	13054.4	1715.33
80	90	65	13054.4	1715.33

The Earthquake Weight does not include the weight of the saddles.

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**Shop/Field Installation Options :**

Insulation is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	1060.818 cm
Center of Gravity of the Liquid	1014.717 cm
Center of Gravity of the Insulation	1019.235 cm
Center of Gravity of the Nozzles	1016.992 cm
Center of Gravity of the Added Weights (Operating)	182.818 cm
Center of Gravity of the Added Weights (Empty)	182.818 cm
Center of Gravity of the Tubesheet(s)	998.357 cm
Center of Gravity of the Tubes	1032.818 cm
Center of Gravity of Bare Shell New and Cold	779.149 cm
Center of Gravity of Bare Shell Corroded	774.814 cm
Vessel CG in the Operating Condition	864.338 cm
Vessel CG in the Fabricated (Shop/Empty) Condition	864.625 cm
Vessel CG in the Test Condition	894.697 cm

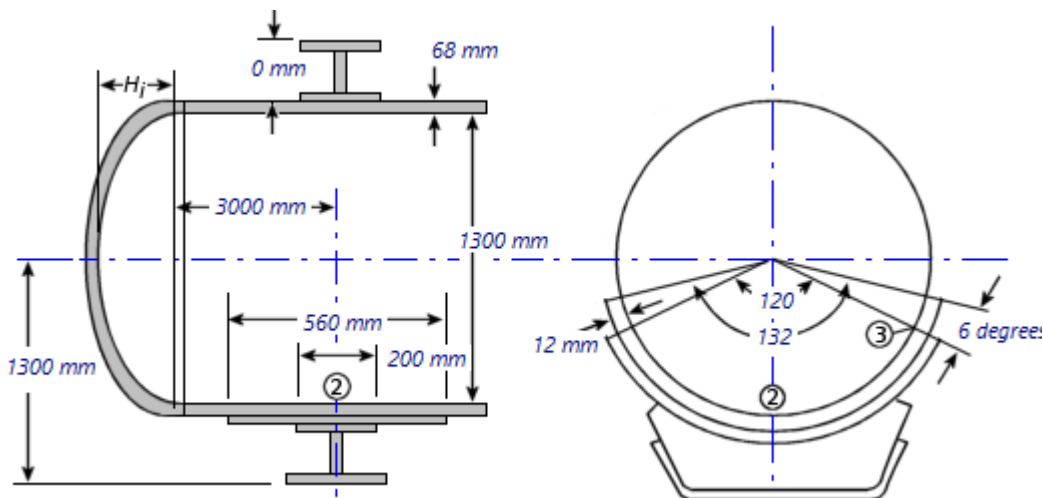
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Input and Calculated Values:

Vessel Mean Radius	Rm	684.75	mm
Shell Thickness used in this Case	t	68.000	mm
Stiffened Vessel Length per 4.15.6	L	1705.08	cm
Distance from Saddle to Vessel tangent	a1 or a	3000.00	mm
Saddle Width	b1 or b	200.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	560.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	12.0	mm
Wear Plate Allowable Stress	fw or Sr	1723.26	kgf/cm ²
Shell Allowable Stress used in Calculation		1723.26	kgf/cm ²
Head Allowable Stress used in Calculation		1521.35	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1300.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case	100548.38	kgf
Pressure used in Analysis	P	163.162 kgf/cm ²

Horizontal Vessel Analysis Results:		
	Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan	709.02	1723.26
Long. Stress at Bottom of Midspan	971.06	1723.26
Long. Stress at Top of Saddles	1335.00	1723.26
Long. Stress at Bottom of Saddles	565.70	1723.26
Tangential Shear in Shell	167.54	1375.03
Circ. Stress at Horn of Saddle	189.45	2154.08
Circ. Compressive Stress in Shell	18.27	1723.26

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:



$$= \text{Ftr} (\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (7095/2 + 0) \cdot 1300/1264$$

$$= 10942.9 \text{ kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(551, 0) \cdot 1300/9200$$

$$= 77.9 \text{ kgf}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(13723, 0) \cdot 1300/9200$$

$$= 1939.1 \text{ kgf}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr} (\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (13723/2 + 0) \cdot 1300/1264$$

$$= 21163.5 \text{ kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 79385 + \max(77.9, 10943, 1939, 21164)$$

$$= 100548.4 \text{ kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1727 (0 + \pi/4 (1.59 \cdot 1.2)^2)$$

$$= 5405.922 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	101226.20	kgf
Transverse Shear Load	6861.31	kgf
Longitudinal Shear Load	13722.61	kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot Rm)$$

$$= \min(2(1300/2 + 68 + 12) \sin(120/2), 2 \cdot 685)$$

$$= 1264.397 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $Rm/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L)) / (1 + (4hm)/3L)]$$

$$= -100548 \cdot 300 [1 - (1 - 300/1705 + (68.5^2 - 0^2) / (2 \cdot 300 \cdot 1705)) / (1 + (4 \cdot 0) / (3 \cdot 1705))]$$

$$= -51690.5 \text{ kgf-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 ((1 + 2(Rm^2 - hm^2)/(L^2)) / (1 + (4hm^2)/(3L)) - 4a/L)$$

$$= 100548 \cdot 1705 / 4 ((1 + 2(68.5^2 - 0^2)/(1705^2)) / (1 + (4 \cdot 0) / (3 \cdot 1705)) - 4 \cdot 300/1705)$$

$$= 128345.1 \text{ kgf-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 t)$$

$$= 163 \cdot 685 / (2 \cdot 66.5) - 128345 / (\pi \cdot 685^2 \cdot 66.5)$$

$$= 709.02 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t)$$



$$= 163 \cdot 685 / (2 \cdot 66.5) + 128345 / (\pi \cdot 685^2 \cdot 66.5)$$

$$= 971.06 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 163 \cdot 685 / (2 \cdot 66.5) - 51690 / (0.11 \cdot \pi \cdot 685^2 \cdot 66.5)$$

$$= 1335.00 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 163 \cdot 685 / (2 \cdot 66.5) + 51690 / (0.19 \cdot \pi \cdot 685^2 \cdot 66.5)$$

$$= 565.70 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 100548 \cdot (1705 - 2 \cdot 300) / (1705 + (4 \cdot 0 / 3))$$

$$= 65166.4 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 65166 / (685 \cdot 66.5)$$

$$= 167.54 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(685 \cdot 66.5)}$$

$$= 166.445 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(200 + 1.56 \cdot \sqrt{(685 \cdot 66.5)}, 2 \cdot 300)$$

$$= 532.89 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(S_r / S, 1)$$

$$= \min(1723 / 1723, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 100548 \cdot 0.1 / (533(66.5 + 1 \cdot 12))$$

$$= -18.27 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -100548 / (4(66.5 + 1 \cdot 12)533) -$$

$$3 \cdot 0.053 \cdot 100548 / (2(66.5 + 1 \cdot 12)^2)$$

$$= -189.45 \text{ kgf/cm}^2$$

Distance between Saddle Supports [Ls]:

$$= 9200.0 \text{ mm}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot L_s(\text{Design Temperature} - \text{Ambient Temperature})$$

$$= 0.13074\text{E-}04 \cdot 9200(250 - 21.1)$$

$$= 27.531 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1406.0000	mm
Baseplate Thickness	Bpthk	36.0000	mm
Baseplate Width	Bpwid	380.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	28.0000	mm
Web Thickness	Webtk	28.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	2579.8	kgf/cm ²
Height of Web at Center	Hw,c	600.0	mm
Friction Coefficient	mu	0.000	



Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(652 \cdot 66.5)} + 560)^3) / 12$$

$$= 2168.119 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 2168 + 588 (172 - 33.2)^2$$

$$= 116033.555 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	884.7	66.5	33.2	588.3	1956199.5	0.116E+06
Wearplate	560.0	12.0	72.5	67.2	487200.0	0.671E+04
Web	28.0	534.0	345.5	149.5	5165915.5	0.803E+05
BasePlate	380.0	36.0	630.5	136.8	8625240.0	0.287E+06
Totals	...	...	...	941.9	16234555.0	0.490E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 16234555/942$$

$$= 172.369 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 100548$$

$$= 20463.7773 \text{ kgf}$$

Tension Stress, St = (Fh/As) = 57.8858 kgf/cm²
 Allowed Stress, Sa = 0.6 * Yield Str = 1547.8756 kgf/cm²

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 1300 - 652 \cdot \sin(120/2) / 1.05$$

$$= 761.214 \text{ mm}$$

Bending Moment, M = Fh * d = 15577.3281 kgf-m.

Bending Stress, Sb = (M * C1 / I) = 54.7564 kgf/cm²
 Allowed Stress, Sa = 2/3 * Yield Str = 1719.8618 kgf/cm²

Minimum Thickness of Baseplate per Moss:

The maximum of 2/3*Yield and the Saddle Allowable Stress value was used to determine the baseplate thickness.

$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(100548 + 678) 380 / (4 \cdot 1406 \cdot 1723)}$$

$$= 34.507 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (650 + 68 + 12)$$

$$= 1264.397 \text{ mm}$$

Distance between Ribs [e]:

$$\begin{aligned} &= \text{Web Length} / (\text{Nribs} - 1) \\ &= 1264 / (5 - 1) \\ &= 316.099 \text{ mm} \end{aligned}$$

Baseplate Pressure Area [Ap]:

$$\begin{aligned} &= e \cdot \text{Bpwid} / 2 \\ &= 316 \cdot 380 / 2 \\ &= 600.589 \text{ cm}^2 \end{aligned}$$

Bearing Pressure [Bp]:

$$\begin{aligned} &= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth}) \\ &= 101226 / (1406 \cdot 380) \\ &= 18.946 \text{ kgf/cm}^2 \end{aligned}$$

Axial Load [P]:

$$\begin{aligned} &= \text{Ap} \cdot \text{Bp} \\ &= 601 \cdot 18.9 \\ &= 11378.923 \text{ kgf} \end{aligned}$$

Area of the Rib and Web [Ar]:

$$\begin{aligned} &= \text{Rib Area} + \text{Web Area} \\ &= 48.2 + 44.3 \\ &= 92.414 \text{ cm}^2 \end{aligned}$$

Compressive Stress [Sc]:

$$\begin{aligned} &= P / \text{Ar} \\ &= 11379 / 92.4 \\ &= 123.130 \text{ kgf/cm}^2 \end{aligned}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	28.0	200.0	...	56.0	...	0.187E+04

Rib dimension [D]:

$$\begin{aligned} &= \text{Saddle Width} - \text{Web Thickness} \\ &= 200 - 28 \\ &= 172.000 \text{ mm} \end{aligned}$$

Distance to Centroid from Datum [ytot]:

$$\begin{aligned} &= \text{AY} / \text{A} \\ &= 0 / 92.4 \\ &= 0.000 \text{ mm} \end{aligned}$$

Distance to Centroid [C1]:

$$\begin{aligned} &= \text{Saddle Width} / 2 \\ &= 200 / 2 \\ &= 100.000 \text{ mm} \end{aligned}$$

Radius of Gyration [r]:

$$\begin{aligned} &= \sqrt{ (\text{Total Inertia} / \text{Total Area}) } \\ &= \sqrt{ (1867 / 92.4) } \\ &= 44.943 \text{ mm} \end{aligned}$$

Length of Outer Rib [L]:

$$\begin{aligned} &= \text{Saddle Height} - \cos(\text{theta} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk} \\ &= 1300 - \cos(120 / 2) (650 + 68 + 12) - 36 \\ &= 899.000 \text{ mm} \end{aligned}$$

Intermediate Term [Cc]:

$$\begin{aligned} &= \sqrt{ (2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}) } \\ &= \sqrt{ (2 \cdot \pi^2 \cdot 2038903 / 2580) } \\ &= 124.902 \end{aligned}$$

Slenderness ratio [KL/r]:

$$= \text{KL} / \text{r}$$

$$= 1 \cdot 899/44.9$$
$$= 20.003$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$
$$= 13723 / (2 \cdot 1406) \cdot 316 \cdot 899 / 2$$
$$= 693.386 \text{ kgf-m.}$$

Compressive Allowable, $KL/r < C_c$ ($20 < 125$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$
$$= (1 - (20)^2 / (2 \cdot 125^2)) 2580 /$$
$$(5/3 + 3 \cdot (20) / (8 \cdot 125) - (20^3) / (8 \cdot 125^3))$$
$$= 1475 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$
$$= 123/1475 + (693 \cdot 100/18666668) / 1720$$
$$= 0.299$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	28.0	172.0	0.0	48.2	0.0	0.186E+04
Web	316.1	28.0	0.0	88.5	0.0	57.8
Totals	...	...	...	136.7	...	0.192E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$
$$= 0/137$$
$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$
$$= 200/2$$
$$= 100.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$
$$= 1300 - \cos(730/2) (36 + 0 + 0) - 0$$
$$= 534.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$
$$= \sqrt{ (1919/137) }$$
$$= 37.475 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$
$$= 1 \cdot 534/37.5$$
$$= 14.249$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$
$$= 13723 / (2 \cdot 1406)$$
$$= 4.880 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$
$$= 4.88 \cdot 316 \cdot 534$$
$$= 823.733 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$
$$= 824/191937$$
$$= 429.169 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < C_c$ ($14.2 < 125$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$



$$= (1 - (14.2)^2 / (2 \cdot 125^2)) 2580 /$$

$$(5/3 + 3 \cdot (14.2) / (8 \cdot 125) - (14.2^3) / (8 \cdot 125^3))$$

$$= 1499 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 126/1499 + (824 \cdot 100/1919) / 1720$$

$$= 0.334$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	1400.02 kgf/cm ²
Bolt Corrosion Allowance	Bca	0.8 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	5.4000 cm ²
BasePlate Allowable Stress	S	1723.26 kgf/cm ²
Area Available in a Single Bolt	BltArea	4.7998 cm ²
Saddle Load QO (Weight)	QO	80062.7 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	1939.1 kgf
Maximum Transverse Force	Ft	6861.3 kgf
Maximum Longitudinal Force	Fl	13722.6 kgf
Saddle Bolted to Steel Foundation		No

Friction force due to thermal growth, [Ff]:
 = 10462.9 kgf

Shear Stress in a Single Bolt, Longitudinal Direction [τ_{b,l}]:
 = max(Ff, Fl) / (Bolt Area • Number of Bolts)
 = 13723 / (4.8 • 4)
 = 714.7 kgf/cm². Must be less than 1100.0 kgf/cm².

Shear Stress in a Single Bolt, Transverse Direction [τ_{b,t}]:
 = Ft / (Bolt Area • Number of Bolts)
 = 6861 / (4.8 • 4)
 = 357.4 kgf/cm². Must be less than 1100.0 kgf/cm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 = 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 = Fl / (BoltShearAllowable • Nbolts)
 = 13723 / (1100 • 4)
 = 3.1189 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 = B • Ft + Sum of X Moments
 = 1300 • 6861 + 0
 = 8919.71 kgf-m.

Eccentricity (e):
 = Rmom / QO
 = 8920/80063
 = 111.41 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 = 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 = max[Bltarearl, Bltarears, Bltareart]
 = max[0, 3.12, 0]
 = 3.1189 cm²



Note: Saddle and wear plate material are both specified as impact tested.

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	684.75	mm
Shell Thickness used in this Case	t	68.000	mm
Stiffened Vessel Length per 4.15.6	L	1705.08	cm
Distance from Saddle to Vessel tangent	a1 or a	3000.00	mm
Saddle Width	b1 or b	200.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	560.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	12.0	mm
Wear Plate Allowable Stress	fw or Sr	1723.26	kgf/cm ²
Inside or Mean Depth of Head	Hi or hm	35.58	cm
Shell Allowable Stress used in Calculation		1723.26	kgf/cm ²
Head Allowable Stress used in Calculation		1723.26	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1300.00	mm
Coefficient of Friction	mu	0.30	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		55362.12	kgf
Pressure used in Analysis	P	163.162	kgf/cm ²

Horizontal Vessel Analysis Results:	Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan	774.64	1723.26
Long. Stress at Bottom of Midspan	905.43	1723.26
Long. Stress at Top of Saddles	1150.15	1723.26
Long. Stress at Bottom of Saddles	668.16	1723.26
Tangential Shear in Shell	89.75	1375.03
Circ. Stress at Horn of Saddle	104.31	2154.08
Circ. Compressive Stress in Shell	10.06	1723.26

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr}(F_t/\text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (7095/2 + 0) \cdot 1300/1264$
 $= 10942.9 \text{ kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l,\text{wind}}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(551, 0) \cdot 1300/9200$
 $= 77.9 \text{ kgf}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(F_{l,\text{eq}}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(13723, 0) \cdot 1300/9200$
 $= 1939.1 \text{ kgf}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= F_{tr}(F_t/\text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (13723/2 + 0) \cdot 1300/1264$
 $= 21163.5 \text{ kgf}$



Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 34199 + \max(77.9, 10943, 1939, 21164)$$

$$= 55362.1 \text{ kgf}$$

Longitudinal Wind Force [FI]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1727 (0 + \pi/4(1.59 \cdot 1.2)^2)$$

$$= 5405.922 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	56039.95	kgf
Transverse Shear Load	6861.31	kgf
Longitudinal Shear Load	13722.61	kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(1300/2 + 68 + 12)\sin(120/2), 2 \cdot 685)$$

$$= 1264.397 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$$

$$= -55362 \cdot 300 [1 - (1 - 300/1705 + (68.5^2 - 35.6^2)/(2 \cdot 300 \cdot 1705))/(1 + (4 \cdot 35.6)/(3 \cdot 1705))]$$

$$= -32385.7 \text{ kgf-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 55362 \cdot 1705/4((1 + 2(68.5^2 - 35.6^2)/(1705^2))/(1 + (4 \cdot 35.6)/(3 \cdot 1705)) - 4 \cdot 300/1705)$$

$$= 64059.2 \text{ kgf-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M_2/(\pi \cdot R_m^2 \cdot t)$$

$$= 163 \cdot 685/(2 \cdot 66.5) - 64059/(\pi \cdot 685^2 \cdot 66.5)$$

$$= 774.64 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M_2/(\pi \cdot R_m^2 \cdot t)$$

$$= 163 \cdot 685/(2 \cdot 66.5) + 64059/(\pi \cdot 685^2 \cdot 66.5)$$

$$= 905.43 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m/(2t) - M_1/(K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 163 \cdot 685/(2 \cdot 66.5) - 32386/(0.11 \cdot \pi \cdot 685^2 \cdot 66.5)$$

$$= 1150.15 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m/(2t) + M_1/(K_1^* \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 163 \cdot 685/(2 \cdot 66.5) + 32386/(0.19 \cdot \pi \cdot 685^2 \cdot 66.5)$$

$$= 668.16 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot h_m/3))$$

$$= 55362 (1705 - 2 \cdot 300)/(1705 + (4 \cdot 35.6/3))$$

$$= 34909.6 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 1.17 \cdot 34910 / (685 \cdot 66.5)$$

$$= 89.75 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x1,x2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(685 \cdot 66.5)}$$

$$= 166.445 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(200 + 1.56 \cdot \sqrt{(685 \cdot 66.5)}, 2 \cdot 300)$$

$$= 532.89 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$= \min(Sr/S, 1)$$

$$= \min(1723/1723, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ6,r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 55362 \cdot 0.1 / (533(66.5 + 1 \cdot 12))$$

$$= -10.06 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, L ≥ 8Rm (4.15.28) [σ7,r]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -55362 / (4(66.5 + 1 \cdot 12)533) -$$

$$3 \cdot 0.053 \cdot 55362 / (2(66.5 + 1 \cdot 12)^2)$$

$$= -104.31 \text{ kgf/cm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1406.0000	mm
Baseplate Thickness	Bpthk	36.0000	mm
Baseplate Width	Bpwid	380.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	28.0000	mm
Web Thickness	Webtk	28.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	2579.8	kgf/cm ²
Height of Web at Center	Hw,c	600.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(652 \cdot 66.5)} + 560)66.5^3) / 12$$

$$= 2168.119 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 2168 + 588(172 - 33.2)^2$$

$$= 116033.555 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	884.7	66.5	33.2	588.3	1956199.5	0.116E+06
Wearplate	560.0	12.0	72.5	67.2	487200.0	0.671E+04
Web	28.0	534.0	345.5	149.5	5165915.5	0.803E+05
BasePlate	380.0	36.0	630.5	136.8	8625240.0	0.287E+06
Totals	...	...	...	941.9	16234555.0	0.490E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 16234555/942$$

$$= 172.369 \text{ mm}$$

Angle [β]:
= 180 - Saddle Angle/2
= 180 - 120/2
= 120.0

Saddle Splitting Coefficient [K1]:
= (1 + cos(β) - 0.5•sin(β)²)/(π - β + sin(β)cos(β))
= (1 + cos(120) - 0.5•sin(120)²)/(π - 2.09 + sin(120)cos(120))
= 0.2035

Saddle Splitting Force [Fh]:
= K1 • Q
= 0.2 • 55362
= 11267.3926 kgf

Tension Stress, St = (Fh/As) = 31.8721 kgf/cm²
Allowed Stress, Sa = 0.6 • Yield Str = 1547.8756 kgf/cm²

Saddle Splitting Dimension [d]:
= B - R • sin(theta/2)/(theta/2 in radians)
= 1300 - 652 • sin(120/2)/1.05
= 761.214 mm

Bending Moment, M = Fh • d = 8576.9043 kgf-m.

Bending Stress, Sb = (M • C1 / I) = 30.1490 kgf/cm²
Allowed Stress, Sa = 2/3 • Yield Str = 1719.8618 kgf/cm²

Minimum Thickness of Baseplate per Moss:

The maximum of 2/3•Yield and the Saddle Allowable Stress value was used to determine the baseplate thickness.

$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$
$$= \sqrt{3(55362 + 678)380 / (4 \cdot 1406 \cdot 1723)}$$
$$= 25.675 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
= 2 • cos(90 - Saddle Angle/2)(Inside Radius + Shell Thk + Wear Plate Thk)
= 2 • cos(90 - 120/2)(650 + 68 + 12)
= 1264.397 mm

Distance between Ribs [e]:
= Web Length / (Nribs - 1)
= 1264/(5 - 1)
= 316.099 mm

Baseplate Pressure Area [Ap]:
= e • Bpwid / 2
= 316 • 380/2
= 600.589 cm²

Bearing Pressure [Bp]:
= (Q + SaddleWt) / (BasePlateLength • BasePlateWidth)
= 56040/(1406 • 380)
= 10.489 kgf/cm²

Axial Load [P]:
= Ap • Bp
= 601 • 10.5
= 6299.497 kgf

Area of the Rib and Web [Ar]:
= Rib Area + Web Area
= 48.2 + 44.3
= 92.414 cm²

Compressive Stress [Sc]:

$$= P/Ar$$

$$= 6299/92.4$$

$$= 68.166 \text{ kgf/cm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	28.0	200.0	...	56.0	...	0.187E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 200 - 28$$

$$= 172.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/92.4$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 200/2$$

$$= 100.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(1867/92.4)}$$

$$= 44.943 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta}/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1300 - \cos(120/2) (650 + 68 + 12) - 36$$

$$= 899.000 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 2038903/2580)}$$

$$= 124.902$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 899/44.9$$

$$= 20.003$$

Bending Moment [Rm]:

$$= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 13723 / (2 \cdot 1406) \cdot 316 \cdot 899/2$$

$$= 693.386 \text{ kgf-m.}$$

Compressive Allowable, $KL/r < Cc$ ($20 < 125$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (20)^2 / (2 \cdot 125^2)) 2580 / (5/3 + 3 \cdot (20) / (8 \cdot 125) - (20^3) / (8 \cdot 125^3))$$

$$= 1475 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 68.2/1475 + (693 \cdot 100/18666668) / 1720$$

$$= 0.262$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	28.0	172.0	0.0	48.2	0.0	0.186E+04
Web	316.1	28.0	0.0	88.5	0.0	57.8



Totals	...	...	...	136.7	...	0.192E+04
--------	-----	-----	-----	-------	-----	-----------

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/137$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 200/2$$

$$= 100.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1300 - \cos(730/2) (36 + 0 + 0) - 0$$

$$= 534.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(1919/137)}$$

$$= 37.475 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 534/37.5$$

$$= 14.249$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 13723 / (2 \cdot 1406)$$

$$= 4.880 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 4.88 \cdot 316 \cdot 534$$

$$= 823.733 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 824/191937$$

$$= 429.169 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < C_c$ ($14.2 < 125$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (14.2)^2 / (2 \cdot 125^2)) 2580 /$$

$$(5/3 + 3 \cdot (14.2) / (8 \cdot 125) - (14.2)^3 / (8 \cdot 125^3))$$

$$= 1499 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 70/1499 + (824 \cdot 100/1919) / 1720$$

$$= 0.296$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	1400.02 kgf/cm ²
Bolt Corrosion Allowance	Bca	0.8 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	5.4000 cm ²
BasePlate Allowable Stress	S	1723.26 kgf/cm ²
Area Available in a Single Bolt	BltArea	4.7998 cm ²
Saddle Load Q0 (Weight)	Q0	34876.4 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	1939.1 kgf
Maximum Transverse Force	Ft	6861.3 kgf
Maximum Longitudinal Force	Fl	13722.6 kgf
Saddle Bolted to Steel Foundation	No	



Shear Stress in a Single Bolt, Longitudinal Direction [τ_b, l]:
= $F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$
= $13723 / (4.8 \cdot 4)$
= 714.7 kgf/cm^2 . Must be less than 1100.0 kgf/cm^2 .

Shear Stress in a Single Bolt, Transverse Direction [τ_b, t]:
= $F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$
= $6861 / (4.8 \cdot 4)$
= 357.4 kgf/cm^2 . Must be less than 1100.0 kgf/cm^2 .

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [$Bltarearl$]:
= 0.0 ($QO > QL \rightarrow$ No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [$Bltarears$]:
= $F_l / (\text{BoltShearAllowable} \cdot \text{Nbols})$
= $13723 / (1100 \cdot 4)$
= 3.1189 cm^2

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [$Rmom$]:
= $B \cdot F_t + \text{Sum of X Moments}$
= $1300 \cdot 6861 + 0$
= 8919.71 kgf-m .

Eccentricity (e):
= $Rmom / QO$
= $8920 / 34876$
= $255.75 \text{ mm} > Bplen / 6 \rightarrow$ Uplift in Transverse direction

$f = Bplen / 2 - Edgedis$
= $1406 / 2 - 50.8$
= 652.20 mm

Modular Ratio Of Steel/Concrete ($n1$):
= ES / EC
= $2074056 / 219500$
= 9.45

$K1 = 3 (e - 0.5 \cdot Bplen)$
= $3 (256 - 0.5 \cdot 1406)$
= -1341.75 mm

$K2 = 6 \cdot n1 \cdot At / Bpwid \cdot (f + e)$
= $6 \cdot 9.45 \cdot 9.6 / 380 \cdot (652 + 256)$
= 130037.97 mm^2

$K3 = -K2 \cdot (0.5 \cdot Bplen + f)$
= $-130038 \cdot (0.5 \cdot 1406 + 652)$
= $-176227452.43 \text{ mm}^3$

Iteratively Solving for the Effective Bearing Length:
 $Y^3 + K1 \cdot Y^2 + K2 \cdot Y + K3 = 0$
 $Y^3 + -1342 \cdot Y^2 + 130038 \cdot Y + -176227440 = 0$
 $Y = 1342.65 \text{ mm}$

$Num = (Bplen / 2 - Y / 3 - e)$
= $(1406 / 2 - 1343 / 3 - 256)$
= -0.30

$Denom = (Bplen / 2 - Y / 3 + f)$
= $(1406 / 2 - 1343 / 3 + 652)$
= 907.65

Total Bolt Tension Force [$Tforce$]:
= $-QO \cdot Num / Denom$
= $-34876 \cdot -0.3 / 908$



= 11.59 kgf

Bolt Area Required due to Transverse Load [Bltareart]:

$$\begin{aligned} &= Tforce / (Stba \cdot Nbt) \\ &= 11.6 / (1400 \cdot 2) \\ &= 0.0041 \text{ cm}^2 \end{aligned}$$

Required Area of a Single Bolt [Bltarear]:

$$\begin{aligned} &= \max[Bltarearl, Bltarears, Bltareart] \\ &= \max[0, 3.12, 0.0041] \\ &= 3.1189 \text{ cm}^2 \end{aligned}$$

Baseplate Thickness Calculation per D. Moss:

Bearing Pressure (fc)

$$\begin{aligned} &= 2(QO + Tforce) / (Y \cdot Bpwid) \\ &= 2(34876 + 11.6) / (1343 \cdot 380) \\ &= 13.68 \text{ kgf/cm}^2 \end{aligned}$$

Distance from Baseplate Edge to the Web [ADIST]:

$$\begin{aligned} &= (Bplen - WebIngh) / 2 \\ &= (1406 - 1355) / 2 \\ &= 25.4000 \text{ mm} \end{aligned}$$

Overturning Moment due To Bolt Tension [Mt]:

$$\begin{aligned} &= Tforce \cdot Adist \\ &= 11.6 \cdot 2.54 \\ &= 0.29 \text{ kgf-m.} \end{aligned}$$

Equivalent Bearing Pressure (f1):

$$\begin{aligned} &= fc \cdot (Y - Adist) / Y \\ &= 13.7 \cdot (1343 - 25.4) / 1343 \\ &= 13.42 \text{ kgf/cm}^2 \end{aligned}$$

Overturning Moment due to Bearing Pressure [Mc]:

$$\begin{aligned} &= (Adist^2 \cdot Bpwid / 6) \cdot (f1 + 2 \cdot fc) \\ &= (25.4^2 \cdot 380 / 6) \cdot (13.4 + 2 \cdot 13.7) \\ &= 16.66 \text{ kgf-m.} \end{aligned}$$

Baseplate Required Thickness [Treq]:

$$\begin{aligned} &= \sqrt{(6 \cdot \max(Mt, Mc) / (Bpwid \cdot Sba))} \\ &= \sqrt{(6 \cdot \max(0.29, 16.7) / (380 \cdot 2585))} \\ &= 3.1899 \text{ mm} \end{aligned}$$

Note: Saddle and wear plate material are both specified as impact tested.

Review notes about nozzle loadings and supports in the Warnings report.

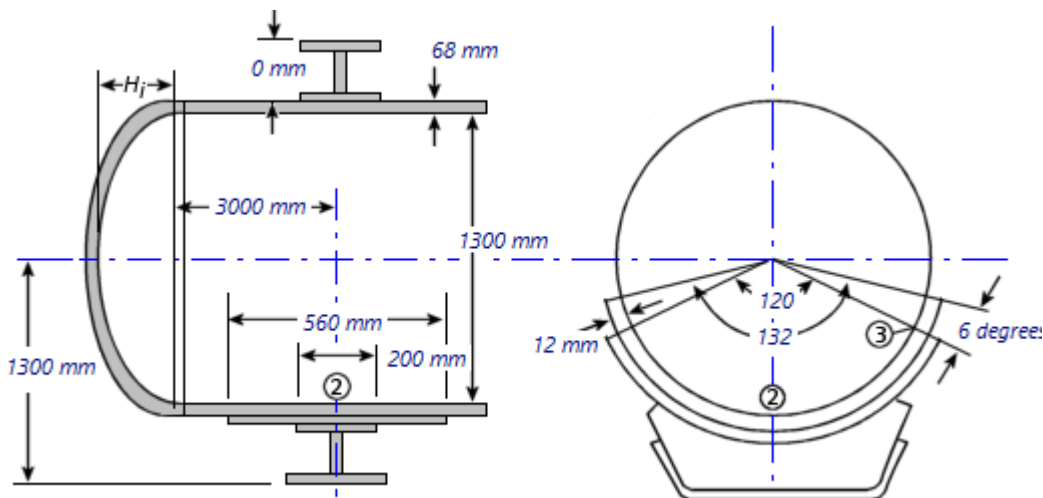
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Input and Calculated Values:

Vessel Mean Radius	Rm	684.00	mm
Shell Thickness used in this Case	t	68.000	mm
Stiffened Vessel Length per 4.15.6	L	1705.08	cm
Distance from Saddle to Vessel tangent	a1 or a	3000.00	mm
Saddle Width	b1 or b	200.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	560.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	12.0	mm
Wear Plate Allowable Stress	fw or Sr	1723.26	kgf/cm ²
Shell Allowable Stress used in Calculation		3002.91	kgf/cm ²
Head Allowable Stress used in Calculation		3002.91	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1300.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		90552.58	kgf
Pressure used in Analysis	P	256.488	kgf/cm ²

Horizontal Vessel Analysis Results:		Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan		1174.34	3002.91
Long. Stress at Bottom of Midspan		1405.63	3002.91
Long. Stress at Top of Saddles		1726.89	3002.91
Long. Stress at Bottom of Saddles		1047.82	3002.91
Tangential Shear in Shell		147.72	1684.79
Circ. Stress at Horn of Saddle		164.92	4504.36
Circ. Compressive Stress in Shell		16.04	3002.91

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft}/\text{Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (2342/2 + 0) \cdot 1300/1264$$

$$= 3611.2 \text{ kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{F}_{\text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(182, 0) \cdot 1300/9200$$

$$= 25.7 \text{ kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 86941 + \max(25.7, 3611, 0, 0)$$

$$= 90552.6 \text{ kgf}$$

Longitudinal Wind Force [FI]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1727 (0 + \pi/4(1.59 \cdot 1.2)^2)$$

$$= 5405.922 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	91230.40	kgf
Transverse Shear Load	1170.75	kgf
Longitudinal Shear Load	181.91	kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(1300/2 + 68 + 12)\sin(120/2), 2 \cdot 684)$$

$$= 1264.397 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$$

$$= -90553 \cdot 300 [1 - (1 - 300/1705 + (68.4^2 - 0^2)/(2 \cdot 300 \cdot 1705))/(1 + (4 \cdot 0)/(3 \cdot 1705))]$$

$$= -46554.5 \text{ kgf-m}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 90553 \cdot 1705/4((1 + 2(68.4^2 - 0^2)/(1705^2))/(1 + (4 \cdot 0)/(3 \cdot 1705)) - 4 \cdot 300/1705)$$

$$= 115583.2 \text{ kgf-m}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M_2/(\pi \cdot R_m^2 t)$$

$$= 256 \cdot 684/(2 \cdot 68) - 115583/(\pi \cdot 684^2 \cdot 68)$$

$$= 1174.34 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M_2/(\pi \cdot R_m^2 \cdot t)$$

$$= 256 \cdot 684/(2 \cdot 68) + 115583/(\pi \cdot 684^2 \cdot 68)$$

$$= 1405.63 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m/(2t) - M_1/(K_1 \cdot \pi \cdot R_m^2 t)$$

$$= 256 \cdot 684/(2 \cdot 68) - 46554/(0.11 \cdot \pi \cdot 684^2 \cdot 68)$$

$$= 1726.89 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m/(2t) + M_1/(K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 256 \cdot 684 / (2 \cdot 68) + 46554 / (0.19 \cdot \pi \cdot 684^2 \cdot 68)$$

$$= 1047.82 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a)/(L+(4 \cdot hm/3))$$

$$= 90553 (1705 - 2 \cdot 300) / (1705 + (4 \cdot 0/3))$$

$$= 58688.1 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 58688 / (684 \cdot 68)$$

$$= 147.72 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(684 \cdot 68)}$$

$$= 168.220 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(200 + 1.56 \cdot \sqrt{(684 \cdot 68)}, 2 \cdot 300)$$

$$= 536.44 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, test case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 90553 \cdot 0.1 / (536(68 + 1 \cdot 12))$$

$$= -16.04 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -90553 / (4(68 + 1 \cdot 12)536) -$$

$$3 \cdot 0.053 \cdot 90553 / (2(68 + 1 \cdot 12)^2)$$

$$= -164.92 \text{ kgf/cm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1406.0000	mm
Baseplate Thickness	Bpthk	36.0000	mm
Baseplate Width	Bpwid	380.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	28.0000	mm
Web Thickness	Webtk	28.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	3161.0	kgf/cm ²
Height of Web at Center	Hw,c	600.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(650 \cdot 68)} + 560)68^3) / 12$$

$$= 2326.719 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 2327 + 604(171 - 34)^2$$

$$= 115897.195 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	888.0	68.0	34.0	603.8	2052989.4	0.116E+06
Wearplate	560.0	12.0	74.0	67.2	497280.0	0.635E+04
Web	28.0	534.0	347.0	149.5	5188343.0	0.818E+05
BasePlate	380.0	36.0	632.0	136.8	8645760.0	0.291E+06
Totals	...	...	...	957.3	16384372.0	0.495E+06

Distance to Centroid [C1]:

$$\begin{aligned} &= AY / A \\ &= 16384372/957 \\ &= 171.145 \text{ mm} \end{aligned}$$

Angle [β]:

$$\begin{aligned} &= 180 - \text{Saddle Angle}/2 \\ &= 180 - 120/2 \\ &= 120.0 \end{aligned}$$

Saddle Splitting Coefficient [K1]:

$$\begin{aligned} &= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta)) \\ &= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120)) \\ &= 0.2035 \end{aligned}$$

Saddle Splitting Force [Fh]:

$$\begin{aligned} &= K1 \cdot Q \\ &= 0.2 \cdot 90553 \\ &= 18429.4121 \text{ kgf} \end{aligned}$$

$$\begin{aligned} \text{Tension Stress, } St &= (Fh/As) = 52.1313 \text{ kgf/cm}^2 \\ \text{Allowed Stress, } Sa &= 0.6 \cdot \text{Yield Str} = 1896.5723 \text{ kgf/cm}^2 \end{aligned}$$

Saddle Splitting Dimension [d]:

$$\begin{aligned} &= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians}) \\ &= 1300 - 650 \cdot \sin(120/2) / 1.05 \\ &= 762.454 \text{ mm} \end{aligned}$$

$$\text{Bending Moment, } M = Fh \cdot d = 14051.5996 \text{ kgf-m.}$$

$$\begin{aligned} \text{Bending Stress, } Sb &= (M \cdot C1 / I) = 48.6114 \text{ kgf/cm}^2 \\ \text{Allowed Stress, } Sa &= 2/3 \cdot \text{Yield Str} = 2107.3025 \text{ kgf/cm}^2 \end{aligned}$$

Minimum Thickness of Baseplate per Moss:

$$\begin{aligned} &= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})) \\ &= \sqrt{(3(90553 + 678) \cdot 380) / (4 \cdot 1406 \cdot 2107)} \\ &= 29.624 \text{ mm} \end{aligned}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$\begin{aligned} &= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\ &= 2 \cdot \cos(90 - 120/2) (650 + 68 + 12) \\ &= 1264.397 \text{ mm} \end{aligned}$$

Distance between Ribs [e]:

$$\begin{aligned} &= \text{Web Length} / (\text{Nr ribs} - 1) \\ &= 1264 / (5 - 1) \\ &= 316.099 \text{ mm} \end{aligned}$$

Baseplate Pressure Area [Ap]:

$$\begin{aligned} &= e \cdot \text{Bpwid} / 2 \\ &= 316 \cdot 380 / 2 \\ &= 600.589 \text{ cm}^2 \end{aligned}$$

Bearing Pressure [Bp]:

$$\begin{aligned} &= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth}) \\ &= 91230 / (1406 \cdot 380) \\ &= 17.075 \text{ kgf/cm}^2 \end{aligned}$$

Axial Load [P]:

$$\begin{aligned} &= Ap \cdot Bp \\ &= 601 \cdot 17.1 \\ &= 10255.285 \text{ kgf} \end{aligned}$$

Area of the Rib and Web [Ar]:

$$\begin{aligned} &= \text{Rib Area} + \text{Web Area} \\ &= 48.2 + 44.3 \\ &= 92.414 \text{ cm}^2 \end{aligned}$$

Compressive Stress [Sc]:

$$= P/A_r$$

$$= 10255/92.4$$

$$= 110.971 \text{ kgf/cm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	28.0	200.0	...	56.0	...	0.187E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 200 - 28$$

$$= 172.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/92.4$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 200/2$$

$$= 100.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(1867/92.4)}$$

$$= 44.943 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1300 - \cos(120/2) (650 + 68 + 12) - 36$$

$$= 899.000 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 2038903/3161)}$$

$$= 112.838$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 899/44.9$$

$$= 20.003$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{pln}) \cdot e \cdot L / 2$$

$$= 182 / (2 \cdot 1406) \cdot 316 \cdot 899/2$$

$$= 9.192 \text{ kgf-m.}$$

Compressive Allowable, $KL/r < C_c$ ($20 < 113$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (20)^2 / (2 \cdot 113^2)) 3161 / (5/3 + 3 \cdot (20) / (8 \cdot 113) - (20^3) / (8 \cdot 113^3))$$

$$= 1796 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 111/1796 + (9.19 \cdot 100/18666668) / 2107$$

$$= 0.064$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	28.0	172.0	0.0	48.2	0.0	0.186E+04



Web	316.1	28.0	0.0	88.5	0.0	57.8
Totals	...	...	...	136.7	...	0.192E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/137$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 200/2$$

$$= 100.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1300 - \cos(730/2)(36 + 0 + 0) - 0$$

$$= 534.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(1919/137)}$$

$$= 37.475 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 534/37.5$$

$$= 14.249$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 182/(2 \cdot 1406)$$

$$= 0.065 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.065 \cdot 316 \cdot 534$$

$$= 10.920 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 10.9/191937$$

$$= 5.689 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < C_c$ ($14.2 < 113$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (14.2)^2 / (2 \cdot 113^2)) 3161 /$$

$$(5/3 + 3 \cdot (14.2) / (8 \cdot 113) - (14.2^3) / (8 \cdot 113^3))$$

$$= 1830 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 114/1830 + (10.9 \cdot 100/1919) / 2107$$

$$= 0.065$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	1753.82 kgf/cm ²
Bolt Corrosion Allowance	Bca	0.8 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	5.4000 cm ²
BasePlate Allowable Stress	S	2110.70 kgf/cm ²
Area Available in a Single Bolt	BltArea	4.7998 cm ²
Saddle Load Q0 (Weight)	Q0	87619.2 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	25.7 kgf
Maximum Transverse Force	Ft	1170.8 kgf
Maximum Longitudinal Force	Fl	181.9 kgf
Saddle Bolted to Steel Foundation	No	

Shear Stress in a Single Bolt, Longitudinal Direction [τ_{b,l}]:
 $= F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 182 / (4.8 \cdot 4)$
 $= 9.5 \text{ kgf/cm}^2$. Must be less than 1052.3 kgf/cm².

Shear Stress in a Single Bolt, Transverse Direction [τ_{b,t}]:
 $= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 1171 / (4.8 \cdot 4)$
 $= 61.0 \text{ kgf/cm}^2$. Must be less than 1052.3 kgf/cm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 $= 0.0$ (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 $= F_l / (\text{BoltShearAllowable} \cdot \text{Nbols})$
 $= 182 / (1052 \cdot 4)$
 $= 0.0432 \text{ cm}^2$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 $= B \cdot F_t + \text{Sum of X Moments}$
 $= 1300 \cdot 1171 + 0$
 $= 1521.98 \text{ kgf-m}$.

Eccentricity (e):
 $= R_{mom} / QO$
 $= 1522 / 87619$
 $= 17.37 \text{ mm} < B_{plen} / 6$ --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 $= 0$ (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 $= \max[Bltarearl, Bltarears, Bltareart]$
 $= \max[0, 0.043, 0]$
 $= 0.0432 \text{ cm}^2$

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	684.00	mm
Shell Thickness used in this Case	t	68.000	mm
Stiffened Vessel Length per 4.15.6	L	1705.08	cm
Distance from Saddle to Vessel tangent	a1 or a	3000.00	mm
Saddle Width	b1 or b	200.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	560.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	12.0	mm
Wear Plate Allowable Stress	fw or Sr	1723.26	kgf/cm ²
Inside or Mean Depth of Head	Hi or hm	35.50	cm
Shell Allowable Stress used in Calculation		3002.91	kgf/cm ²
Head Allowable Stress used in Calculation		3002.91	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1300.00	mm
Coefficient of Friction	mu	0.30	

Results for Horizontal Vessel Analysis:



Saddle Force Q, Test Case, no Ext. Forces 44058.75 kgf
 Pressure used in Analysis P 256.488 kgf/cm²

Horizontal Vessel Analysis Results:	Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan	1238.96	3002.91
Long. Stress at Bottom of Midspan	1341.00	3002.91
Long. Stress at Top of Saddles	1531.81	3002.91
Long. Stress at Bottom of Saddles	1155.95	3002.91
Tangential Shear in Shell	69.93	1684.79
Circ. Stress at Horn of Saddle	80.24	4504.36
Circ. Compressive Stress in Shell	7.81	3002.91

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (2342/2 + 0) \cdot 1300/1264$
 $= 3611.2 \text{ kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l, \text{wind}}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(182, 0) \cdot 1300/9200$
 $= 25.7 \text{ kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 40448 + \max(25.7, 3611, 0, 0)$
 $= 44058.8 \text{ kgf}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1727 (0 + \pi/4 (1.59 \cdot 1.2)^2)$
 $= 5405.922 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	44736.57	kgf
Transverse Shear Load	1170.75	kgf
Longitudinal Shear Load	181.91	kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$
 $= \min(2(1300/2 + 68 + 12) \sin(120/2), 2 \cdot 684)$
 $= 1264.397 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2) / (2a \cdot L)) / (1 + (4h_m) / (3L))]$
 $= -44059 \cdot 300 [1 - (1 - 300/1705 + (68.4^2 - 35.5^2) / (2 \cdot 300 \cdot 1705)) / (1 + (4 \cdot 35.5) / (3 \cdot 1705))]$
 $= -25768.0 \text{ kgf-m.}$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L / 4 ((1 + 2(R_m^2 - h_m^2) / (L^2)) / (1 + (4h_m^2) / (3L)) - 4a/L)$
 $= 44059 \cdot 1705 / 4 ((1 + 2(68.4^2 - 35.5^2) / (1705^2)) / (1 + (4 \cdot 35.5) / (3 \cdot 1705)) - 4 \cdot 300 / 1705)$



$$(3 \cdot 1705) - 4 \cdot 300 / 1705) \\ = 50990.0 \text{ kgf-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 t) \\ = 256 \cdot 684 / (2 \cdot 68) - 50990 / (\pi \cdot 684^2 \cdot 68) \\ = 1238.96 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t) \\ = 256 \cdot 684 / (2 \cdot 68) + 50990 / (\pi \cdot 684^2 \cdot 68) \\ = 1341.00 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 t) \\ = 256 \cdot 684 / (2 \cdot 68) - 25768 / (0.11 \cdot \pi \cdot 684^2 \cdot 68) \\ = 1531.81 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t) \\ = 256 \cdot 684 / (2 \cdot 68) + 25768 / (0.19 \cdot \pi \cdot 684^2 \cdot 68) \\ = 1155.95 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h_m / 3)) \\ = 44059 (1705 - 2 \cdot 300) / (1705 + (4 \cdot 35.5 / 3)) \\ = 27783.7 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t) \\ = 1.17 \cdot 27784 / (684 \cdot 68) \\ = 69.93 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)} \\ = 0.78 \cdot \sqrt{(684 \cdot 68)} \\ = 168.220 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a) \\ = \min(200 + 1.56 \cdot \sqrt{(684 \cdot 68)}, 2 \cdot 300) \\ = 536.44 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$= 1.0000 \text{ Materials are the same, test case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr)) \\ = -0.76 \cdot 44059 \cdot 0.1 / (536(68 + 1 \cdot 12)) \\ = -7.81 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot tr)^2) \\ = -44059 / (4(68 + 1 \cdot 12)536) - \\ 3 \cdot 0.053 \cdot 44059 / (2(68 + 1 \cdot 12)^2) \\ = -80.24 \text{ kgf/cm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1406.0000	mm
Baseplate Thickness	Bpthk	36.0000	mm
Baseplate Width	Bpwid	380.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	28.0000	mm
Web Thickness	Webtk	28.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	3161.0	kgf/cm ²
Height of Web at Center	Hw,c	600.0	mm
Friction Coefficient	mu	0.300	



Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:
 $= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$
 $= ((1.56 \cdot \sqrt{(650 \cdot 68)} + 560)^3) / 12$
 $= 2326.719 \text{ cm}^4$

Resolved Inertia for the Shell [I,shell]:
 $= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$
 $= 2327 + 604 (171 - 34)^2$
 $= 115897.195 \text{ cm}^4$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	888.0	68.0	34.0	603.8	2052989.4	0.116E+06
Wearplate	560.0	12.0	74.0	67.2	497280.0	0.635E+04
Web	28.0	534.0	347.0	149.5	5188343.0	0.818E+05
BasePlate	380.0	36.0	632.0	136.8	8645760.0	0.291E+06
Totals	...	...	...	957.3	16384372.0	0.495E+06

Distance to Centroid [C1]:
 $= AY / A$
 $= 16384372 / 957$
 $= 171.145 \text{ mm}$

Angle [β]:
 $= 180 - \text{Saddle Angle} / 2$
 $= 180 - 120 / 2$
 $= 120.0$

Saddle Splitting Coefficient [K1]:
 $= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$
 $= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$
 $= 0.2035$

Saddle Splitting Force [Fh]:
 $= K1 \cdot Q$
 $= 0.2 \cdot 44059$
 $= 8966.9102 \text{ kgf}$

Tension Stress, St = (Fh/As) = 25.3647 kgf/cm²
 Allowed Stress, Sa = 0.6 * Yield Str = 1896.5723 kgf/cm²

Saddle Splitting Dimension [d]:
 $= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$
 $= 1300 - 650 \cdot \sin(120/2) / 1.05$
 $= 762.454 \text{ mm}$

Bending Moment, M = Fh * d = 6836.8677 kgf-m.

Bending Stress, Sb = (M * C1 / I) = 23.6521 kgf/cm²
 Allowed Stress, Sa = 2/3 * Yield Str = 2107.3025 kgf/cm²

Minimum Thickness of Baseplate per Moss:

$= \sqrt[3]{(3(Q + \text{SaddleWt}) \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$
 $= \sqrt[3]{(3(44059 + 678)380 / (4 \cdot 1406 \cdot 2107))}$
 $= 20.744 \text{ mm}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 \cdot \cos(90 - 120/2) (650 + 68 + 12)$
 $= 1264.397 \text{ mm}$

Distance between Ribs [e]:
 $= \text{Web Length} / (\text{Nr ribs} - 1)$
 $= 1264 / (5 - 1)$
 $= 316.099 \text{ mm}$



Baseplate Pressure Area [Ap]:

$$\begin{aligned} &= e \cdot Bpwid / 2 \\ &= 316 \cdot 380 / 2 \\ &= 600.589 \text{ cm}^2 \end{aligned}$$

Bearing Pressure [Bp]:

$$\begin{aligned} &= (Q + SaddleWt) / (BasePlateLength \cdot BasePlateWidth) \\ &= 44737 / (1406 \cdot 380) \\ &= 8.373 \text{ kgf/cm}^2 \end{aligned}$$

Axial Load [P]:

$$\begin{aligned} &= Ap \cdot Bp \\ &= 601 \cdot 8.37 \\ &= 5028.875 \text{ kgf} \end{aligned}$$

Area of the Rib and Web [Ar]:

$$\begin{aligned} &= Rib \text{ Area} + Web \text{ Area} \\ &= 48.2 + 44.3 \\ &= 92.414 \text{ cm}^2 \end{aligned}$$

Compressive Stress [Sc]:

$$\begin{aligned} &= P / Ar \\ &= 5029 / 92.4 \\ &= 54.417 \text{ kgf/cm}^2 \end{aligned}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	28.0	200.0	...	56.0	...	0.187E+04

Rib dimension [D]:

$$\begin{aligned} &= Saddle \text{ Width} - Web \text{ Thickness} \\ &= 200 - 28 \\ &= 172.000 \text{ mm} \end{aligned}$$

Distance to Centroid from Datum [ytot]:

$$\begin{aligned} &= AY / A \\ &= 0 / 92.4 \\ &= 0.000 \text{ mm} \end{aligned}$$

Distance to Centroid [C1]:

$$\begin{aligned} &= Saddle \text{ Width} / 2 \\ &= 200 / 2 \\ &= 100.000 \text{ mm} \end{aligned}$$

Radius of Gyration [r]:

$$\begin{aligned} &= \sqrt{ (Total \text{ Inertia} / Total \text{ Area}) } \\ &= \sqrt{ (1867 / 92.4) } \\ &= 44.943 \text{ mm} \end{aligned}$$

Length of Outer Rib [L]:

$$\begin{aligned} &= Saddle \text{ Height} - \cos(\theta / 2) (radius + shlthk + wpdthk) - bpthk \\ &= 1300 - \cos(120 / 2) (650 + 68 + 12) - 36 \\ &= 899.000 \text{ mm} \end{aligned}$$

Intermediate Term [Cc]:

$$\begin{aligned} &= \sqrt{ (2 \cdot \pi^2 \cdot Elastic \text{ Modulus} / Yield \text{ Stress}) } \\ &= \sqrt{ (2 \cdot \pi^2 \cdot 2038903 / 3161) } \\ &= 112.838 \end{aligned}$$

Slenderness ratio [KL/r]:

$$\begin{aligned} &= KL / r \\ &= 1 \cdot 899 / 44.9 \\ &= 20.003 \end{aligned}$$

Bending Moment [Rm]:

$$= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 182 / (2 \cdot 1406) \cdot 316 \cdot 899 / 2$$

$$= 9.192 \text{ kgf-m.}$$

Compressive Allowable, $KL/r < Cc$ ($20 < 113$) per AISC E2-1 [Sca]:

$$= (1 - (Klr)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (Klr) / (8 \cdot Cc) - (Klr^3) / (8 \cdot Cc^3))$$

$$= (1 - (20)^2 / (2 \cdot 113^2)) 3161 / (5/3 + 3 \cdot (20) / (8 \cdot 113) - (20^3) / (8 \cdot 113^3))$$

$$= 1796 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 54.4/1796 + (9.19 \cdot 100/18666668) / 2107$$

$$= 0.033$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	28.0	172.0	0.0	48.2	0.0	0.186E+04
Web	316.1	28.0	0.0	88.5	0.0	57.8
Totals	...	...	...	136.7	...	0.192E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/137$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 200/2$$

$$= 100.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - Bpthk$$

$$= 1300 - \cos(730/2) (36 + 0 + 0) - 0$$

$$= 534.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (1919/137) }$$

$$= 37.475 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 534/37.5$$

$$= 14.249$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 182 / (2 \cdot 1406)$$

$$= 0.065 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.065 \cdot 316 \cdot 534$$

$$= 10.920 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [$\sigma B,base,c$]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 10.9/191937$$

$$= 5.689 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < Cc$ ($14.2 < 113$) per AISC E2-1 [Sca]:

$$= (1 - (Klr)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (Klr) / (8 \cdot Cc) - (Klr^3) / (8 \cdot Cc^3))$$

$$= (1 - (14.2)^2 / (2 \cdot 113^2)) 3161 / (5/3 + 3 \cdot (14.2) / (8 \cdot 113) - (14.2^3) / (8 \cdot 113^3))$$

$$= 1830 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1/I)/Sba$$

$$= 55.9/1830 + (10.9 \cdot 100/1919)/2107$$

$$= 0.033$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	1753.82 kgf/cm ²
Bolt Corrosion Allowance	Bca	0.8 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	5.4000 cm ²
BasePlate Allowable Stress	S	2110.70 kgf/cm ²
Area Available in a Single Bolt	BlArea	4.7998 cm ²
Saddle Load QO (Weight)	QO	41125.4 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	25.7 kgf
Maximum Transverse Force	Ft	1170.8 kgf
Maximum Longitudinal Force	Fl	181.9 kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b, l]:

$$= Fl / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 182 / (4.8 \cdot 4)$$

$$= 9.5 \text{ kgf/cm}^2. \text{ Must be less than } 1052.3 \text{ kgf/cm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b, t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 1171 / (4.8 \cdot 4)$$

$$= 61.0 \text{ kgf/cm}^2. \text{ Must be less than } 1052.3 \text{ kgf/cm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} QO > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= Fl / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 182 / (1052 \cdot 4)$$

$$= 0.0432 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 1300 \cdot 1171 + 0$$

$$= 1521.98 \text{ kgf-m.}$$

Eccentricity (e):

$$= Rmom / QO$$

$$= 1522 / 41125$$

$$= 37.01 \text{ mm} < Bplen/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0, 0.043, 0]$$

$$= 0.0432 \text{ cm}^2$$

Review notes about nozzle loadings and supports in the Warnings report.

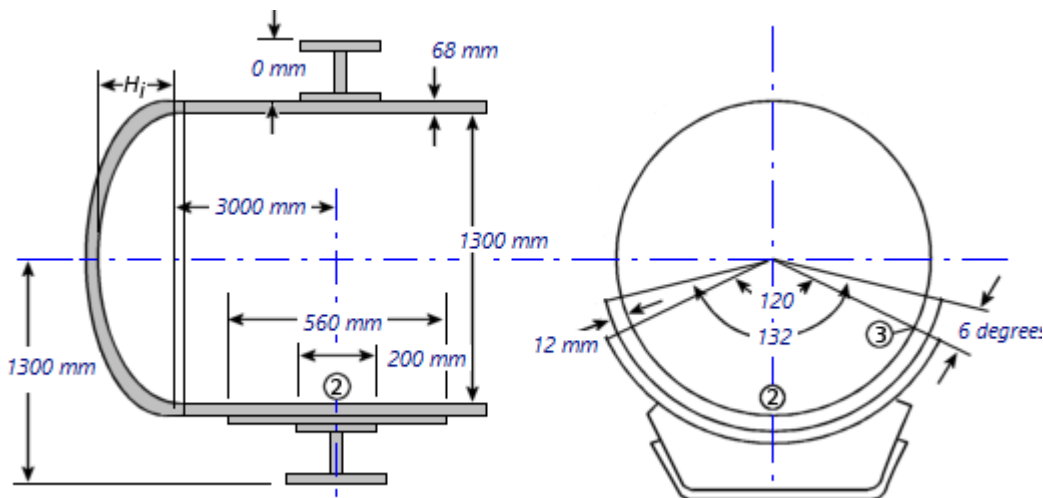
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : External Pressure

Input and Calculated Values:

Vessel Mean Radius	Rm	684.75	mm
Shell Thickness used in this Case	t	68.000	mm
Stiffened Vessel Length per 4.15.6	L	1705.08	cm
Distance from Saddle to Vessel tangent	a1 or a	3000.00	mm
Saddle Width	b1 or b	200.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	560.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	12.0	mm
Wear Plate Allowable Stress	fw or Sr	1723.26	kgf/cm ²
Shell Allowable Stress used in Calculation		1723.26	kgf/cm ²
Head Allowable Stress used in Calculation		1521.35	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1300.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case	100548.38	kgf
Pressure used in Analysis	P	-0.000 kgf/cm ²

Horizontal Vessel Analysis Results:		Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan		-131.02	-2321.81
Long. Stress at Top of Midspan		131.02	1723.26
Long. Stress at Bottom of Midspan		131.02	1723.26
Long. Stress at Top of Saddles		494.96	1723.26
Long. Stress at Bottom of Saddles		-274.34	-2321.81
Long. Stress at Bottom of Saddles		274.34	1723.26
Tangential Shear in Shell		167.54	1375.03
Circ. Stress at Horn of Saddle		189.45	2154.08
Circ. Compressive Stress in Shell		18.27	1723.26

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:



Transverse Saddle Reaction Force [Fwt]:
 $= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$
 $= 3 (7095/2 + 0) \cdot 1300/1264$
 $= 10942.9 \text{ kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls}$
 $= \max(551, 0) \cdot 1300/9200$
 $= 77.9 \text{ kgf}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / \text{Ls}$
 $= \max(13723, 0) \cdot 1300/9200$
 $= 1939.1 \text{ kgf}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$
 $= 3 (13723/2 + 0) \cdot 1300/1264$
 $= 21163.5 \text{ kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$
 $= 79385 + \max(77.9, 10943, 1939, 21164)$
 $= 100548.4 \text{ kgf}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1727 (0 + \pi/4(1.59 \cdot 1.2)^2)$
 $= 5405.922 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	101226.20	kgf
Transverse Shear Load	6861.31	kgf
Longitudinal Shear Load	13722.61	kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot \text{Rm})$
 $= \min(2(1300/2 + 68 + 12) \sin(120/2), 2 \cdot 685)$
 $= 1264.397 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $\text{Rm}/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (\text{Rm}^2 - \text{hm}^2)/(\text{L}^2)) / (1 + (4\text{hm})/3\text{L})]$
 $= -100548 \cdot 300 [1 - (1 - 300/1705 + (68.5^2 - 0^2)/(\text{L}^2)) / (1 + (4 \cdot 0)/(3 \cdot 1705))]$
 $= -51690.5 \text{ kgf-m}$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L/4 ((1 + 2(\text{Rm}^2 - \text{hm}^2)/(\text{L}^2)) / (1 + (4\text{hm})/3\text{L}) - 4a/L)$
 $= 100548 \cdot 1705/4 ((1 + 2(68.5^2 - 0^2)/(1705^2)) / (1 + (4 \cdot 0)/(3 \cdot 1705)) - 4 \cdot 300/1705)$
 $= 128345.1 \text{ kgf-m}$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 $= P \cdot \text{Rm}/(2t) - \text{M2}/(\pi \cdot \text{Rm}^2 t)$
 $= -0 \cdot 685/(2 \cdot 66.5) - 128345/(\pi \cdot 685^2 \cdot 66.5)$
 $= -131.02 \text{ kgf/cm}^2$



Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= -0 \cdot 685 / (2 \cdot 66.5) + 128345 / (\pi \cdot 685^2 \cdot 66.5)$$

$$= 131.02 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= -0 \cdot 685 / (2 \cdot 66.5) - 51690 / (0.11 \cdot \pi \cdot 685^2 \cdot 66.5)$$

$$= 494.96 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= -0 \cdot 685 / (2 \cdot 66.5) + 51690 / (0.19 \cdot \pi \cdot 685^2 \cdot 66.5)$$

$$= -274.34 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h_m / 3))$$

$$= 100548 \cdot (1705 - 2 \cdot 300) / (1705 + (4 \cdot 0 / 3))$$

$$= 65166.4 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 65166 / (685 \cdot 66.5)$$

$$= 167.54 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(685 \cdot 66.5)}$$

$$= 166.445 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(200 + 1.56 \cdot \sqrt{(685 \cdot 66.5)}, 2 \cdot 300)$$

$$= 532.89 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(S_r / S, 1)$$

$$= \min(1723 / 1723, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 100548 \cdot 0.1 / (533(66.5 + 1 \cdot 12))$$

$$= -18.27 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -100548 / (4(66.5 + 1 \cdot 12)533) -$$

$$3 \cdot 0.053 \cdot 100548 / (2(66.5 + 1 \cdot 12)^2)$$

$$= -189.45 \text{ kgf/cm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1406.0000	mm
Baseplate Thickness	Bpthk	36.0000	mm
Baseplate Width	Bpwid	380.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	28.0000	mm
Web Thickness	Webtk	28.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	2579.8	kgf/cm ²
Height of Web at Center	Hw,c	600.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(652 \cdot 66.5)} + 560)^3) / 12$$

$$= 2168.119 \text{ cm}^4$$



Resolved Inertia for the Shell [I,shell]:
 = ShellInertia + AShell(C1 - Y)²
 = 2168 + 588 (172 - 33.2)²
 = 116033.555 cm⁴

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	884.7	66.5	33.2	588.3	1956199.5	0.116E+06
Wearplate	560.0	12.0	72.5	67.2	487200.0	0.671E+04
Web	28.0	534.0	345.5	149.5	5165915.5	0.803E+05
BasePlate	380.0	36.0	630.5	136.8	8625240.0	0.287E+06
Totals	...	...	...	941.9	16234555.0	0.490E+06

Distance to Centroid [C1]:
 = AY / A
 = 16234555/942
 = 172.369 mm

Angle [β]:
 = 180 - Saddle Angle/2
 = 180 - 120/2
 = 120.0

Saddle Splitting Coefficient [K1]:
 = (1 + cos(β) - 0.5•sin(β)²)/(π - β + sin(β)cos(β))
 = (1 + cos(120) - 0.5•sin(120)²)/(π - 2.09 + sin(120)cos(120))
 = 0.2035

Saddle Splitting Force [Fh]:
 = K1 • Q
 = 0.2 • 100548
 = 20463.7773 kgf

Tension Stress, St = (Fh/As) = 57.8858 kgf/cm²
 Allowed Stress, Sa = 0.6 * Yield Str = 1547.8756 kgf/cm²

Saddle Splitting Dimension [d]:
 = B - R • sin(theta/2)/(theta/2 in radians)
 = 1300 - 652 • sin(120/2)/1.05
 = 761.214 mm

Bending Moment, M = Fh * d = 15577.3281 kgf-m.

Bending Stress, Sb = (M * C1 / I) = 54.7564 kgf/cm²
 Allowed Stress, Sa = 2/3 * Yield Str = 1719.8618 kgf/cm²

Minimum Thickness of Baseplate per Moss:

The maximum of 2/3*Yield and the Saddle Allowable Stress value was used to determine the baseplate thickness.

$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(100548 + 678)380 / (4 \cdot 1406 \cdot 1723)}$$
 = 34.507 mm

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 = 2 • cos(90 - Saddle Angle/2)(Inside Radius + Shell Thk + Wear Plate Thk)
 = 2 • cos(90 - 120/2)(650 + 68 + 12)
 = 1264.397 mm

Distance between Ribs [e]:
 = Web Length / (Nribs - 1)
 = 1264/(5 - 1)
 = 316.099 mm

Baseplate Pressure Area [Ap]:



$$= e \cdot Bpwid / 2$$

$$= 316 \cdot 380 / 2$$

$$= 600.589 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + SaddleWt) / (BasePlateLength \cdot BasePlateWidth)$$

$$= 101226 / (1406 \cdot 380)$$

$$= 18.946 \text{ kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 601 \cdot 18.9$$

$$= 11378.923 \text{ kgf}$$

Area of the Rib and Web [Ar]:

$$= Rib \text{ Area} + Web \text{ Area}$$

$$= 48.2 + 44.3$$

$$= 92.414 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 11379 / 92.4$$

$$= 123.130 \text{ kgf/cm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	28.0	200.0	...	56.0	...	0.187E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 200 - 28$$

$$= 172.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 92.4$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 200 / 2$$

$$= 100.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ \text{Total Inertia} / \text{Total Area} }$$

$$= \sqrt{ 1867 / 92.4 }$$

$$= 44.943 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1300 - \cos(120 / 2) (650 + 68 + 12) - 36$$

$$= 899.000 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{ 2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress} }$$

$$= \sqrt{ 2 \cdot \pi^2 \cdot 2038903 / 2580 }$$

$$= 124.902$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 899 / 44.9$$

$$= 20.003$$

Bending Moment [Rm]:

$$= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 13723 / (2 \cdot 1406) \cdot 316 \cdot 899 / 2$$



= 693.386 kgf-m.

Compressive Allowable, $KL/r < Cc$ ($20 < 125$) per AISC E2-1 [Sca]:

$$= \left(1 - \frac{(KL/r)^2}{(2 \cdot Cc^2)} \right) F_y / \left(\frac{5}{3} + 3 \cdot \frac{(KL/r)}{(8 \cdot Cc)} - \frac{(KL/r)^3}{(8 \cdot Cc^3)} \right)$$

$$= \left(1 - \frac{(20)^2}{(2 \cdot 125^2)} \right) 2580 / \left(\frac{5}{3} + 3 \cdot \frac{(20)}{(8 \cdot 125)} - \frac{(20^3)}{(8 \cdot 125^3)} \right)$$

$$= 1475 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 123/1475 + (693 \cdot 100/18666668) / 1720$$

$$= 0.299$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	28.0	172.0	0.0	48.2	0.0	0.186E+04
Web	316.1	28.0	0.0	88.5	0.0	57.8
Totals	...	...	...	136.7	...	0.192E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/137$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 200/2$$

$$= 100.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - Bpthk$$

$$= 1300 - \cos(730/2) (36 + 0 + 0) - 0$$

$$= 534.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(1919/137)}$$

$$= 37.475 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 534/37.5$$

$$= 14.249$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 13723 / (2 \cdot 1406)$$

$$= 4.880 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 4.88 \cdot 316 \cdot 534$$

$$= 823.733 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 824/191937$$

$$= 429.169 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < Cc$ ($14.2 < 125$) per AISC E2-1 [Sca]:

$$= \left(1 - \frac{(KL/r)^2}{(2 \cdot Cc^2)} \right) F_y / \left(\frac{5}{3} + 3 \cdot \frac{(KL/r)}{(8 \cdot Cc)} - \frac{(KL/r)^3}{(8 \cdot Cc^3)} \right)$$

$$= \left(1 - \frac{(14.2)^2}{(2 \cdot 125^2)} \right) 2580 / \left(\frac{5}{3} + 3 \cdot \frac{(14.2)}{(8 \cdot 125)} - \frac{(14.2^3)}{(8 \cdot 125^3)} \right)$$

$$= 1499 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1/I) / Sba$$



$$= 126/1499 + (824 \cdot 100/1919)/1720$$

$$= 0.334$$

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	684.75	mm
Shell Thickness used in this Case	t	68.000	mm
Stiffened Vessel Length per 4.15.6	L	1705.08	cm
Distance from Saddle to Vessel tangent	a1 or a	3000.00	mm
Saddle Width	b1 or b	200.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	560.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	12.0	mm
Wear Plate Allowable Stress	fw or Sr	1723.26	kgf/cm ²
Inside or Mean Depth of Head	Hi or hm	35.58	cm
Shell Allowable Stress used in Calculation		1723.26	kgf/cm ²
Head Allowable Stress used in Calculation		1723.26	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1300.00	mm
Coefficient of Friction	mu	0.30	

Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case		55362.12	kgf
Pressure used in Analysis	P	-0.000	kgf/cm ²

Horizontal Vessel Analysis Results:		Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan		-65.40	-2321.81
Long. Stress at Top of Midspan		65.40	1723.26
Long. Stress at Bottom of Midspan		65.40	1723.26
Long. Stress at Top of Saddles		310.11	1723.26
Long. Stress at Bottom of Saddles		-171.88	-2321.81
Long. Stress at Bottom of Saddles		171.88	1723.26
Tangential Shear in Shell		89.75	1375.03
Circ. Stress at Horn of Saddle		104.31	2154.08
Circ. Compressive Stress in Shell		10.06	1723.26

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr}(F_t/\text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (7095/2 + 0) \cdot 1300/1264$
 $= 10942.9 \text{ kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l,\text{wind}}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(551, 0) \cdot 1300/9200$
 $= 77.9 \text{ kgf}$

Saddle Reaction Force due to Earthquake FI [Fsl]:
 $= \max(F_{l,\text{eq}}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(13723, 0) \cdot 1300/9200$
 $= 1939.1 \text{ kgf}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= F_{tr}(F_t/\text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (13723/2 + 0) \cdot 1300/1264$

= 21163.5 kgf

Load Combination Results for Q + Wind or Seismic [Q]:

= Saddle Load + max(Fwl, Fwt, Fsl, Fst)
 = 34199 + max(77.9, 10943, 1939, 21164)
 = 55362.1 kgf

Longitudinal Wind Force [FI]:

= WindScalar • WindPress(Platform Area + ($\pi/4$ (OD • WindDiaMult)²))
 = 1.1 • 1727 (0 + $\pi/4$ (1.59 • 1.2)²)
 = 5405.922 N

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	56039.95	kgf
Transverse Shear Load	6861.31	kgf
Longitudinal Shear Load	13722.61	kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

= min(2(ShellID/2 + t + WearPadThickness)sin($\theta/2$), 2•Rm)
 = min(2(1300/2 + 68 + 12)sin(120/2), 2•685)
 = 1264.397 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

= -Q•a [1 - (1 - a/L + (Rm²-hm²)/(2a•L))/(1+(4hm)/3L)]
 = -55362•300 [1 - (1 - 300/1705 + (68.5²-35.6²)/(2•300•1705))/(1+(4•35.6)/(3•1705))]
 = -32385.7 kgf-m.

Moment per Equation 4.15.2 [M2]:

= Q•L/4 ((1+2(Rm²-hm²)/(L²))/(1+(4hm²)/(3L)) - 4a/L)
 = 55362•1705/4 ((1+2(68.5²-35.6²)/(1705²))/(1+(4•35.6)/(3•1705)) - 4•300/1705)
 = 64059.2 kgf-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

= P • Rm/(2t) - M2/(π •Rm²t)
 = -0 • 685/(2•66.5) - 64059/(π •685²•66.5)
 = -65.40 kgf/cm²

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

= P • Rm/(2t) + M2/(π • Rm² • t)
 = -0 • 685/(2 • 66.5) + 64059/(π • 685² • 66.5)
 = 65.40 kgf/cm²

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_3]:

= P • Rm/(2t) - M1/(K1• π •Rm²t)
 = -0•685/(2•66.5)--32386/(0.11• π •685²•66.5)
 = 310.11 kgf/cm²

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_4]:

= P • Rm/(2t) + M1/(K1* • π • Rm² • t)
 = -0•685/(2•66.5)+-32386/(0.19• π •685²•66.5)
 = -171.88 kgf/cm²

Maximum Shear Force in the Saddle (4.15.3) [T]:

= Q(L-2a)/(L+(4•hm/3))
 = 55362 (1705 - 2 • 300)/(1705 + (4 • 35.6/3))
 = 34909.6 kgf



Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 34910 / (685 \cdot 66.5)$$

$$= 89.75 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(685 \cdot 66.5)}$$

$$= 166.445 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(200 + 1.56 \cdot \sqrt{(685 \cdot 66.5)}, 2 \cdot 300)$$

$$= 532.89 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(S_r/S, 1)$$

$$= \min(1723/1723, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 55362 \cdot 0.1 / (533(66.5 + 1 \cdot 12))$$

$$= -10.06 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -55362 / (4(66.5 + 1 \cdot 12)533) -$$

$$3 \cdot 0.053 \cdot 55362 / (2(66.5 + 1 \cdot 12)^2)$$

$$= -104.31 \text{ kgf/cm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1406.0000	mm
Baseplate Thickness	Bpthk	36.0000	mm
Baseplate Width	Bpwid	380.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	28.0000	mm
Web Thickness	Webtk	28.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	2579.8	kgf/cm ²
Height of Web at Center	Hw,c	600.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(652 \cdot 66.5)} + 560)^3) / 12$$

$$= 2168.119 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 2168 + 588(172 - 33.2)^2$$

$$= 116033.555 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	884.7	66.5	33.2	588.3	1956199.5	0.116E+06
Wearplate	560.0	12.0	72.5	67.2	487200.0	0.671E+04
Web	28.0	534.0	345.5	149.5	5165915.5	0.803E+05
BasePlate	380.0	36.0	630.5	136.8	8625240.0	0.287E+06
Totals	...	...	...	941.9	16234555.0	0.490E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 16234555/942$$



$$= 172.369 \text{ mm}$$

Angle [β]:

$$\begin{aligned} &= 180 - \text{Saddle Angle}/2 \\ &= 180 - 120/2 \\ &= 120.0 \end{aligned}$$

Saddle Splitting Coefficient [K1]:

$$\begin{aligned} &= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta)) \\ &= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120)) \\ &= 0.2035 \end{aligned}$$

Saddle Splitting Force [Fh]:

$$\begin{aligned} &= K1 \cdot Q \\ &= 0.2 \cdot 55362 \\ &= 11267.3926 \text{ kgf} \end{aligned}$$

$$\begin{aligned} \text{Tension Stress, } St &= (Fh/As) = 31.8721 \text{ kgf/cm}^2 \\ \text{Allowed Stress, } Sa &= 0.6 \cdot \text{Yield Str} = 1547.8756 \text{ kgf/cm}^2 \end{aligned}$$

Saddle Splitting Dimension [d]:

$$\begin{aligned} &= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians}) \\ &= 1300 - 652 \cdot \sin(120/2) / 1.05 \\ &= 761.214 \text{ mm} \end{aligned}$$

$$\text{Bending Moment, } M = Fh \cdot d = 8576.9043 \text{ kgf-m.}$$

$$\begin{aligned} \text{Bending Stress, } Sb &= (M \cdot C1 / I) = 30.1490 \text{ kgf/cm}^2 \\ \text{Allowed Stress, } Sa &= 2/3 \cdot \text{Yield Str} = 1719.8618 \text{ kgf/cm}^2 \end{aligned}$$

Minimum Thickness of Baseplate per Moss:

The maximum of $2/3 \cdot \text{Yield}$ and the Saddle Allowable Stress value was used to determine the baseplate thickness.

$$\begin{aligned} &= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})} \\ &= \sqrt{3(55362 + 678)380 / (4 \cdot 1406 \cdot 1723)} \\ &= 25.675 \text{ mm} \end{aligned}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$\begin{aligned} &= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\ &= 2 \cdot \cos(90 - 120/2) (650 + 68 + 12) \\ &= 1264.397 \text{ mm} \end{aligned}$$

Distance between Ribs [e]:

$$\begin{aligned} &= \text{Web Length} / (\text{Nribs} - 1) \\ &= 1264 / (5 - 1) \\ &= 316.099 \text{ mm} \end{aligned}$$

Baseplate Pressure Area [Ap]:

$$\begin{aligned} &= e \cdot \text{Bpwid} / 2 \\ &= 316 \cdot 380 / 2 \\ &= 600.589 \text{ cm}^2 \end{aligned}$$

Bearing Pressure [Bp]:

$$\begin{aligned} &= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth}) \\ &= 56040 / (1406 \cdot 380) \\ &= 10.489 \text{ kgf/cm}^2 \end{aligned}$$

Axial Load [P]:

$$\begin{aligned} &= Ap \cdot Bp \\ &= 601 \cdot 10.5 \\ &= 6299.497 \text{ kgf} \end{aligned}$$

Area of the Rib and Web [Ar]:

$$\begin{aligned} &= \text{Rib Area} + \text{Web Area} \\ &= 48.2 + 44.3 \\ &= 92.414 \text{ cm}^2 \end{aligned}$$

Compressive Stress [Sc]:

$$= P/A_r$$

$$= 6299/92.4$$

$$= 68.166 \text{ kgf/cm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

B	D	Y	A	AY	Io
Rib+Web 28.0	200.0	...	56.0	...	0.187E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 200 - 28$$

$$= 172.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/92.4$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 200/2$$

$$= 100.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(1867/92.4)}$$

$$= 44.943 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1300 - \cos(120/2) (650 + 68 + 12) - 36$$

$$= 899.000 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 2038903/2580)}$$

$$= 124.902$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 899/44.9$$

$$= 20.003$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 13723 / (2 \cdot 1406) \cdot 316 \cdot 899/2$$

$$= 693.386 \text{ kgf-m.}$$

Compressive Allowable, $KL/r < C_c$ ($20 < 125$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (20)^2 / (2 \cdot 125^2)) 2580 / (5/3 + 3 \cdot (20) / (8 \cdot 125) - (20^3) / (8 \cdot 125^3))$$

$$= 1475 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 68.2/1475 + (693 \cdot 100/18666668) / 1720$$

$$= 0.262$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

B	D	Y	A	AY	Io
---	---	---	---	----	----



Rib	28.0	172.0	0.0	48.2	0.0	0.186E+04
Web	316.1	28.0	0.0	88.5	0.0	57.8
Totals	...	...	...	136.7	...	0.192E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/137$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 200/2$$

$$= 100.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1300 - \cos(730/2)(36 + 0 + 0) - 0$$

$$= 534.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(1919/137)}$$

$$= 37.475 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 534/37.5$$

$$= 14.249$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 13723/(2 \cdot 1406)$$

$$= 4.880 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 4.88 \cdot 316 \cdot 534$$

$$= 823.733 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 824/191937$$

$$= 429.169 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < C_c$ ($14.2 < 125$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (14.2)^2 / (2 \cdot 125^2)) 2580 /$$

$$(5/3 + 3 \cdot (14.2) / (8 \cdot 125) - (14.2^3) / (8 \cdot 125^3))$$

$$= 1499 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1/I) / Sba$$

$$= 70/1499 + (824 \cdot 100/1919) / 1720$$

$$= 0.296$$

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: N1-Tube Inlet

From: 30

Pressure for Reinforcement Calculations	P	163.158	kgf/cm ²
Temperature for Internal Pressure	Temp	260	°C
Parent Material		SA-387 11	+IT
Parent Allowable Stress at Temperature	Sv	1711.02	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	2110.70	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	1300.00	mm
Shell Finished (Minimum) Thickness	t	68.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		182.82	cm
User Entered Minimum Design Metal Temperature		-28.89	°C

Type of Element Connected to the Parent : Nozzle

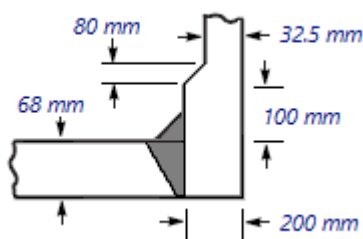
Material		SA-336 F11	+IT
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1521.35	kgf/cm ²
Allowable Stress At Ambient	Sna	1876.18	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		500.0000	mm
Size and Thickness Basis		Nominal	
Nominal Thickness		100	
Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	80.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		200.0000	mm
Corrosion Allowance	can	1.5000	mm
Outside Projection	ho	350.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	72.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: N1-Tube Inlet

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	442.925	mm.
Actual Thickness Used in Calculation	32.537	mm



Nozzle input data check completed without errors.



Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 446 \cdot (\exp(163/(1521 \cdot 1)) - 1) + 1.5 + 0$
 $= 25.2 + 1.5 + 0 = 26.7 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 1521 \cdot 1 \cdot \ln((2 \cdot 31 + 446) / 446) - 0$
 $= 198 - 0 = 198 \text{ kgf/cm}^2$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 9.840 mm
Min. required shell thickness, internal p + c : 66.736 mm

Nozzle neck minimum required thickness : 9.840 mm

Minimum nozzle neck thickness 28.473 mm, is greater than required thickness 9.840 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 620.3,	Allowable	: 1521.4 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 3626.6 kgf/cm ²	Passed
Occasional	: 636.0,	Allowable	: 2023.4 kgf/cm ²	Passed
Shear	: 38.3,	Allowable	: 1064.9 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
 $= \min[S_n/S, 1]$
 $= \min[1521/1711, 1]$
 $= 0.889$

Thickness of Nozzle at Shell [tn]:
 $= \text{hub thickness} - \text{corrosion allowance}$
 $= 200 - 1.5$
 $= 198.500 \text{ mm}$

Thickness of Nozzle at Top [tn2]:
 $= \text{thickness} - \text{corrosion allowance}$
 $= 32.5 - 1.5$
 $= 31.037 \text{ mm}$

Shell Diameter to Thickness ratio [D/t]:
 $= D_i/t$
 $= 1303/66.5$
 $= 19.594 \text{ must be less than } 400.$

Effective Pressure Radius [Reff]:
 $= D_i/2 + \text{corrosion allowance}$
 $= 1300/2 + 1.5$
 $= 651.500 \text{ mm}$

Effective Length of Vessel Wall [LR]:
 $= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$
 $= \min(\sqrt{652 \cdot 66.5}, 2 \cdot 223)$
 $= 208.146 \text{ mm}$

Thickness Limit Candidate [LH1]:
 $= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$



$$= \min[1.5 \cdot 66.5, 0] + \sqrt{(223 \cdot 198)}$$
$$= 210.376 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$
$$= 350$$
$$= 350.000 \text{ mm}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$
$$= 8(66.5 + 0)$$
$$= 532.000 \text{ mm}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles}$$
$$= \min[210, 350, 532] + 66.5$$
$$= 276.876 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$
$$= 66.5 + ((0 \cdot 1) / 208)$$
$$= 66.500 \text{ mm}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}}))$$
$$= \min(12, (2 \cdot 223 + 198) / (\sqrt{(1303 + 66.5) \cdot 66.5}))$$
$$= 2.135$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$
$$= 66.5 \cdot 208 \cdot \max((2.14/5)^{0.85}, 1)$$
$$= 138.417 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= A_{2A} + A_{2C}$$
$$= 331 + 20.1$$
$$= 350.642 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$
$$= 0.5 \cdot 10^2$$
$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43$$
$$= 138 + 0.89(351 + 0) + 0.5 + 0 + 0$$
$$= 450.689 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$
$$= 198 / \ln[1 + 198/223]$$
$$= 311.751 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$
$$= 66.5 / \ln[1 + 66.5/652]$$
$$= 684.212 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$
$$= 1.5 \cdot 1711 \cdot 1$$
$$= 2566.5 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$
$$= 163 \cdot 312 \cdot 277$$
$$= 140832.1 \text{ kgf}$$



Force acting on the Shell [fS]:

$$\begin{aligned} &= P \cdot Rxs (LR + tn) \\ &= 163 \cdot 684 (208 + 198) \\ &= 453958.2 \text{ kgf} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= P \cdot Rxs \cdot Rnc \\ &= 163 \cdot 684 \cdot 223 \\ &= 248903.7 \text{ kgf} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (fS + fY + fN) / P \\ &= (453958 + 248904 + 140832) / 163 \\ &= 5171.0 \text{ cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= Sallow / (2 \cdot Ap / AT - Rxs / teff) \\ &= 2567 / (2 \cdot 5171 / 451 - 684 / 66.5) \\ &= 202.8 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S[t / Rxs] \\ &= 1711 [66.5 / 684] \\ &= 166.3 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(Pmax1, Pmax2) \\ &= \min(203, 166) \\ &= 166.298 \text{ kgf/cm}^2 \end{aligned}$$

Average Primary Membrane Stress [σAvg]:

$$\begin{aligned} &= (fN + fS + fY) / AT \\ &= (140832 + 453958 + 248904) / 451 \\ &= 1872.011 \text{ kgf/cm}^2 \end{aligned}$$

General Primary Membrane Stress [σCirc]:

$$\begin{aligned} &= P \cdot Rxs / teff \\ &= 163 \cdot 684 / 66.5 \\ &= 1678.7 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigmaAvg - \sigmaCirc, \sigmaCirc) \\ &= \max(2 \cdot 1872 - 1679, 1679) \\ &= 2065.3 \text{ kgf/cm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2566.54	kgf/cm ²
Local Primary Membrane Stress	PL	2065.30	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	166.30	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned} &= (Rnc + tn) / Rnc \\ &= (223 + 198) / 223 \\ &= 1.890 \text{ For set-in Nozzles} \end{aligned}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$\begin{aligned} &= \pi / 2 \cdot (Rn + tn) \\ &= \pi / 2 \cdot (223 + 198) \\ &= 662.032 \text{ mm} \end{aligned}$$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 198.500, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi / 4 \cdot P \cdot Rn^2 \cdot ky^2)$$



$$= \min(248904 \cdot 1.89, 1.5 \cdot 1521(351+0), \pi/4 \cdot 163 \cdot 223^2 \cdot 1.89^2)$$

$$= 227623.359 \text{ kgf}$$

Weld Stress Value [τ]:

$$= \text{fwelds}/(L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$$

$$= 227623/(662(0.49 \cdot 7.07 + 0.6 \cdot 66.5 + 0.49 \cdot 0))$$

$$= 792.869 < \text{or} = \text{to } 1711.025 \text{ Weld Stress Passed}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

The Drop for this Nozzle is : 155.1573 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 573.1573 mm

Input Echo, WRC297 Item 1, Description: N1-Tube Inlet :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1300.000	mm
Shell Thickness	68.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-387 11	
Vessel Cold S.I. Allowable	Smc 2110.70	kgf/cm ²
Vessel Hot S.I. Allowable	Smh 1711.02	kgf/cm ²
Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.5000	mm
Nozzle Diameter	442.925	mm
Nozzle Thickness	200.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-336 F11	
Nozzle Cold S.I. Allowable	SNmc 1876.18	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh 1521.35	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSstained loads.

Design Internal Pressure	Dp	163.16	kgf/cm ²
Radial Load	P	-6373.56	kgf
Circumferential Shear	Vc	5098.83	kgf
Longitudinal Shear	Vl	5098.83	kgf
Circumferential Moment	Mc	7.65	kgf-m.
Longitudinal Moment	Ml	7.65	kgf-m.
Torsional Moment	Mt	8.92	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Dv (.587) is > 0.5
 check the Limitations of WRC bulletin 297.

Warning - The ratio, Dn/Tn (4.215) is < 10
 check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - c_a))}$$

$$= 842.925 + 2 \cdot 1.65 \cdot \sqrt{(684.75(68.0 - 1.5))}$$

$$= 1547.117 \text{ mm}$$



WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 1369.500 mm
 Nozzle Outside Diameter (d) = 842.925 mm
 Vessel Thickness used (T) = 66.500 mm
 Nozzle Thickness used (t) = 198.500 mm
 T / t = 0.335
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.793$$

Nr/P = 0.061
 Mr/P = 0.111
 M0/P = 0.035
 N0/P = 0.075
 MrD/Mc = 0.269
 NrDL/Mc = 0.095
 M0d/Mc = 0.081
 N0DL/Mc = 0.084
 MrD/Ml = 0.092
 NrDL/Ml = 0.060
 M0D/Ml = 0.026
 N0DL/Ml = 0.089

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	10	10	10	10
Outplane Bending (P)	29	-29	29	-29
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	1520	1683	1520	1683
Outplane Stress Summary	1561	1664	1561	1664

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	8	8	8	8
Inplane Bending (P)	96	-96	96	-96
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	760	760	760	760
Inplane Stress Summary	865	672	865	672

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	5	5	-5	-5
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	5	5	-5	-5

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Two * Max Shear Stress 1561 1664 1561 1664

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Membrane (P)	10	10	10	10
Outplane Bending (P)	29	-29	29	-29
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	760	760	760	760
Outplane Stress Summary	800	741	801	741

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	8	8	8	8
Inplane Bending (P)	96	-96	96	-96
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	1520	1683	1520	1683
Inplane Stress Summary	1625	1596	1625	1596

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-5	-5	5	5
Torsional Shear (Mt)	0	0	0	0
Shear Stress Summary	-5	-5	5	5

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	1625	1596	1625	1596

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	10	10	10	10
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	152	347	152	347
Hoop Stress Summary	162	358	162	358

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside



Axial Membrane (P)	1	1	1	1
Axial Bending (P)	7	-7	7	-7
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	76	76	76	76

Axial Stress Summary	85	69	85	69
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	1	1	-1	-1

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Outside	Inside	Outside	Inside
Two * Max Shear Stress	162	358	162	358

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside
Hoop Membrane (P)	10	10	10	10
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	152	347	152	347
Hoop Stress Summary	162	358	162	358

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	7	-7	7	-7
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	76	76	76	76
Axial Stress Summary	85	69	85	69

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	0	0	0	0
Shear Stress Summary	-1	-1	1	1

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Outside	Inside	Outside	Inside

Two * Max Shear Stress 162 358 162 358

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1520	1683	1520	1683	1520	1683	1520	1683
Circ. Pl (SUS)	10	10	10	10	8	8	8	8
Circ. Q (SUS)	29	-29	29	-29	96	-96	96	-96
Long. Pm (SUS)	760	760	760	760	760	760	760	760
Long. Pl (SUS)	8	8	8	8	10	10	10	10
Long. Q (SUS)	96	-96	96	-96	29	-29	29	-29
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	5	5	-5	-5	-5	-5	5	5
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	1520	1683	1520	1683	1520	1683	1520	1683
Pm+Pl (SUS)	1530	1693	1530	1693	1528	1691	1528	1691
Pm+Pl+Q (Total)	1559	1664	1559	1664	1624	1595	1624	1595

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1683.00	1711.02	Passed
Pm+Pl (SUS)	1693.03	2566.54	Passed
Pm+Pl+Q (TOTAL)	1664.03	5133.07	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	152	347	152	347	152	347	152	347
Circ. Pl (SUS)	10	10	10	10	10	10	10	10
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	76	76	76	76	76	76	76	76
Long. Pl (SUS)	1	1	1	1	1	1	1	1
Long. Q (SUS)	7	-7	7	-7	7	-7	7	-7
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	152.0	347.0	152.0	347.0	152.0	347.0	152.0	347.0
Pm+Pl (SUS)	162.0	357.0	162.0	357.0	162.0	357.0	162.0	357.0
Pm+Pl+Q (Total)	162.0	357.0	162.0	357.0	162.0	357.0	162.0	357.0

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
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Pm (SUS)	347.00	1521.35	Passed
Pm+Pl (SUS)	357.00	2282.03	Passed
Pm+Pl+Q (TOTAL)	357.00	4564.06	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: N3-Shell Inlet

From: 40

Pressure for Reinforcement Calculations	P	163.165	kgf/cm ²
Temperature for Internal Pressure	Temp	250	°C
Parent Material		SA-387 11	+IT
Parent Allowable Stress at Temperature	Sv	1723.26	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	2110.70	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	1300.00	mm
Shell Finished (Minimum) Thickness	t	68.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		450.82	cm
User Entered Minimum Design Metal Temperature		-28.89	°C

Type of Element Connected to the Parent : Nozzle

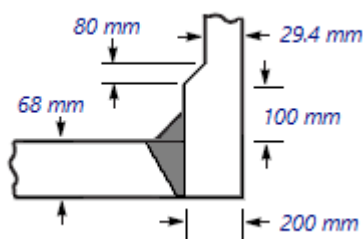
Material		SA-336 F11	+IT
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1529.52	kgf/cm ²
Allowable Stress At Ambient	Sna	1876.18	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		450.0000	mm
Size and Thickness Basis		Nominal	
Nominal Thickness		100	
Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	80.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		200.0000	mm
Corrosion Allowance	can	1.5000	mm
Outside Projection	ho	350.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	72.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: N3-Shell Inlet

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	398.475	mm.
Actual Thickness Used in Calculation	29.362	mm



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 401 \cdot (\exp(163/(1530 \cdot 1)) - 1) + 1.5 + 0$
 $= 22.6 + 1.5 + 0 = 24.1 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 1530 \cdot 1 \cdot \ln((2 \cdot 27.9 + 401) / 401) - 0$
 $= 199 - 0 = 199 \text{ kgf/cm}^2$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 9.840 mm
Min. required shell thickness, internal p + c : 66.252 mm

Nozzle neck minimum required thickness : 9.840 mm

Minimum nozzle neck thickness 25.705 mm, is greater than required thickness 9.840 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 621.0,	Allowable	: 1529.5 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 3636.1 kgf/cm ²	Passed
Occasional	: 638.5,	Allowable	: 2034.3 kgf/cm ²	Passed
Shear	: 42.7,	Allowable	: 1070.7 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
 $= \min[S_n/S, 1]$
 $= \min[1530/1723, 1]$
 $= 0.888$

Thickness of Nozzle at Shell [tn]:
 $= \text{hub thickness} - \text{corrosion allowance}$
 $= 200 - 1.5$
 $= 198.500 \text{ mm}$

Thickness of Nozzle at Top [tn2]:
 $= \text{thickness} - \text{corrosion allowance}$
 $= 29.4 - 1.5$
 $= 27.862 \text{ mm}$

Shell Diameter to Thickness ratio [D/t]:
 $= D_i/t$
 $= 1303/66.5$
 $= 19.594 \text{ must be less than } 400.$

Effective Pressure Radius [Reff]:
 $= D_i/2 + \text{corrosion allowance}$
 $= 1300/2 + 1.5$
 $= 651.500 \text{ mm}$

Effective Length of Vessel Wall [LR]:
 $= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$
 $= \min(\sqrt{652 \cdot 66.5}, 2 \cdot 201)$
 $= 208.146 \text{ mm}$

Thickness Limit Candidate [LH1]:
 $= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$



$$= \min[1.5 \cdot 66.5, 0] + \sqrt{(201 \cdot 198)}$$
$$= 199.616 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$
$$= 350$$
$$= 350.000 \text{ mm}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$
$$= 8(66.5 + 0)$$
$$= 532.000 \text{ mm}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles}$$
$$= \min[200, 350, 532] + 66.5$$
$$= 266.116 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$
$$= 66.5 + ((0 \cdot 1) / 208)$$
$$= 66.500 \text{ mm}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}}))$$
$$= \min(12, (2 \cdot 201 + 198) / (\sqrt{(1303 + 66.5) \cdot 66.5}))$$
$$= 1.988$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$
$$= 66.5 \cdot 208 \cdot \max((1.99/5)^{0.85}, 1)$$
$$= 138.417 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= A_{2A} + A_{2C}$$
$$= 331 + 16.3$$
$$= 346.756 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$
$$= 0.5 \cdot 10^2$$
$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43$$
$$= 138 + 0.89(347 + 0) + 0.5 + 0 + 0$$
$$= 446.688 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$
$$= 198 / \ln[1 + 198/201]$$
$$= 288.703 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$
$$= 66.5 / \ln[1 + 66.5/652]$$
$$= 684.212 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$
$$= 1.5 \cdot 1723 \cdot 1$$
$$= 2584.9 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$
$$= 163 \cdot 289 \cdot 266$$
$$= 125356.8 \text{ kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot Rxs (LR + tn)$$

$$= 163 \cdot 684 (208 + 198)$$

$$= 453976.3 \text{ kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot Rxs \cdot Rnc$$

$$= 163 \cdot 684 \cdot 201$$

$$= 224101.8 \text{ kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN) / P$$

$$= (453976 + 224102 + 125357) / 163$$

$$= 4924.1 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot Ap / AT - Rxs / teff)$$

$$= 2585 / (2 \cdot 4924 / 447 - 684 / 66.5)$$

$$= 219.8 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t / Rxs]$$

$$= 1723 [66.5 / 684]$$

$$= 167.5 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(Pmax1, Pmax2)$$

$$= \min(220, 167)$$

$$= 167.487 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σAvg]:

$$= (fN + fS + fY) / AT$$

$$= (125357 + 453976 + 224102) / 447$$

$$= 1798.650 \text{ kgf/cm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot Rxs / teff$$

$$= 163 \cdot 684 / 66.5$$

$$= 1678.8 \text{ kgf/cm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ)$$

$$= \max(2 \cdot 1799 - 1679, 1679)$$

$$= 1918.5 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2584.89	kgf/cm ²
Local Primary Membrane Stress	PL	1918.51	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	167.49	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (Rnc + tn) / Rnc$$

$$= (201 + 198) / 201$$

$$= 1.989 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi / 2 \cdot (Rn + tn)$$

$$= \pi / 2 \cdot (201 + 198)$$

$$= 627.121 \text{ mm}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 198.500, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi / 4 \cdot P \cdot Rn^2 \cdot ky^2)$$



$$= \min(224102 \cdot 1.99, 1.5 \cdot 1530(347+0), \pi/4 \cdot 163 \cdot 201^2 \cdot 1.99^2)$$

$$= 204257.969 \text{ kgf}$$

Weld Stress Value [τ]:
 $= \text{fwelds} / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$
 $= 204258 / (627(0.49 \cdot 7.07 + 0.6 \cdot 66.5 + 0.49 \cdot 0))$
 $= 751.088 < \text{or} = \text{to } 1723.261 \text{ Weld Stress Passed}$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

The Drop for this Nozzle is : 137.0582 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 555.0582 mm

Input Echo, WRC297 Item 1, Description: N3-Shell Inlet:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1300.000	mm
Shell Thickness	68.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-387 11	
Vessel Cold S.I. Allowable	Smc 2110.70	kgf/cm ²
Vessel Hot S.I. Allowable	Smh 1723.26	kgf/cm ²
Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.5000	mm
Nozzle Diameter	398.475	mm
Nozzle Thickness	200.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-336 F11	
Nozzle Cold S.I. Allowable	SNmc 1876.18	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh 1529.52	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSstained loads.

Design Internal Pressure	Dp	163.16	kgf/cm ²
Radial Load	P	-5736.17	kgf
Circumferential Shear	Vc	4588.95	kgf
Longitudinal Shear	Vl	4588.95	kgf
Circumferential Moment	Mc	6.19	kgf-m.
Longitudinal Moment	Ml	6.19	kgf-m.
Torsional Moment	Mt	7.22	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Dv (.556) is > 0.5
 check the Limitations of WRC bulletin 297.

Warning - The ratio, Dn/Tn (3.992) is < 10
 check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - c_a))}$
 $= 798.475 + 2 \cdot 1.65 \cdot \sqrt{(684.75(68.0 - 1.5))}$
 $= 1502.667 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 1369.500 mm
 Nozzle Outside Diameter (d) = 798.475 mm
 Vessel Thickness used (T) = 66.500 mm
 Nozzle Thickness used (t) = 198.500 mm
 T / t = 0.335
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.646$$

Nr/P = 0.062
 Mr/P = 0.116
 M0/P = 0.036
 N0/P = 0.077
 MrD/Mc = 0.273
 NrDL/Mc = 0.096
 M0d/Mc = 0.082
 N0DL/Mc = 0.083
 MrD/Ml = 0.095
 NrDL/Ml = 0.063
 M0D/Ml = 0.027
 N0DL/Ml = 0.094

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	9	9	9	9
Outplane Bending (P)	27	-27	27	-27
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	1520	1684	1520	1684
Outplane Stress Summary	1558	1666	1558	1666

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	8	8	8	8
Inplane Bending (P)	90	-90	90	-90
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	760	760	760	760
Inplane Stress Summary	858	678	858	678

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	5	5	-5	-5
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	5	5	-5	-5

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Two * Max Shear Stress 1558 1666 1558 1666

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Membrane (P)	9	9	9	9
Outplane Bending (P)	27	-27	27	-27
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	760	760	760	760
Outplane Stress Summary	797	742	798	742

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	8	8	8	8
Inplane Bending (P)	90	-90	90	-90
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	1520	1684	1520	1684
Inplane Stress Summary	1618	1602	1619	1601

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-5	-5	5	5
Torsional Shear (Mt)	0	0	0	0
Shear Stress Summary	-5	-5	5	5

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	1618	1602	1619	1601

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	9	9	9	9
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	132	328	132	328
Hoop Stress Summary	142	338	142	338

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside



Axial Membrane (P)	1	1	1	1
Axial Bending (P)	7	-7	7	-7
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	66	66	66	66

Axial Stress Summary	75	60	75	60
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	1	1	-1	-1

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Outside	Inside	Outside	Inside
Two * Max Shear Stress	142	338	142	338

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside
Hoop Membrane (P)	9	9	9	9
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	132	328	132	328
Hoop Stress Summary	142	338	142	338

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	7	-7	7	-7
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	66	66	66	66
Axial Stress Summary	75	60	75	60

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	0	0	0	0
Shear Stress Summary	-1	-1	1	1

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Outside	Inside	Outside	Inside

Two * Max Shear Stress 142 338 142 338

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1520	1684	1520	1684	1520	1684	1520	1684
Circ. Pl (SUS)	9	9	9	9	8	8	8	8
Circ. Q (SUS)	27	-27	27	-27	90	-90	90	-90
Long. Pm (SUS)	760	760	760	760	760	760	760	760
Long. Pl (SUS)	8	8	8	8	9	9	9	9
Long. Q (SUS)	90	-90	90	-90	27	-27	27	-27
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	5	5	-5	-5	-5	-5	5	5
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	1520	1684	1520	1684	1520	1684	1520	1684
Pm+Pl (SUS)	1529	1693	1529	1693	1528	1692	1528	1692
Pm+Pl+Q (Total)	1556	1666	1556	1666	1618	1602	1618	1602

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1684.00	1723.26	Passed
Pm+Pl (SUS)	1693.03	2584.89	Passed
Pm+Pl+Q (TOTAL)	1666.03	5169.78	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	132	328	132	328	132	328	132	328
Circ. Pl (SUS)	9	9	9	9	9	9	9	9
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	66	66	66	66	66	66	66	66
Long. Pl (SUS)	1	1	1	1	1	1	1	1
Long. Q (SUS)	7	-7	7	-7	7	-7	7	-7
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	132.0	328.0	132.0	328.0	132.0	328.0	132.0	328.0
Pm+Pl (SUS)	141.0	337.0	141.0	337.0	141.0	337.0	141.0	337.0
Pm+Pl+Q (Total)	141.0	337.0	141.0	337.0	141.0	337.0	141.0	337.0

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
---------------------	-----------	----------------	--------



Pm (SUS)	328.00	1529.52	Passed
Pm+Pl (SUS)	337.00	2294.28	Passed
Pm+Pl+Q (TOTAL)	337.00	4588.56	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: N4-Shell outlet

From: 40

Pressure for Reinforcement Calculations	P	163.158	kgf/cm ²
Temperature for Internal Pressure	Temp	250	°C
Parent Material		SA-387 11	+IT
Parent Allowable Stress at Temperature	Sv	1723.26	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	2110.70	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	1300.00	mm
Shell Finished (Minimum) Thickness	t	68.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1600.82	cm
User Entered Minimum Design Metal Temperature		-28.89	°C

Type of Element Connected to the Parent : Nozzle

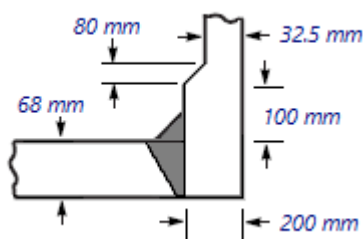
Material		SA-336 F11	+IT
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1529.52	kgf/cm ²
Allowable Stress At Ambient	Sna	1876.18	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		500.0000	mm
Size and Thickness Basis		Nominal	
Nominal Thickness		100	
Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	80.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		200.0000	mm
Corrosion Allowance	can	1.5000	mm
Outside Projection	ho	350.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	72.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: N4-Shell outlet

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	442.925	mm.
Actual Thickness Used in Calculation	32.537	mm



Nozzle input data check completed without errors.



Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 446 \cdot (\exp(163/(1530 \cdot 1)) - 1) + 1.5 + 0$
 $= 25.1 + 1.5 + 0 = 26.6 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 1530 \cdot 1 \cdot \ln((2 \cdot 31 + 446) / 446) - 0$
 $= 199 - 0 = 199 \text{ kgf/cm}^2$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 9.840 mm
Min. required shell thickness, internal p + c : 66.249 mm

Nozzle neck minimum required thickness : 9.840 mm

Minimum nozzle neck thickness 28.473 mm, is greater than required thickness 9.840 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 620.3,	Allowable	: 1529.5 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 3636.8 kgf/cm ²	Passed
Occasional	: 636.0,	Allowable	: 2034.3 kgf/cm ²	Passed
Shear	: 38.3,	Allowable	: 1070.7 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
 $= \min[S_n/S, 1]$
 $= \min[1530/1723, 1]$
 $= 0.888$

Thickness of Nozzle at Shell [tn]:
 $= \text{hub thickness} - \text{corrosion allowance}$
 $= 200 - 1.5$
 $= 198.500 \text{ mm}$

Thickness of Nozzle at Top [tn2]:
 $= \text{thickness} - \text{corrosion allowance}$
 $= 32.5 - 1.5$
 $= 31.037 \text{ mm}$

Shell Diameter to Thickness ratio [D/t]:
 $= D_i/t$
 $= 1303/66.5$
 $= 19.594 \text{ must be less than } 400.$

Effective Pressure Radius [Reff]:
 $= D_i/2 + \text{corrosion allowance}$
 $= 1300/2 + 1.5$
 $= 651.500 \text{ mm}$

Effective Length of Vessel Wall [LR]:
 $= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$
 $= \min(\sqrt{652 \cdot 66.5}, 2 \cdot 223)$
 $= 208.146 \text{ mm}$

Thickness Limit Candidate [LH1]:
 $= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$



$$= \min[1.5 \cdot 66.5, 0] + \sqrt{(223 \cdot 198)} \\ = 210.376 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1} \\ = 350 \\ = 350.000 \text{ mm}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e) \\ = 8(66.5 + 0) \\ = 532.000 \text{ mm}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles} \\ = \min[210, 350, 532] + 66.5 \\ = 276.876 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR) \\ = 66.5 + ((0 \cdot 1) / 208) \\ = 66.500 \text{ mm}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}})) \\ = \min(12, (2 \cdot 223 + 198) / (\sqrt{(1303 + 66.5) \cdot 66.5})) \\ = 2.135$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\ = 66.5 \cdot 208 \cdot \max((2.14/5)^{0.85}, 1) \\ = 138.417 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= A_{2A} + A_{2C} \\ = 331 + 20.1 \\ = 350.642 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2 \\ = 0.5 \cdot 10^2 \\ = 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 \\ = 138 + 0.89(351 + 0) + 0.5 + 0 + 0 \\ = 450.137 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n] \\ = 198 / \ln[1 + 198/223] \\ = 311.751 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ = 66.5 / \ln[1 + 66.5/652] \\ = 684.212 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E \\ = 1.5 \cdot 1723 \cdot 1 \\ = 2584.9 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H \\ = 163 \cdot 312 \cdot 277 \\ = 140832.1 \text{ kgf}$$



Force acting on the Shell [fS]:

$$\begin{aligned} &= P \cdot Rxs (LR + tn) \\ &= 163 \cdot 684 (208 + 198) \\ &= 453958.2 \text{ kgf} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= P \cdot Rxs \cdot Rnc \\ &= 163 \cdot 684 \cdot 223 \\ &= 248903.7 \text{ kgf} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (fS + fY + fN) / P \\ &= (453958 + 248904 + 140832) / 163 \\ &= 5171.0 \text{ cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= Sallow / (2 \cdot Ap / AT - Rxs / teff) \\ &= 2585 / (2 \cdot 5171 / 450 - 684 / 66.5) \\ &= 203.8 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S[t / Rxs] \\ &= 1723 [66.5 / 684] \\ &= 167.5 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(Pmax1, Pmax2) \\ &= \min(204, 167) \\ &= 167.487 \text{ kgf/cm}^2 \end{aligned}$$

Average Primary Membrane Stress [σAvg]:

$$\begin{aligned} &= (fN + fS + fY) / AT \\ &= (140832 + 453958 + 248904) / 450 \\ &= 1874.306 \text{ kgf/cm}^2 \end{aligned}$$

General Primary Membrane Stress [σCirc]:

$$\begin{aligned} &= P \cdot Rxs / teff \\ &= 163 \cdot 684 / 66.5 \\ &= 1678.7 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ) \\ &= \max(2 \cdot 1874 - 1679, 1679) \\ &= 2069.9 \text{ kgf/cm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2584.89	kgf/cm ²
Local Primary Membrane Stress	PL	2069.89	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	167.49	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned} &= (Rnc + tn) / Rnc \\ &= (223 + 198) / 223 \\ &= 1.890 \text{ For set-in Nozzles} \end{aligned}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$\begin{aligned} &= \pi / 2 \cdot (Rn + tn) \\ &= \pi / 2 \cdot (223 + 198) \\ &= 662.032 \text{ mm} \end{aligned}$$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 198.500, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi / 4 \cdot P \cdot Rn^2 \cdot ky^2)$$



$$= \min(248904 \cdot 1.89, 1.5 \cdot 1530(351+0), \pi/4 \cdot 163 \cdot 223^2 \cdot 1.89^2)$$

$$= 227623.359 \text{ kgf}$$

Weld Stress Value [τ]:

$$= \text{fwelds}/(L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$$

$$= 227623/(662(0.49 \cdot 7.07 + 0.6 \cdot 66.5 + 0.49 \cdot 0))$$

$$= 792.869 < \text{or} = \text{to } 1723.261 \text{ Weld Stress Passed}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

The Drop for this Nozzle is : 155.1573 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 573.1573 mm

Input Echo, WRC297 Item 1, Description: N4-Shell outle:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1300.000	mm
Shell Thickness	68.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-387 11	
Vessel Cold S.I. Allowable	Smc 2110.70	kgf/cm ²
Vessel Hot S.I. Allowable	Smh 1723.26	kgf/cm ²
Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.5000	mm
Nozzle Diameter	442.925	mm
Nozzle Thickness	200.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-336 F11	
Nozzle Cold S.I. Allowable	SNmc 1876.18	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh 1529.52	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	163.16	kgf/cm ²
Radial Load	P	-6373.56	kgf
Circumferential Shear	Vc	5098.83	kgf
Longitudinal Shear	Vl	5098.83	kgf
Circumferential Moment	Mc	7.65	kgf-m.
Longitudinal Moment	Ml	7.65	kgf-m.
Torsional Moment	Mt	8.92	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Dv (.587) is > 0.5
check the Limitations of WRC bulletin 297.

Warning - The ratio, Dn/Tn (4.215) is < 10
check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - c_a))}$$

$$= 842.925 + 2 \cdot 1.65 \cdot \sqrt{(684.75(68.0 - 1.5))}$$

$$= 1547.117 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 1369.500 mm
 Nozzle Outside Diameter (d) = 842.925 mm
 Vessel Thickness used (T) = 66.500 mm
 Nozzle Thickness used (t) = 198.500 mm
 T / t = 0.335
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.793$$

Nr/P = 0.061
 Mr/P = 0.111
 M0/P = 0.035
 N0/P = 0.075
 MrD/Mc = 0.269
 NrDL/Mc = 0.095
 M0d/Mc = 0.081
 N0DL/Mc = 0.084
 MrD/Ml = 0.092
 NrDL/Ml = 0.060
 M0D/Ml = 0.026
 N0DL/Ml = 0.089

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	10	10	10	10
Outplane Bending (P)	29	-29	29	-29
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	1520	1683	1520	1683
Outplane Stress Summary	1561	1664	1561	1664

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	8	8	8	8
Inplane Bending (P)	96	-96	96	-96
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	760	760	760	760
Inplane Stress Summary	865	672	865	672

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	5	5	-5	-5
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	5	5	-5	-5

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Two * Max Shear Stress 1561 1664 1561 1664

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Membrane (P)	10	10	10	10
Outplane Bending (P)	29	-29	29	-29
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	760	760	760	760
Outplane Stress Summary	800	741	801	741

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	8	8	8	8
Inplane Bending (P)	96	-96	96	-96
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	1520	1683	1520	1683
Inplane Stress Summary	1625	1596	1625	1596

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-5	-5	5	5
Torsional Shear (Mt)	0	0	0	0
Shear Stress Summary	-5	-5	5	5

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	1625	1596	1625	1596

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	10	10	10	10
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	152	347	152	347
Hoop Stress Summary	162	358	162	358

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside



Axial Membrane (P)	1	1	1	1
Axial Bending (P)	7	-7	7	-7
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	76	76	76	76

Axial Stress Summary	85	69	85	69
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	1	1	-1	-1

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Outside	Inside	Outside	Inside
Two * Max Shear Stress	162	358	162	358

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside
Hoop Membrane (P)	10	10	10	10
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	152	347	152	347
Hoop Stress Summary	162	358	162	358

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	7	-7	7	-7
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	76	76	76	76
Axial Stress Summary	85	69	85	69

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	0	0	0	0
Shear Stress Summary	-1	-1	1	1

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Outside	Inside	Outside	Inside

Two * Max Shear Stress 162 358 162 358

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1520	1683	1520	1683	1520	1683	1520	1683
Circ. Pl (SUS)	10	10	10	10	8	8	8	8
Circ. Q (SUS)	29	-29	29	-29	96	-96	96	-96
Long. Pm (SUS)	760	760	760	760	760	760	760	760
Long. Pl (SUS)	8	8	8	8	10	10	10	10
Long. Q (SUS)	96	-96	96	-96	29	-29	29	-29
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	5	5	-5	-5	-5	-5	5	5
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	1520	1683	1520	1683	1520	1683	1520	1683
Pm+Pl (SUS)	1530	1693	1530	1693	1528	1691	1528	1691
Pm+Pl+Q (Total)	1559	1664	1559	1664	1624	1595	1624	1595

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1683.00	1723.26	Passed
Pm+Pl (SUS)	1693.03	2584.89	Passed
Pm+Pl+Q (TOTAL)	1664.03	5169.78	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	152	347	152	347	152	347	152	347
Circ. Pl (SUS)	10	10	10	10	10	10	10	10
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	76	76	76	76	76	76	76	76
Long. Pl (SUS)	1	1	1	1	1	1	1	1
Long. Q (SUS)	7	-7	7	-7	7	-7	7	-7
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	152.0	347.0	152.0	347.0	152.0	347.0	152.0	347.0
Pm+Pl (SUS)	162.0	357.0	162.0	357.0	162.0	357.0	162.0	357.0
Pm+Pl+Q (Total)	162.0	357.0	162.0	357.0	162.0	357.0	162.0	357.0

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
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Pm (SUS)	347.00	1529.52	Passed
Pm+Pl (SUS)	357.00	2294.28	Passed
Pm+Pl+Q (TOTAL)	357.00	4588.56	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: N2-Tube Outlet

From: 70

Pressure for Reinforcement Calculations	P	163.158	kgf/cm ²
Temperature for Internal Pressure	Temp	250	°C
Parent Material		SA-387 11	+IT
Parent Allowable Stress at Temperature	Sv	1723.26	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	2110.70	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	1300.00	mm
Shell Finished (Minimum) Thickness	t	68.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1910.82	cm
User Entered Minimum Design Metal Temperature		-28.89	°C

Type of Element Connected to the Parent : Nozzle

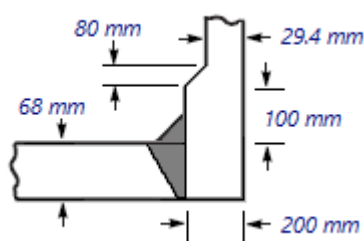
Material		SA-336 F11	+IT
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1529.52	kgf/cm ²
Allowable Stress At Ambient	Sna	1876.18	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		450.0000	mm
Size and Thickness Basis		Nominal	
Nominal Thickness		100	
Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	80.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		200.0000	mm
Corrosion Allowance	can	1.5000	mm
Outside Projection	ho	350.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	72.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: N2-Tube Outlet

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	398.475	mm.
Actual Thickness Used in Calculation	29.362	mm



Nozzle input data check completed without errors.



Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$\begin{aligned} &= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o \\ &= 0.5 \cdot 401 \cdot (\exp(163/(1530 \cdot 1)) - 1) + 1.5 + 0 \\ &= 22.6 + 1.5 + 0 = 24.1 \text{ mm Equation 4.3.1} \end{aligned}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$\begin{aligned} &= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq} \\ &= 1530 \cdot 1 \cdot \ln((2 \cdot 27.9 + 401) / 401) - 0 \\ &= 199 - 0 = 199 \text{ kgf/cm}^2 \end{aligned}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 9.840 mm
Min. required shell thickness, internal p + c : 66.249 mm

Nozzle neck minimum required thickness : 9.840 mm

Minimum nozzle neck thickness 25.705 mm, is greater than required thickness 9.840 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 621.0,	Allowable	: 1529.5 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 3636.1 kgf/cm ²	Passed
Occasional	: 638.4,	Allowable	: 2034.3 kgf/cm ²	Passed
Shear	: 42.7,	Allowable	: 1070.7 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$\begin{aligned} &= \min[S_n/S, 1] \\ &= \min[1530/1723, 1] \\ &= 0.888 \end{aligned}$$

Thickness of Nozzle at Shell [tn]:

$$\begin{aligned} &= \text{hub thickness} - \text{corrosion allowance} \\ &= 200 - 1.5 \\ &= 198.500 \text{ mm} \end{aligned}$$

Thickness of Nozzle at Top [tn2]:

$$\begin{aligned} &= \text{thickness} - \text{corrosion allowance} \\ &= 29.4 - 1.5 \\ &= 27.862 \text{ mm} \end{aligned}$$

Shell Diameter to Thickness ratio [D/t]:

$$\begin{aligned} &= D_i/t \\ &= 1303/66.5 \\ &= 19.594 \text{ must be less than 400.} \end{aligned}$$

Effective Pressure Radius [Reff]:

$$\begin{aligned} &= D_i/2 + \text{corrosion allowance} \\ &= 1300/2 + 1.5 \\ &= 651.500 \text{ mm} \end{aligned}$$

Effective Length of Vessel Wall [LR]:

$$\begin{aligned} &= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n) \\ &= \min(\sqrt{652 \cdot 66.5}, 2 \cdot 201) \\ &= 208.146 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$



$$= \min[1.5 \cdot 66.5, 0] + \sqrt{(201 \cdot 198)}$$
$$= 199.616 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$
$$= 350$$
$$= 350.000 \text{ mm}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$
$$= 8(66.5 + 0)$$
$$= 532.000 \text{ mm}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles}$$
$$= \min[200, 350, 532] + 66.5$$
$$= 266.116 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$
$$= 66.5 + ((0 \cdot 1) / 208)$$
$$= 66.500 \text{ mm}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}}))$$
$$= \min(12, (2 \cdot 201 + 198) / (\sqrt{(1303 + 66.5) \cdot 66.5}))$$
$$= 1.988$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$
$$= 66.5 \cdot 208 \cdot \max((1.99/5)^{0.85}, 1)$$
$$= 138.417 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= A_{2A} + A_{2C}$$
$$= 331 + 16.3$$
$$= 346.756 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$
$$= 0.5 \cdot 10^2$$
$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43$$
$$= 138 + 0.89(347 + 0) + 0.5 + 0 + 0$$
$$= 446.688 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$
$$= 198 / \ln[1 + 198/201]$$
$$= 288.703 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$
$$= 66.5 / \ln[1 + 66.5/652]$$
$$= 684.212 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$
$$= 1.5 \cdot 1723 \cdot 1$$
$$= 2584.9 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$
$$= 163 \cdot 289 \cdot 266$$
$$= 125351.8 \text{ kgf}$$



Force acting on the Shell [fS]:

$$\begin{aligned} &= P \cdot Rxs (LR + tn) \\ &= 163 \cdot 684 (208 + 198) \\ &= 453958.2 \text{ kgf} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= P \cdot Rxs \cdot Rnc \\ &= 163 \cdot 684 \cdot 201 \\ &= 224092.9 \text{ kgf} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (fS + fY + fN) / P \\ &= (453958 + 224093 + 125352) / 163 \\ &= 4924.1 \text{ cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= Sallow / (2 \cdot Ap / AT - Rxs / teff) \\ &= 2585 / (2 \cdot 4924 / 447 - 684 / 66.5) \\ &= 219.8 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S[t / Rxs] \\ &= 1723 [66.5 / 684] \\ &= 167.5 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(Pmax1, Pmax2) \\ &= \min(220, 167) \\ &= 167.487 \text{ kgf/cm}^2 \end{aligned}$$

Average Primary Membrane Stress [σAvg]:

$$\begin{aligned} &= (fN + fS + fY) / AT \\ &= (125352 + 453958 + 224093) / 447 \\ &= 1798.578 \text{ kgf/cm}^2 \end{aligned}$$

General Primary Membrane Stress [σCirc]:

$$\begin{aligned} &= P \cdot Rxs / teff \\ &= 163 \cdot 684 / 66.5 \\ &= 1678.7 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ) \\ &= \max(2 \cdot 1799 - 1679, 1679) \\ &= 1918.4 \text{ kgf/cm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2584.89	kgf/cm ²
Local Primary Membrane Stress	PL	1918.44	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	167.49	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned} &= (Rnc + tn) / Rnc \\ &= (201 + 198) / 201 \\ &= 1.989 \text{ For set-in Nozzles} \end{aligned}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$\begin{aligned} &= \pi / 2 \cdot (Rn + tn) \\ &= \pi / 2 \cdot (201 + 198) \\ &= 627.121 \text{ mm} \end{aligned}$$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 198.500, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi / 4 \cdot P \cdot Rn^2 \cdot ky^2)$$

$$= \min(224093 \cdot 1.99, 1.5 \cdot 1530(347+0), \pi/4 \cdot 163 \cdot 201^2 \cdot 1.99^2)$$

$$= 204249.812 \text{ kgf}$$

Weld Stress Value [τ]:
 $= \text{fwelds}/(L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$
 $= 204250/(627(0.49 \cdot 7.07 + 0.6 \cdot 66.5 + 0.49 \cdot 0))$
 $= 751.058 < \text{or} = \text{to } 1723.261 \text{ Weld Stress Passed}$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

The Drop for this Nozzle is : 137.0582 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 555.0582 mm

Input Echo, WRC297 Item 1, Description: N2-Tube Outlet:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1300.000	mm
Shell Thickness	68.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-387 11	
Vessel Cold S.I. Allowable	Smc 2110.70	kgf/cm ²
Vessel Hot S.I. Allowable	Smh 1723.26	kgf/cm ²
Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.5000	mm
Nozzle Diameter	398.475	mm
Nozzle Thickness	200.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-336 F11	
Nozzle Cold S.I. Allowable	SNmc 1876.18	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh 1529.52	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:
 External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSstained loads.

Design Internal Pressure	Dp	163.16	kgf/cm ²
Radial Load	P	-5736.17	kgf
Circumferential Shear	Vc	4588.95	kgf
Longitudinal Shear	Vl	4588.95	kgf
Circumferential Moment	Mc	6.19	kgf-m.
Longitudinal Moment	Ml	6.19	kgf-m.
Torsional Moment	Mt	7.22	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Dv (.556) is > 0.5
 check the Limitations of WRC bulletin 297.
 Warning - The ratio, Dn/Tn (3.992) is < 10
 check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - c_a))}$
 $= 798.475 + 2 \cdot 1.65 \cdot \sqrt{(684.75(68.0 - 1.5))}$
 $= 1502.667 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 1369.500 mm
 Nozzle Outside Diameter (d) = 798.475 mm
 Vessel Thickness used (T) = 66.500 mm
 Nozzle Thickness used (t) = 198.500 mm
 T / t = 0.335
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.646$$

Nr/P = 0.062
 Mr/P = 0.116
 M0/P = 0.036
 N0/P = 0.077
 MrD/Mc = 0.273
 NrDL/Mc = 0.096
 M0d/Mc = 0.082
 N0DL/Mc = 0.083
 MrD/Ml = 0.095
 NrDL/Ml = 0.063
 M0D/Ml = 0.027
 N0DL/Ml = 0.094

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	9	9	9	9
Outplane Bending (P)	27	-27	27	-27
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	1520	1683	1520	1683
Outplane Stress Summary	1558	1666	1558	1666

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	8	8	8	8
Inplane Bending (P)	90	-90	90	-90
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	760	760	760	760
Inplane Stress Summary	858	678	858	677

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	5	5	-5	-5
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	5	5	-5	-5

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Two * Max Shear Stress 1558 1666 1558 1666

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Membrane (P)	9	9	9	9
Outplane Bending (P)	27	-27	27	-27
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	760	760	760	760
Outplane Stress Summary	797	742	798	742

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	8	8	8	8
Inplane Bending (P)	90	-90	90	-90
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	1520	1683	1520	1683
Inplane Stress Summary	1618	1601	1619	1601

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-5	-5	5	5
Torsional Shear (Mt)	0	0	0	0
Shear Stress Summary	-5	-5	5	5

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	1618	1601	1619	1601

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	9	9	9	9
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	132	328	132	328
Hoop Stress Summary	142	338	142	338

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside



Axial Membrane (P)	1	1	1	1
Axial Bending (P)	7	-7	7	-7
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	66	66	66	66

Axial Stress Summary	75	60	75	60
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	1	1	-1	-1

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Outside	Inside	Outside	Inside
Two * Max Shear Stress	142	338	142	338

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside
Hoop Membrane (P)	9	9	9	9
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	132	328	132	328
Hoop Stress Summary	142	338	142	338

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	7	-7	7	-7
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	66	66	66	66
Axial Stress Summary	75	60	75	60

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	0	0	0	0
Shear Stress Summary	-1	-1	1	1

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Outside	Inside	Outside	Inside

Two * Max Shear Stress 142 338 142 338

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1520	1683	1520	1683	1520	1683	1520	1683
Circ. Pl (SUS)	9	9	9	9	8	8	8	8
Circ. Q (SUS)	27	-27	27	-27	90	-90	90	-90
Long. Pm (SUS)	760	760	760	760	760	760	760	760
Long. Pl (SUS)	8	8	8	8	9	9	9	9
Long. Q (SUS)	90	-90	90	-90	27	-27	27	-27
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	5	5	-5	-5	-5	-5	5	5
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	1520	1683	1520	1683	1520	1683	1520	1683
Pm+Pl (SUS)	1529	1692	1529	1692	1528	1691	1528	1691
Pm+Pl+Q (Total)	1556	1665	1556	1665	1618	1601	1618	1601

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1683.00	1723.26	Passed
Pm+Pl (SUS)	1692.03	2584.89	Passed
Pm+Pl+Q (TOTAL)	1665.03	5169.78	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	132	328	132	328	132	328	132	328
Circ. Pl (SUS)	9	9	9	9	9	9	9	9
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	66	66	66	66	66	66	66	66
Long. Pl (SUS)	1	1	1	1	1	1	1	1
Long. Q (SUS)	7	-7	7	-7	7	-7	7	-7
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	132.0	328.0	132.0	328.0	132.0	328.0	132.0	328.0
Pm+Pl (SUS)	141.0	337.0	141.0	337.0	141.0	337.0	141.0	337.0
Pm+Pl+Q (Total)	141.0	337.0	141.0	337.0	141.0	337.0	141.0	337.0

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
---------------------	-----------	----------------	--------



Pm (SUS)	328.00	1529.52	Passed
Pm+Pl (SUS)	337.00	2294.28	Passed
Pm+Pl+Q (TOTAL)	337.00	4588.56	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input Echo, Tubesheet Number 1, Description:

Shell Data:

Main Shell Description: Main Shell

Shell Maximum Design Pressure	Psd,max	163.16	kgf/cm ²
Shell Maximum Operating Pressure	Psox,max	163.16	kgf/cm ²
Shell Minimum Operating Pressure	Psox,min	0.00	kgf/cm ²
Shell Thickness	ts	68.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	casext	0.0000	mm
Inside Diameter of Shell	Ds	1300.000	mm
Shell Temperature for Internal Pressure	Ts	260.00	°C
Shell Material		SA-387 11	
Shell Material UNS Number		K11789	
Shell Allowable Stress at Temperature	Ss	1711.02	kgf/cm ²
Shell Allowable Stress at Ambient		2110.70	kgf/cm ²

Channel Data:

Channel Description: LHS Ch Shell

Channel Type:		Cylinder	
Channel Maximum Design Pressure	Ptd,max	163.16	kgf/cm ²
Channel Maximum Operating Pressure	Ptox,max	163.16	kgf/cm ²
Channel Minimum Operating Pressure	Ptox,min	0.00	kgf/cm ²
Channel Thickness	tc	68.0000	mm
Channel Internal Corrosion Allowance	cac	1.5000	mm
Channel External Corrosion Allowance	cacext	0.0000	mm
Inside Diameter of Channel	Dc	1300.000	mm
Channel Design Temperature	TEMPC	260.00	°C
Channel Flange Matl		SA-387 11	
Channel Material UNS Number		K11789	
Channel Allowable Stress at Temperature	Sc	1711.02	kgf/cm ²
Channel Allowable Stress at Ambient		2110.70	kgf/cm ²

Tube Data:

Number of Tube Holes	Nt	2615	
Tube Wall Thickness	et	2.0000	mm
Tube Outside Diameter	D	14.0000	mm
Total Straight Tube Length	Lt	15200.00	mm
Straight Tube Length (bet. inner tubsht faces) L		14640.00	mm
Design Temperature of the Tubes		260.00	°C
Tube Material		SA-213 T11	
Tube Material UNS Number		K11597	
Is this a Welded Tube		No	
Tube Material Specification used		Smls. tube	
Tube Allowable Stress at Temperature		1144.08	kgf/cm ²
Tube Allowable Stress At Ambient		1407.13	kgf/cm ²
Tube Yield Stress At design Temperature	Syt	1711.02	kgf/cm ²
Tube Pitch (Center to Center Spacing)	P	19.0000	mm
Tube Layout Pattern		Triangular	
Fillet Weld Leg	af	2.0000	mm
Groove Weld Leg	ag	2.0000	mm
Tube-Tubesheet Joint Weld Type		Full Strength	
Method for Tube-Tubesheet Jt. Allowable		4.21.2	
Is the Tube Inset		No	
Radius to Outermost Tube Hole Center	ro	577.862	mm
Largest Center-to-Center Tube Distance	Ul	0.0000	mm
Length of Expanded Portion of Tube	ltx	0.0000	mm
Tube-side pass partition groove depth	hg	6.0000	mm

Tubesheet Data:

Tubesheet TYPE: Stationary, Integral both Sides, Conf. a
Exchanger TYPE: With an Immersed Floating Head, Conf. a

Tubesheet Design Metal Temperature	T	260.00	°C
Tubesheet Material		SA-336 F11	



See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Tubesheet Material UNS Number		K11572
Tubesheet Allowable Stress at Temperature	S	1521.35 kgf/cm ²
Tubesheet Allowable Stress at Ambient	Tt	1876.18 kgf/cm ²
Thickness of Tubesheet	h	280.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	1.5000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	1.5000 mm
Tubesheet Outside Diameter	A	1444.000 mm

Area of the Untubed Lanes	AL	0.0 cm ²
---------------------------	----	---------------------

Additional Data for Fixed/Floating Tubesheet Exchangers:

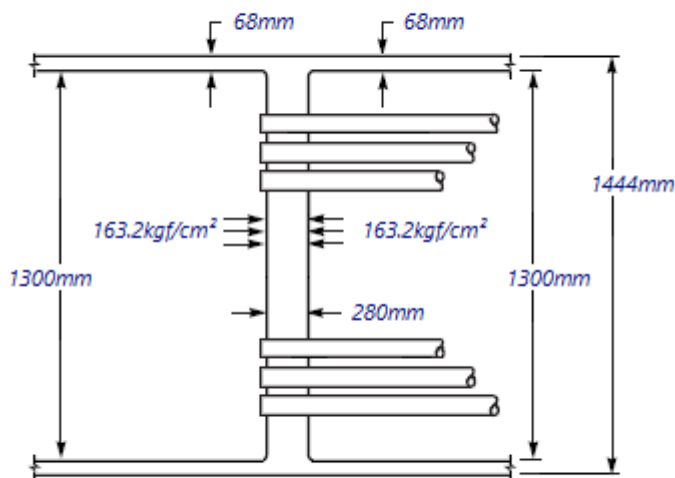
Unsupported Tube Span under consideration	l	0.000 mm
Tube End condition corresponding to Span (l)	k	0.00

Ignore Radial Thermal Exp. effects (4.18.8/4.18.9)	Yes
Perform Differential Pressure Design	No
Run Multiple Load Cases	Yes
Junction Stress Reduction option	Perform Plastic Calculation

Tubesheet Integral with	Both
Tubesheet Extended as Flange	No

Is Exchanger in Creep range (skip EP, Use 3S for Sps)	No
---	----

Tubesheet Integral w/Shell and Channel



Configuration a:

Messages regarding the Tubesheet & Exchanger Input Data:

Warning - Tube end condition k should be bet 0 - 1 (.00).

ASME Division 2 Tubesheet Results per Part 4.18, 2025

Elasticity/Expansion Material Properties:

Shell	- TE-1	Carbon & Low Alloy Steels, Group 1
Shell	- TM-1	Material Group C

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000112153 /°C
Elastic Mod. at Design Temperature	260.0 °C	0.19264E+07 kgf/cm ²
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.20952E+07 kgf/cm ²



Elastic Mod. at Ambient Temperature 21.1 °C 0.20811E+07 kgf/cm²

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
Channel - TM-1 Material Group C

Th. Exp. Coeff. Metal Temp. at Tubsht 21.1 °C 0.0000115190 /°C
Elastic Mod. at Design Temperature 260.0 °C 0.19264E+07 kgf/cm²
Elastic Mod. at Ambient Temperature 21.1 °C 0.20811E+07 kgf/cm²

Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
Tubes - TM-1 Material Group C

Th. Exp. Coeff. Metal Temp. along Len 0.0 °C 0.0000112153 /°C
Elastic Mod. at Design Temperature 260.0 °C 0.19264E+07 kgf/cm²
Elastic Mod. at Metal Temp. along Len 0.0 °C 0.20952E+07 kgf/cm²
Elastic Mod. at Tubsht. Design Temp. 260.0 °C 0.19264E+07 kgf/cm²
Elastic Mod. at Ambient Temperature 21.1 °C 0.20811E+07 kgf/cm²

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
TubeSheet - TM-1 Material Group C

Elastic Mod. at Design Temperature 260.0 °C 0.19264E+07 kgf/cm²
Elastic Mod. at Ambient Temperature 21.1 °C 0.20811E+07 kgf/cm²

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.
Please insure these properties are consistent with the
type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Computed Minimum Required Thickness [t]:
= $0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + ci$ Equation 4.3.1
= $0.5 \cdot 10 \cdot (\exp(163/(1144 \cdot 1)) - 1) + 0$
= 0.77 + 0
= 0.77 mm

Tube Required Thickness under External Pressure (Shellside pressure) :

Determine Parameter [Mx]:

= $L / \sqrt{(Ro \cdot t)}$
= $14640 / \sqrt{(7 \cdot 2)}$
= 3912.705

Compute the Factor [Ch]:

Since $Mx > 2 \cdot (Do/t)^{0.94}$, $Ch = 0.55 / (Do/t)$
= $0.55 \cdot (14/2)$
= 0.0786

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

= $1.6 \cdot Ch \cdot Ey(t / Do)$
= $1.6 \cdot 0.079 \cdot 1926428 (2/14)$
= 34597.086 kgf/cm²

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

= Fhe / Fy
= $34597.1 / 1711.0$
= 20.220

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

= $\sigma_{ys} / \sigma_{uts}$
= $1711 / 4218$
= 0.406

Ratio [AE]:

= Fhe / E
= $34597 / 1926428$
= 0.017959

Factor [m1]:

$$= (\ln(r) + (\text{epsp} - \text{epsys})) / \ln(\ln(1 + \text{epsp}) / \ln(1 + \text{epsys}))$$

$$= (\ln(0.41) + (0.20000\text{E-}04 - 0.002)) / \ln(\ln(1 + 0.20000\text{E-}04) / \ln(1 + 0.002))$$

$$= 0.196$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \text{epsys}) / \ln(1 + \text{epsys})^{m1}$$

$$= 1711 (1 + 0.0020) / \ln(1 + 0.0020)^{0.19641939}$$

$$= 5812.170 \text{ kgf/cm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6(1 - 0.41)$$

$$= 0.357$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$

$$= (4218 \cdot \exp(0.36)) / (0.36^{0.36})$$

$$= 8704.171 \text{ kgf/cm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.411.5 - 0.5 \cdot 0.412.5 - 0.413.5$$

$$= 0.293$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	2109	-0.914	0.693E-05	0.128E-05	0.125E-04	0.318E-05	0.410E+05	0.335E-01
2	1054	-3.789	0.406E-06	0.405E-06	0.358E-05	-0.357E-05	0.745E+06	0.165E-01
3	1581	-2.352	0.213E-05	0.203E-05	0.743E-05	-0.689E-05	0.191E+06	0.969E-02
4	1845	-1.633	0.401E-05	0.315E-05	0.981E-05	-0.660E-05	0.918E+05	0.215E-02
5	1713	-1.993	0.296E-05	0.266E-05	0.859E-05	-0.725E-05	0.134E+06	0.514E-02
6	1779	-1.813	0.346E-05	0.294E-05	0.919E-05	-0.711E-05	0.111E+06	0.194E-02
7	1812	-1.723	0.373E-05	0.306E-05	0.950E-05	-0.691E-05	0.101E+06	0.177E-04
8	1829	-1.678	0.387E-05	0.311E-05	0.966E-05	-0.677E-05	0.963E+05	0.103E-02
9	1820	-1.701	0.380E-05	0.309E-05	0.958E-05	-0.684E-05	0.986E+05	0.500E-03
10	1816	-1.712	0.376E-05	0.308E-05	0.954E-05	-0.688E-05	0.998E+05	0.239E-03
11	1814	-1.717	0.374E-05	0.307E-05	0.952E-05	-0.689E-05	0.100E+06	0.110E-03
12	1813	-1.720	0.373E-05	0.307E-05	0.951E-05	-0.690E-05	0.101E+06	0.462E-04
13	1813	-1.722	0.373E-05	0.306E-05	0.951E-05	-0.691E-05	0.101E+06	0.142E-04
14	1812	-1.722	0.373E-05	0.306E-05	0.950E-05	-0.691E-05	0.101E+06	0.175E-05
15	1812	-1.722	0.373E-05	0.306E-05	0.950E-05	-0.691E-05	0.101E+06	0.623E-05
16	1812	-1.722	0.373E-05	0.306E-05	0.950E-05	-0.691E-05	0.101E+06	0.224E-05

Factor of Safety [FS]:

$$\text{Since } F_{ic}[1813] \geq F_y[1711], FS = 1.667$$

Allowable Circumferential Compressive Stress [Fha]:

$$= F_{ic}/FS$$

$$= 1813/1.67$$

$$= 1087.537 \text{ kgf/cm}^2$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot F_{ha}(t/D_o)$$

$$= 2 \cdot 1088(2/14)$$

$$= 310.725 \text{ kgf/cm}^2$$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	1.2819 mm
Allowable Internal Pressure at Corroded thickness	384.95 kgf/cm ²
Required Internal Design Pressure	163.16 kgf/cm ²
Allowable External Pressure at Corroded thickness	310.72 kgf/cm ²
Required External Design Pressure	163.16 kgf/cm ²
Required Thickness due to Shell Side pressure	1.2819 mm



Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Results for ASME Stationary Tubesheet Calculations for Configuration a:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.9.4 Step 1:

Tube expansion depth ratio [p]:

$$= ltx / h \text{ (modified for corrosion if present)}$$

$$= 0/280$$

$$= 0.0000 \text{ (must be } 0 \leq p \leq 1 \text{)}$$

Effective tube hole diameter [d*]:

$$= \max(dt - 2 \cdot tt(EtT/E) \cdot (StT/S) \cdot (\rho); dt - 2 \cdot tt)$$

$$= \max(14 - 2 \cdot 2 (1926428/1926428) \cdot$$

$$(1144/1521) \cdot (0); 14 - 2 \cdot 2)$$

$$= 14.0000 \text{ mm}$$

Equivalent outer tube limit circle diameter [Do]:

$$= 2 \cdot ro + dt$$

$$= 2 \cdot 578 + 14$$

$$= 1169.724 \text{ mm}$$

Determine the basic ligament efficiency for shear [μ]:

$$= (p - dt) / p$$

$$= (19 - 14) / 19$$

$$= 0.263$$

Equivalent outer tube limit radius [ao]:

$$= Do / 2$$

$$= 1170/2$$

$$= 584.8620 \text{ mm}$$

Effective tube pitch [p*]:

$$= p / \sqrt{ (1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot Do \cdot p) / (\pi \cdot Do^2)) }$$

$$= 19 / \sqrt{ (1 - 4 \cdot \min(0 \cdot 100 ; 4 \cdot 1170 \cdot 19) / (\pi \cdot 1170^2)) }$$

$$= 19.0000 \text{ mm}$$

Effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$

$$= (19 - 14) / 19$$

$$= 0.2632$$

Ratio [ps]:

$$= as / ao$$

$$= 650/585$$

$$= 1.1114$$

Ratio [pc]:

$$= ac / ao$$

$$= 650/585$$

$$= 1.1114$$

Compute parameter [xt]:

$$= 1 - Nt((dt - 2 \cdot tt) / (2 \cdot ao))^2$$

$$= 1 - 2615 ((14 - 2 \cdot 2) / (2 \cdot 585))^2$$

$$= 0.8089$$

Determine parameter [xs]:

$$= 1 - Nt(dt / (2 \cdot ao))^2$$

$$= 1 - 2615 (14 / (2 \cdot 585))^2$$

$$= 0.6254$$

Determine the value [h'g]:

$$= \max((hg - CATC); 0) \text{ (For pressure only cases)}$$

$$= \max((6 - 0); 0)$$

$$= 6.000 \text{ mm}$$



ASME Section VIII-2 Part 4.18.9.4 Step 2:

Shell coefficient [β_s]:

$$= \left(\left(12(1 - \nu_s^2) \right)^{1/4} \right) / \left((Ds + ts)ts \right)$$

$$= \left(\left(12(1 - 0.3^2) \right)^{1/4} \right) / \left((1300 + 68)68 \right)$$

$$= 0.0060 \text{ 1/mm}$$

Shell coefficient [k_s]:

$$= \beta_s \cdot E_s \cdot ts^3 / (6(1 - \nu_s^2))$$

$$= 0.006 \cdot 1926428 \cdot 68^3 / (6(1 - 0.3^2))$$

$$= 661218944.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Shell coefficient [λ_s]:

$$= 6(Ds + ts) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(1300 + 68) \cdot 661218944 / 280^3 \cdot (1 + 280 \cdot 0.006 + (280 \cdot 0.006)^2 / 2)$$

$$= 1004108.8750 \text{ kgf/cm}^2$$

Shell coefficient [δ_s]:

$$= Ds^2 / (4 \cdot E_s \cdot ts) \cdot (1 - Ds \cdot \nu_s / (2(Ds + ts)))$$

$$= 1300^2 / (4 \cdot 1926428 \cdot 68) \cdot (1 - 1300 \cdot 0.3 / (2(1300 + 68)))$$

$$= 0.27655\text{E-}02 \text{ mm/kgf/cm}^2$$

Channel coefficient [β_c]:

$$= \left(\left(12(1 - \nu_c^2) \right)^{1/4} \right) / \left((Dc + tc)tc \right)$$

$$= \left(\left(12(1 - 0.3^2) \right)^{1/4} \right) / \left((1300 + 68)68 \right)$$

$$= 0.0060 \text{ 1/mm}$$

Channel coefficient [k_c]:

$$= \beta_c \cdot E_c \cdot tc^3 / (6(1 - \nu_c^2))$$

$$= 0.006 \cdot 1926428 \cdot 68^3 / (6(1 - 0.3^2))$$

$$= 661218944.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(Dc + tc / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (CmDelta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot CqDelta)$$

$$= 6(1300 + 68 / \cos(0)) \cdot 661218944 / 280^3 \cdot (1 + 280 \cdot 0.006 \cdot (1 + 1) / 2 + (280 \cdot 0.006)^2 / 2 \cdot 1)$$

$$= 1004108.9 \text{ kgf/cm}^2$$

Channel coefficient [δ_c]:

$$= Dc^2 / (4 \cdot E_c \cdot Tc) \cdot (1 - Dc \cdot \nu_c / (2(Dc + tc)))$$

$$= 1300^2 / (4 \cdot 1926428 \cdot 68) \cdot (1 - 1300 \cdot 0.3 / (2(1300 + 68)))$$

$$= 0.27655\text{E-}02 \text{ mm/kgf/cm}^2$$

Channel coefficient [zetac]:

$$zetac = 0$$

ASME Section VIII-2 Part 4.18.9.4 Step 3:

$$h/p = 14.736844 ; \mu^* = 0.263158 ; E^* = 449550 \text{ kgf/cm}^2$$

$$E^*/E = 0.233360 ; \nu^* = 0.383525$$

Note: As h/p (14.737) is > 2 , data values for $h/p = 2$ were used.

Tube Bundle Stiffness Factor [X_a]:

$$= \left(\left(24(1 - \nu^*)Nt \cdot Et \cdot tt(dt - tt)ao^2 \right) / (E^* \cdot L \cdot H^3) \right)^{1/4}$$

$$= \left(\left(24(1 - 0.38^2)2615 \cdot 1926428 \cdot 2 \right. \right.$$

$$\left. \left(14 - 2 \right)585^2 \right) / (449551 \cdot 14640 \cdot 280^3) \right)^{1/4}$$

$$= 1.5558$$

Values from Table 4.18.3

$$Z_d = 0.393525 ; Z_v = 0.170662 ; Z_m = 0.692948$$

$$Z_a = 0.578496\text{E}+00 ; Z_w = 0.170662$$

ASME Section VIII-2 Part 4.18.9.4 Step 4:



Diameter ratio [K]:

$$\begin{aligned} &= A / D_o \\ &= 1444/1170 \\ &= 1.2345 \end{aligned}$$

Compute coefficient [F]:

$$\begin{aligned} &= (1 - v^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K)) \\ &= (1 - 0.38) / 449551 \cdot (1004108 + 1004108 + 1926428 \cdot \ln(1.23)) \\ &= 3.3104 \end{aligned}$$

Compute parameter [ϕ]:

$$\begin{aligned} &= (1 + v^*) F \\ &= (1 + 0.38) 3.31 \\ &= 4.5800 \end{aligned}$$

Compute parameter [Q1]:

$$\begin{aligned} &= (\rho_s - 1 - \phi \cdot Z_v) / (1 + \phi \cdot Z_m) \\ &= (1.11 - 1 - 4.58 \cdot 0.17) / (1 + 4.58 \cdot 0.69) \\ &= -0.160590 \end{aligned}$$

ASME Section VIII-2 Part 4.18.9.4 Step 5:

Determine factor [γ_b]:

$$= 0 \text{ (config a)}$$

Calculate parameter [ω_s]:

$$\begin{aligned} &= \rho_s \cdot k_s \cdot \beta_s \cdot \delta S (1 + h \cdot \beta_s) \\ &= 1.11 \cdot 661218944 \cdot 0.006 \cdot 0.00276552 (1 + 280 \cdot 0.006) \\ &= 32326.8633 \text{ mm}^2 \end{aligned}$$

Calculate parameter [ω^*s]:

$$\begin{aligned} &= a o^2 (\rho_s^2 - 1) \cdot (\rho_s - 1) / 4 - \omega_s \\ &= 585^2 (1.11^2 - 1) \cdot (1.11 - 1) / 4 - 32327 \\ &= -30087.2480 \text{ mm}^2 \end{aligned}$$

Parameter [ω_c]:

$$\begin{aligned} &= \rho_c \cdot k_c [\delta C (C_m \Theta + \beta C + C_q \Delta \cdot h \cdot \beta C^2) - \zeta c (C_m \Theta + h \cdot C_q \Theta \cdot h \cdot \beta C / 2)] \\ &= 1.11 \cdot 661218944 [0.00276552 (1 + 0.006 \cdot 1 \cdot 280) - 0.000 (1/280 + 1 \cdot 0.006/2)] \\ &= 32326.8633 \text{ mm}^2 \end{aligned}$$

Parameter [ω^*c]:

$$\begin{aligned} &= a o^2 [(\rho_c^2 + 1) \cdot (\rho_c - 1) / 4 - (\rho_s - 1) / 2] - \omega_c \\ &= 585^2 [(1.11^2 + 1) \cdot (1.11 - 1) / 4 - (1.11 - 1) / 2] - 32327 \\ &= -30087.2480 \text{ mm}^2 \end{aligned}$$

Pressure [P*S]:

$$= 0 \text{ For Pressure only cases or Configurations d,e,f,A,B,C,D}$$

Pressure [P*C]:

$$= 0 \text{ For Pressure only cases or Configurations b,c,d,B,C,D}$$

ASME Section VIII-2 Part 4.18.9.4 Step 6:

$$\begin{aligned} P_e &= P_s - P_t \\ &= 163 - 163 = 0 \text{ kgf/cm}^2 \end{aligned}$$

ASME Section VIII-2 Part 4.18.9.4 Step 7:

Determine factor [Q2]:

$$\begin{aligned} &= [((\omega^*s \cdot P_s - \omega^*c \cdot P_t) - (\omega_s \cdot P^*s - \omega_c \cdot P^*c)) + W^* \cdot \gamma_b / (2 \cdot \pi)] / (1 + \phi \cdot Z_m) \\ &= [((-30087 \cdot 163 - -30087 \cdot 163) - (32327 \cdot 0 - 32327 \cdot 0)) \cdot 0.01 + 0 \cdot 0 / (2 \cdot \pi)] / (1 + 4.58 \cdot 0.69) \\ &= 0.0000 \text{ kgf} \end{aligned}$$



Tubesheet Bending Stress - original thickness [σ]:

$$\begin{aligned} &= 6 \cdot Q2 / (\mu \cdot (H - h'g)^2) \\ &= 6 \cdot 0 / (0.26 \cdot (280 - 6)^2) \\ &= 0.0000 \text{ kgf/cm}^2 \end{aligned}$$

Allowable Tubesheet Bending Stress [σ allowed]:

$$= 1.5 \cdot S = 1.5 \cdot 1521 = 2282 \text{ kgf/cm}^2$$

Tubesheet Bending Stress - final thickness [σ_f]:

$$= 0.0000 \text{ kgf/cm}^2$$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:

$$\begin{aligned} &= h + \text{Cats} + \text{Catc} \\ &= 3.81 + 0 + 0 \\ &= 3.8100 \text{ mm} \end{aligned}$$

ASME Section VIII-2 Part 4.18.9.4 Step 8:

Shear Stress check [Tlimit]:

$$\begin{aligned} &= (2 \cdot \mu \cdot h / ao) \cdot \min(0.8S; 0.533Sy) \\ &= (2 \cdot 0.26 \cdot 280 / 585) \cdot 1216 \\ &= 306.4789 \text{ kgf/cm}^2 \end{aligned}$$

Shear Stress is not required to be computed; [Pe] <= Taulimit

Tubesheet Shear Stress - final thickness [τ_f]:

$$\begin{aligned} &= (1 / (2 \cdot \mu)) \cdot (ao / h) \cdot Pe \\ &= (1 / (2 \cdot 0.26)) \cdot (585 / 3.81) \cdot 0 \\ &= 0.0000 \text{ kgf/cm}^2 \end{aligned}$$

Reqd Tubesheet Thickness, for Shear Stress (including CA) [HreqS]:

$$\begin{aligned} &= H + \text{Cats} + \text{Catc} \\ &= 3.81 + 0 + 0 \\ &= 3.8100 \text{ mm} \end{aligned}$$

Reqd Tubesheet Thickness for Given Loadings (including CA) [Hreqd] :

$$\begin{aligned} &= \max(HreqB; HreqS) \\ &= \max(3.81; 3.81) \\ &= 3.8100 \text{ mm} \end{aligned}$$

ASME Section VIII-2 Part 4.18.9.4 Step 9:

The F_{tmin} and F_{tmax} Coefficients from Table 4.18.4:

$$F_{tmin} = -0.49, F_{tmax} = 0.5$$

First Extreme Tube Axial Stress from among all the tubes [σ_{t1}]:

$$\begin{aligned} &= ((Ps \cdot xs - Pt \cdot xt) - 2 \cdot Q2 \cdot F_{tmin} / ao^2) / (Xt - Xs) \\ &= ((163 \cdot 0.63 - 163 \cdot 0.81) - 2 \cdot 0 \cdot -0.49 / 585^2) / (0.81 - 0.63) \\ &= -163.1581 \text{ kgf/cm}^2 \end{aligned}$$

Second Extreme value of Tube Axial Stress from among all the tubes [σ_{t2}]:

$$\begin{aligned} &= ((Ps \cdot xs - Pt \cdot xt) - 2 \cdot Q2 \cdot F_{tmax} / ao^2) / (Xt - Xs) \\ &= ((163 \cdot 0.63 - 163 \cdot 0.81) - 2 \cdot 0 \cdot 0.5 / 585^2) / (0.81 - 0.63) \\ &= -163.1581 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Tube Axial Stress [$\sigma_{t,max}$]:

$$= \max(|\sigma_{t,1}|; |\sigma_{t,2}|) = 163 \text{ kgf/cm}^2$$

Allowable Tube Stress [$\sigma_{t,A}$]:

$$= S_{ot} = 1144.0795 \text{ kgf/cm}^2$$

Check for Buckling as some of the tubes are in compression.

Determine the Factor of Safety [Fs]:

$$= 1.2500 \text{ (as Pe is 0.0)}$$

Determine the factor [rt]:

$$= (\sqrt{dt^2 + (dt - 2 \cdot tt)^2}) / 4$$



$$= (\sqrt{14^2 + (14 - 2 \cdot 2)^2})/4$$
$$= 4.3012 \text{ mm}$$

Determine the factor [Ct]:

$$= \sqrt{(2 \cdot \pi^2 \cdot Et/Syt)}$$
$$= \sqrt{(2 \cdot \pi^2 \cdot 1926428/1711)}$$
$$= 149.0780$$

Determine the factor [Ft]:

$$= k \cdot L/r$$
$$= 0 \cdot 0/4.3$$
$$= 0.0000$$

Buckling Allowable Stress [Stb]:

$$= (Sy/t/Fs) \cdot (1 - Ft/(2 \cdot Ct))$$
$$= (1711/1.25) \cdot (1 - 0/(2 \cdot 149))$$
$$= 1144.079 \text{ kgf/cm}^2 \text{ (Never greater than } Sot)$$

Note: The Axial Compressive stress in tubes is within limits.

Largest tube-to-tubesheet Joint Load [Wt]:

$$= \sigma_{t,max} \cdot \text{Tube Area}$$
$$= 163 \cdot 0.75$$
$$= 123.02 \text{ kgf}$$

Tube Metal Area [At]:

$$= \pi (do - t)t$$
$$= \pi (14 - 2)2$$
$$= 0.754 \text{ cm}^2$$

Tube Weld Size Results per VIII-2 4.21.2:

Tube Strength [Ft]:

$$= \pi \cdot t (do - t)S$$
$$= \pi \cdot 2 (14 - 2)1144$$
$$= 862.615 \text{ kgf}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot ag (do + 0.67 \cdot ag)Sw \text{ (but not } > Ft)$$
$$= 0.85 \cdot \pi \cdot 2 (14 + 0.67 \cdot 2)1144$$
$$= 862.6145 \text{ kgf}$$

Weld Strength Factor [fw]:

$$= S / Sw \text{ where } Sw = \min(S \text{ and } St)$$
$$= 1144/1144$$
$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot af (do + 0.67 \cdot af)Sw \text{ (but not } > Ft)$$
$$= 0.55 \cdot \pi \cdot 2 (14 + 0.67 \cdot 2)1144$$
$$= 606.4899 \text{ kgf}$$

Minimum Length of Weld [ar]:

$$= 2(\sqrt{((0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t)fw \cdot fd)} - 0.75 \cdot do)$$
$$= 2(\sqrt{((0.75 \cdot 14)^2 + 1.07 \cdot 2 \cdot (14 - 2)1 \cdot 1)} - 0.75 \cdot 14)$$
$$= 2.318 \text{ mm}$$

$$\text{Minimum Required Fillet Weld Leg [afr]: } 1.2000 \text{ mm}$$

$$\text{Minimum Required Groove Weld Leg [agr]: } 1.2000 \text{ mm}$$

Tube-Tubesheet Jt allowable, 863 is $\geq$ tube strength 863 kgf

Note: This tube-tubesheet joint is a Full Strength joint

ASME Section VIII-2 Part 4.18.9.4 Step 10:



Note:

For a given Shell thickness of 68 mm, the minimum Shell length adjacent to the tubesheet should be 535.178 mm

Shell Membrane Stress due to Joint Interaction $[\sigma_s, m]$:

$$= a_o^2 / ((D_s + t_s) t_s) [P_e + (p_s^2 - 1) (P_s - P_t)] + a_s^2 P_t / ((D_s + t_s) t_s)$$

$$= 585^2 / ((1300 + 68) 68) [0 + (1.11^2 - 1) (163 - 163)] + 650^2 \cdot 163 / ((1300 + 68) 68)$$

$$= 741.0378 \text{ kgf/cm}^2$$

Shell Bending Stress due to Joint Interaction $[\sigma_s, b]$:

$$= 6 \cdot k_s / t_s^2 \{ \beta_s [\delta_s \cdot P_s + a_s^2 / (E_s \cdot t_s) \cdot P \cdot s] + 6(1 - \nu^2) / (E^*) (a_o / h)^3 (1 + h \cdot \beta_s / 2) [P_e (Z_v + Z_m \cdot Q_1) + 2 / a_o^2 \cdot Z_m \cdot Q_2] \}$$

$$= 6 \cdot 661218944 / 68^2 \{ 0.006 [0.00276552 \cdot 163 + 650^2 \cdot 0 / (130997160)] + 6(1 - 0.38) / (449551) \cdot (585 / 280)^3 (1 + 280 \cdot 0.006 / 2) \cdot [0(0.17 + 0.69 \cdot -0.16) + 2 / 585^2 \cdot 0.69 \cdot 0] \}$$

$$= 2307.3967 \text{ kgf/cm}^2$$

Shell Stress Summation vs. Allowable

$$:: |\sigma_s, m| + |\sigma_s, b| \leq 1.5 \cdot S_s$$

$$:: |741| + |2307| \leq 2567 \text{ kgf/cm}^2$$

:: 3048, must be $\leq 2567 \text{ kgf/cm}^2$ (See Elastic-Plastic iteration below)

Note:

For a given Channel thickness of 68 mm, the minimum Channel length adjacent to the tubesheet should be 535.178 mm

Channel Membrane Stress due to Joint Interaction $[\sigma_c, m]$:

$$= a_c^2 \cdot P_t / ((D_c + t_c) t_c \cdot \cos(\theta))$$

$$= 650^2 \cdot 163 / ((1300 + 68) 68 \cdot \cos(0))$$

$$= 741.038 \text{ kgf/cm}^2$$

Channel Bending Stress due to Joint Interaction $[\sigma_c, b]$:

$$= 6 \cdot k_c / t_c^2 \cdot \{ (\delta_c \cdot C_m \Delta \cdot \beta_c - \zeta_c \cdot C_m \theta) \cdot P_t - 6(1 - \nu^2) / E^* \cdot (a_o / h)^3 \cdot (C_m \theta + h \cdot \beta_c / 2 \cdot C_m \Delta) \cdot [P_e (Z_v + Z_m \cdot Q_1) + 2 / a_o^2 \cdot Z_m \cdot Q_2] \}$$

$$= 6 \cdot 661218944 / 68^2 \cdot \{ (0.00276552 \cdot 1 \cdot 0.006 - 0.000 \cdot 1) \cdot 163 - 6(1 - 0.38^2) / 449551 \cdot (585 / 280)^3 \cdot (1 + 280 \cdot 0.006 / 2 \cdot 1) \cdot [0(0.17 + 0.69 \cdot -0.16) + 2 / 585^2 \cdot 0.69 \cdot 0] \}$$

$$= 2307.3967 \text{ kgf/cm}^2$$

Channel Stress Summation vs. Allowable

$$:: |\sigma_c, m| + |\sigma_c, b| \leq 1.5 \cdot S_c$$

$$:: |741| + |2307| \leq 2567 \text{ kgf/cm}^2$$

:: 3048 must be $\leq 2567 \text{ kgf/cm}^2$ (See Elast-Plast iteration below)

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	0.0 <=	2282.0 kgf/cm ²	Pass
Tubesheet Shear Stress	0.0 <=	1216.3 kgf/cm ²	Pass
Maximum Tube Stress	163.2 <=	1144.1 kgf/cm ²	Pass
Minimum Tube Stress (Buckling)	-163.2 <=	-1144.1 kgf/cm ²	Pass
Maximum Force on any one Tube	123.0 <=	862.6 kgf	Pass
Shell Stress (jt. inter.)	3048.4 <=	2566.5 kgf/cm ²	See below
Channel Stress (jt. inter.)	3048.4 <=	2566.5 kgf/cm ²	See below

Additional Results for the 2nd Elastic-Plastic Iteration.



These calculations are not intended to be used in the creep range. Check Part 4.18 in Section VIII Division 2.



Compute Allowable [Sstars]:
 = min(Sys, SPS,s/2)
 = min(2564, 5133/2) = 2564 kgf/cm²

Compute Allowable [S*c]:
 = min(Syc, SPS,c/2)
 = min(2564, 5133/2) = 2564 kgf/cm²

Compute Factor [facts]:
 = min([1.4 - 0.4* abs(σs,b) / Sstars], 1)
 = min([1.4 - 0.4* abs(2307)/2564], 1) = 1

Compute Factor [factc]:
 = min([1.4 - 0.4* abs(σc,b) / Sstarc], 1)
 = min([1.4 - 0.4* abs(2307)/2564], 1) = 1

Note: Since facts & factc are both >= 1, the design Passes.

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Shell Stress (jt. inter.)	3048.4 <=	5133.1 kgf/cm ²	Pass
Channel Stress (jt. inter.)	3048.4 <=	5133.1 kgf/cm ²	Pass

Thickness Results for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	3.810	280.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.200	2.000	Ok
Tube-Tubesheet Groove Weld Leg :	1.200	2.000	Ok

Stationary Tubesheet results per ASME Div. 2 Part 4.18.9 2025

Results for 8 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Stresses				Case Type	Pass/Fail
		Bend	Allwd	Shear	Allwd			
D1uc	244.853	...	1757	2282	648	1216	Ps+Pt-Th	D1 Ok
D2uc	244.853	...	1757	2282	648	1216	Ps+Pt-Th	D2 Ok
D3uc	3.810	...	...	2282	...	1216	Ps+Pt-Th	D3 Ok
D4uc	3.810	...	...	2282	...	1216	Ps+Pt-Th	D4 Ok
D1c	245.147	...	1793	2282	655	1216	Ps+Pt-Th-c	D1 Ok
D2c	245.147	...	1793	2282	655	1216	Ps+Pt-Th-c	D2 Ok
D3c	6.810	...	...	2282	...	1216	Ps+Pt-Th-c	D3 Ok
D4c	6.810	...	...	2282	...	1216	Ps+Pt-Th-c	D4 Ok
Max:	245.1469	...	mm	0.786	0.538	(Str. Ratio)		

Load Case Definitions:

[Ps & Pt]:
 Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[(+)-Th]:
 With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:
 With or Without Corrosion Allowance

[D1, D2, D3]:
 Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:
 Design Load Case using the Minimum (Vacuum) Pressures (if specified)



[O1, O2, O3, O4]:

Operating Load Cases using the Maximum and Minimum Operating Pressures and Operating Temperatures

Tube, Shell and Channel Stress Summary:

Case#	---- Tube Stresses				Tube Loads		Shell Stress		Channel Stress		Pass
	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	
D1uc	216	1144	...	...	163	862	804	5133	3852	5133	Ok
D2uc	379	1144	-379	-855	286	862	3852	5133	804	5133	Ok
D3uc	163	1144	-163	-1144	123	862	3048	5133	3048	5133	Ok
D4uc	...	1144	...	...	...	862	...	2566	...	2566	Ok
D1c	218	1144	...	...	165	862	861	5133	3988	5133	Ok
D2c	381	1144	-381	-855	288	862	3988	5133	861	5133	Ok
D3c	163	1144	-163	-1144	123	862	3128	5133	3128	5133	Ok
D4c	...	1144	...	...	...	862	...	2566	...	2566	Ok
Max RATIO	0.333			0.446		0.333		0.777		0.777	

Summary of Thickness Comparisons for 8 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	245.147	280.000	Ok
Tube thickness :	1.282	2.000	Ok
Tube-tubesheet fillet weld leg :	1.200	2.000	Ok
Tube-tubesheet groove weld leg :	1.200	2.000	Ok

Min Shell length of thk, (68.000) adj. to tubesheet: 535.178 mm

Min Channel length of thk, (68.000) adj. to tubesheet: 535.178 mm

Note: This is a full strength Tube to Tubesheet Joint.

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	205.668	0.999	205.668	0.999
Tubesheet Shear Stress	303.194	1.000	303.194	1.000
Tube Tensile Stress	855.279	1.000	489.403	1.000
Tube Compressive Stress	No Calc	No Calc	365.961	1.000
Tube-Tubesheet Joint load	855.279	1.000	489.403	1.000
Shell Stress (Axial, Junction)	205.668	0.999	205.668	0.999
Tubesheet-Channel Junction	205.668	0.999	205.668	0.999
Tube Pressure Stress	384.950	1.000	310.724	1.000
Minimum MAWP	205.668		205.668	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	258.814	0.999	258.814	0.999
Tubesheet Shear Stress	376.588	1.000	376.588	1.000
Tube Tensile Stress	1063.82	1.000	605.813	1.000
Tube Compressive Stress	No Calc	No Calc	449.970	1.000
Tube-Tubesheet Joint load	1063.82	1.000	605.814	1.000
Shell Stress (Axial, Junction)	258.814	0.999	258.814	0.999
Tubesheet-Channel Junction	258.814	0.999	258.814	0.999
Tube Pressure Stress	473.460	1.000	363.930	1.000
Minimum MAPnc	258.814		258.814	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.



Tubesheet MDMT Calculations:

Shell Side MDMT calculation:

Governing thickness on the shell side per figure 3.10:

$$\begin{aligned} &= \max(\text{tubesheet thk}/4, \min(\text{tubesheet thk}, \text{shell thickness})) \\ &= \max(280/4, \min(280, 68)) \\ &= 70.000 \text{ mm} \end{aligned}$$

MDMT reduction ratio:

$$\begin{aligned} &= \min(\text{ps}/\text{Shellside MAPnc}, 1) \\ &= \min(163/259, 1) \\ &= 0.630 \end{aligned}$$

Note: This Element/Detail was specified as being Impact Tested.

Channel Side MDMT calculation:

Governing thickness on the shell side per figure 3.10:

$$\begin{aligned} &= \max(\text{tubesheet thk}/4, \min(\text{tubesheet thk}, \text{channel thickness})) \\ &= \max(280/4, \min(280, 68)) \\ &= 70.000 \text{ mm} \end{aligned}$$

MDMT reduction ratio:

$$\begin{aligned} &= \min(\text{pt}/\text{Tubeside MAPnc}, 1) \\ &= \min(163/259, 1) \\ &= 0.630 \end{aligned}$$

Note: This Element/Detail was specified as being Impact Tested.

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Input Echo, Tubesheet Number 1, Description:

Shell Data:

Main Shell Description: Main Shell

Shell Maximum Design Pressure	Psd,max	163.16	kgf/cm ²
Shell Maximum Operating Pressure	Psox,max	163.16	kgf/cm ²
Shell Minimum Operating Pressure	Psox,min	0.00	kgf/cm ²
Shell Thickness	ts	68.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	casext	0.0000	mm
Inside Diameter of Shell	Ds	1300.000	mm
Shell Temperature for Internal Pressure	Ts	260.00	°C
Shell Material		SA-387 11	
Shell Material UNS Number		K11789	
Shell Allowable Stress at Temperature	Ss	1711.02	kgf/cm ²
Shell Allowable Stress at Ambient		2110.70	kgf/cm ²

Channel Data:

Channel Description: LHS Ch Shell

Channel Type:		Cylinder	
Channel Maximum Design Pressure	Ptd,max	163.16	kgf/cm ²
Channel Maximum Operating Pressure	Ptox,max	163.16	kgf/cm ²
Channel Minimum Operating Pressure	Ptox,min	0.00	kgf/cm ²
Channel Thickness	tc	80.0000	mm
Channel Internal Corrosion Allowance	cac	1.5000	mm
Channel External Corrosion Allowance	cacext	0.0000	mm
Inside Diameter of Channel	Dc	1226.000	mm
Channel Design Temperature	TEMPC	260.00	°C
Channel Flange Matl		SA-387 11	
Channel Material UNS Number		K11789	
Channel Allowable Stress at Temperature	Sc	1711.02	kgf/cm ²
Channel Allowable Stress at Ambient		2110.70	kgf/cm ²

Tube Data:

Number of Tube Holes	Nt	2615	
Tube Wall Thickness	et	2.0000	mm
Tube Outside Diameter	D	14.0000	mm
Total Straight Tube Length	Lt	15200.00	mm
Straight Tube Length (bet. inner tubsht faces) L		14640.00	mm
Design Temperature of the Tubes		260.00	°C
Tube Material		SA-213 T11	
Tube Material UNS Number		K11597	
Is this a Welded Tube		No	
Tube Material Specification used		Smls. tube	
Tube Allowable Stress at Temperature		1144.08	kgf/cm ²
Tube Allowable Stress At Ambient		1407.13	kgf/cm ²
Tube Yield Stress At design Temperature	Syt	1711.02	kgf/cm ²
Tube Pitch (Center to Center Spacing)	P	19.0000	mm
Tube Layout Pattern		Triangular	
Fillet Weld Leg	af	2.0000	mm
Groove Weld Leg	ag	2.0000	mm
Tube-Tubesheet Joint Weld Type		Full Strength	
Method for Tube-Tubesheet Jt. Allowable		4.21.2	
Is the Tube Inset		No	
Radius to Outermost Tube Hole Center	ro	577.862	mm
Largest Center-to-Center Tube Distance	Ul	0.0000	mm
Length of Expanded Portion of Tube	ltx	0.0000	mm
Tube-side pass partition groove depth	hg	5.0000	mm

Tubesheet Data:

Tubesheet TYPE: Floating, Tubesheet Integral, Conf A
Exchanger TYPE: With an Immersed Floating Head, Conf. a

Tubesheet Design Metal Temperature	T	260.00	°C
Tubesheet Material		SA-336 F11	



See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Tubesheet Material UNS Number		K11572
Tubesheet Allowable Stress at Temperature	S	1521.35 kgf/cm ²
Tubesheet Allowable Stress at Ambient	Tt	1876.18 kgf/cm ²
Thickness of Tubesheet	h	280.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	1.5000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	1.5000 mm
Tubesheet Outside Diameter	A	1386.000 mm

Area of the Untubed Lanes	AL	0.0 cm ²
---------------------------	----	---------------------

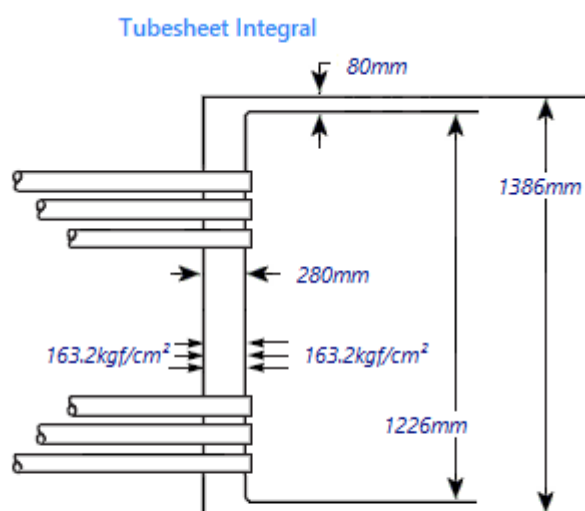
Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration	l	0.000 mm
Tube End condition corresponding to Span (l)	k	0.00

Ignore Radial Thermal Exp. effects (4.18.8/4.18.9)	Yes
Perform Differential Pressure Design	No
Run Multiple Load Cases	Yes
Junction Stress Reduction option	Perform Plastic Calculation

Tubesheet Integral with	Both
Tubesheet Extended as Flange	No

Is Exchanger in Creep range (skip EP, Use 3S for Sps)	No
---	----



Messages regarding the Tubesheet & Exchanger Input Data:

Warning - Tube end condition k should be bet 0 - 1 (.00).

ASME Division 2 Tubesheet Results per Part 4.18, 2025

Elasticity/Expansion Material Properties:

Shell	- TE-1	Carbon & Low Alloy Steels, Group 1
Shell	- TM-1	Material Group C

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000112153 /°C
Elastic Mod. at Design Temperature	260.0 °C	0.19264E+07 kgf/cm ²
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.20952E+07 kgf/cm ²



Elastic Mod. at Ambient Temperature 21.1 °C 0.20811E+07 kgf/cm²

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
Channel - TM-1 Material Group C

Th. Exp. Coeff. Metal Temp. at Tubsht 21.1 °C 0.0000115190 /°C
Elastic Mod. at Design Temperature 260.0 °C 0.19264E+07 kgf/cm²
Elastic Mod. at Ambient Temperature 21.1 °C 0.20811E+07 kgf/cm²

Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
Tubes - TM-1 Material Group C

Th. Exp. Coeff. Metal Temp. along Len 0.0 °C 0.0000112153 /°C
Elastic Mod. at Design Temperature 260.0 °C 0.19264E+07 kgf/cm²
Elastic Mod. at Metal Temp. along Len 0.0 °C 0.20952E+07 kgf/cm²
Elastic Mod. at Tubsht. Design Temp. 260.0 °C 0.19264E+07 kgf/cm²
Elastic Mod. at Ambient Temperature 21.1 °C 0.20811E+07 kgf/cm²

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
TubeSheet - TM-1 Material Group C

Elastic Mod. at Design Temperature 260.0 °C 0.19264E+07 kgf/cm²
Elastic Mod. at Ambient Temperature 21.1 °C 0.20811E+07 kgf/cm²

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.
Please insure these properties are consistent with the
type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Computed Minimum Required Thickness [t]:
= $0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + ci$ Equation 4.3.1
= $0.5 \cdot 10 \cdot (\exp(163/(1144 \cdot 1)) - 1) + 0$
= 0.77 + 0
= 0.77 mm

Tube Required Thickness under External Pressure (Shellside pressure) :

Determine Parameter [Mx]:

= $L / \sqrt{(Ro \cdot t)}$
= $14640 / \sqrt{(7 \cdot 2)}$
= 3912.705

Compute the Factor [Ch]:

Since $Mx > 2 \cdot (Do/t)^{0.94}$, $Ch = 0.55 / (Do/t)$
= $0.55 \cdot (14/2)$
= 0.0786

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

= $1.6 \cdot Ch \cdot Ey(t / Do)$
= $1.6 \cdot 0.079 \cdot 1926428 (2/14)$
= 34597.086 kgf/cm²

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

= Fhe / Fy
= $34597.1 / 1711.0$
= 20.220

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

= $\sigma_{ys} / \sigma_{uts}$
= $1711 / 4218$
= 0.406

Ratio [AE]:

= Fhe / E
= $34597 / 1926428$
= 0.017959

Factor [m1]:

$$= (\ln(r) + (\text{epsp} - \text{epsys})) / \ln(\ln(1 + \text{epsp}) / \ln(1 + \text{epsys}))$$

$$= (\ln(0.41) + (0.20000\text{E-}04 - 0.002)) / \ln(\ln(1 + 0.20000\text{E-}04) / \ln(1 + 0.002))$$

$$= 0.196$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys}(1 + \text{epsys}) / \ln(1 + \text{epsys})^{m1}$$

$$= 1711 (1 + 0.0020) / \ln(1 + 0.0020)^{0.19641939}$$

$$= 5812.170 \text{ kgf/cm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6(1 - 0.41)$$

$$= 0.357$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$

$$= (4218 \cdot \exp(0.36)) / (0.36^{0.36})$$

$$= 8704.171 \text{ kgf/cm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.411.5 - 0.5 \cdot 0.412.5 - 0.413.5$$

$$= 0.293$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	2109	-0.914	0.693E-05	0.128E-05	0.125E-04	0.318E-05	0.410E+05	0.335E-01
2	1054	-3.789	0.406E-06	0.405E-06	0.358E-05	-0.357E-05	0.745E+06	0.165E-01
3	1581	-2.352	0.213E-05	0.203E-05	0.743E-05	-0.689E-05	0.191E+06	0.969E-02
4	1845	-1.633	0.401E-05	0.315E-05	0.981E-05	-0.660E-05	0.918E+05	0.215E-02
5	1713	-1.993	0.296E-05	0.266E-05	0.859E-05	-0.725E-05	0.134E+06	0.514E-02
6	1779	-1.813	0.346E-05	0.294E-05	0.919E-05	-0.711E-05	0.111E+06	0.194E-02
7	1812	-1.723	0.373E-05	0.306E-05	0.950E-05	-0.691E-05	0.101E+06	0.177E-04
8	1829	-1.678	0.387E-05	0.311E-05	0.966E-05	-0.677E-05	0.963E+05	0.103E-02
9	1820	-1.701	0.380E-05	0.309E-05	0.958E-05	-0.684E-05	0.986E+05	0.500E-03
10	1816	-1.712	0.376E-05	0.308E-05	0.954E-05	-0.688E-05	0.998E+05	0.239E-03
11	1814	-1.717	0.374E-05	0.307E-05	0.952E-05	-0.689E-05	0.100E+06	0.110E-03
12	1813	-1.720	0.373E-05	0.307E-05	0.951E-05	-0.690E-05	0.101E+06	0.462E-04
13	1813	-1.722	0.373E-05	0.306E-05	0.951E-05	-0.691E-05	0.101E+06	0.142E-04
14	1812	-1.722	0.373E-05	0.306E-05	0.950E-05	-0.691E-05	0.101E+06	0.175E-05
15	1812	-1.722	0.373E-05	0.306E-05	0.950E-05	-0.691E-05	0.101E+06	0.623E-05
16	1812	-1.722	0.373E-05	0.306E-05	0.950E-05	-0.691E-05	0.101E+06	0.224E-05

Factor of Safety [FS]:

$$\text{Since } F_{ic}[1813] \geq F_y[1711], FS = 1.667$$

Allowable Circumferential Compressive Stress [Fha]:

$$= F_{ic}/FS$$

$$= 1813/1.67$$

$$= 1087.537 \text{ kgf/cm}^2$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot F_{ha}(t / D_o)$$

$$= 2 \cdot 1088 (2/14)$$

$$= 310.725 \text{ kgf/cm}^2$$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	1.2819 mm
Allowable Internal Pressure at Corroded thickness	384.95 kgf/cm ²
Required Internal Design Pressure	163.16 kgf/cm ²
Allowable External Pressure at Corroded thickness	310.72 kgf/cm ²
Required External Design Pressure	163.16 kgf/cm ²
Required Thickness due to Shell Side pressure	1.2819 mm



Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Results for ASME Floating Tubesheet Calculations for Configuration A:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.9.4 Step 1:

Tube expansion depth ratio [p]:

$$= l_{tx} / h \text{ (modified for corrosion if present)}$$

$$= 0/280$$

$$= 0.0000 \text{ (must be } 0 \leq p \leq 1 \text{)}$$

Effective tube hole diameter [d*]:

$$= \max(dt - 2 \cdot tt(Et/E) \cdot (St/S) \cdot (\rho); dt - 2 \cdot tt)$$

$$= \max(14 - 2 \cdot 2 (1926428/1926428) \cdot$$

$$(1144/1521) \cdot (0); 14 - 2 \cdot 2)$$

$$= 14.0000 \text{ mm}$$

Equivalent outer tube limit circle diameter [Do]:

$$= 2 \cdot r_o + dt$$

$$= 2 \cdot 578 + 14$$

$$= 1169.724 \text{ mm}$$

Determine the basic ligament efficiency for shear [μ]:

$$= (p - dt) / p$$

$$= (19 - 14) / 19$$

$$= 0.263$$

Equivalent outer tube limit radius [ao]:

$$= Do / 2$$

$$= 1170/2$$

$$= 584.8620 \text{ mm}$$

Effective tube pitch [p*]:

$$= p / \sqrt{ (1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot Do \cdot p) / (\pi \cdot Do^2)) }$$

$$= 19 / \sqrt{ (1 - 4 \cdot \min(0 \cdot 100 ; 4 \cdot 1170 \cdot 19) / (\pi \cdot 1170^2)) }$$

$$= 19.0000 \text{ mm}$$

Effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$

$$= (19 - 14) / 19$$

$$= 0.2632$$

Ratio [ps]:

$$= a_s / a_o$$

$$= 613/585$$

$$= 1.0481$$

Ratio [pc]:

$$= a_c / a_o$$

$$= 613/585$$

$$= 1.0481$$

Compute parameter [xt]:

$$= 1 - Nt((dt - 2 \cdot tt) / (2 \cdot a_o))^2$$

$$= 1 - 2615 ((14 - 2 \cdot 2) / (2 \cdot 585))^2$$

$$= 0.8089$$

Determine parameter [xs]:

$$= 1 - Nt(dt / (2 \cdot a_o))^2$$

$$= 1 - 2615 (14 / (2 \cdot 585))^2$$

$$= 0.6254$$

Determine the value [h'g]:

$$= \max((h_g - CATC); 0) \text{ (For pressure only cases)}$$

$$= \max((5 - 0); 0)$$

$$= 5.000 \text{ mm}$$



ASME Section VIII-2 Part 4.18.9.4 Step 2:

Intermediate parameters for Tubesheet Gasketed on the Shell Side:

β_s , k_s , λ_s , δ_s are all = 0

Channel coefficient [β_c]:

$$= ((12(1 - \nu_c^2))^{1/4}) / \sqrt{((D_c + t_c)t_c)} \\ = ((12(1 - 0.3^2))^{1/4}) / \sqrt{(1226 + 80)80} \\ = 0.0056 \text{ 1/mm}$$

Channel coefficient [k_c]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - \nu_c^2)) \\ = 0.0056 \cdot 1926428 \cdot 80^3 / (6(1 - 0.3^2)) \\ = 1015942912.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + \\ h \cdot \beta_c \cdot (CmDelta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot CqDelta) \\ = 6(1226 + 80 / \cos(0)) \cdot 1015942912 / 280^3 \cdot (1 + \\ 280 \cdot 0.0056 \cdot (1 + 1) / 2 + (280 \cdot 0.0056)^2 / 2 \cdot 1) \\ = 1383346.2 \text{ kgf/cm}^2$$

Channel coefficient [δ_c]:

$$= D_c^2 / (4 \cdot E_c \cdot T_c) \cdot (1 - D_c \cdot \nu_c / (2(D_c + t_c))) \\ = 1226^2 / (4 \cdot 1926428 \cdot 80) \cdot (1 - 1226 \cdot 0.3 / (2(1226 + 80))) \\ = 0.20949E-02 \text{ mm/kgf/cm}^2$$

Channel coefficient [zetac]:

$$zetac = 0$$

ASME Section VIII-2 Part 4.18.9.4 Step 3:

$$h/p = 14.736844 ; \mu^* = 0.263158 ; E^* = 449550 \text{ kgf/cm}^2$$

$$E^*/E = 0.233360 ; \nu^* = 0.383525$$

Note: As h/p (14.737) is > 2, data values for $h/p = 2$ were used.

Tube Bundle Stiffness Factor [X_a]:

$$= ((24(1 - \nu^*)Nt \cdot Et \cdot t(d_t - t_t)ao^2) / (E^* \cdot L \cdot H^3))^{1/4} \\ = ((24(1 - 0.38^2)2615 \cdot 1926428 \cdot 2 \\ (14 - 2)585^2) / (449551 \cdot 14640 \cdot 280^3))^{1/4} \\ = 1.5558$$

Values from Table 4.18.3

$$Z_d = 0.393525 ; Z_v = 0.170662 ; Z_m = 0.692948 \\ Z_a = 0.578496E+00 ; Z_w = 0.170662$$

ASME Section VIII-2 Part 4.18.9.4 Step 4:

Diameter ratio [K]:

$$= A / D_o \\ = 1386 / 1170 \\ = 1.1849$$

Compute coefficient [F]:

$$= (1 - \nu^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K)) \\ = (1 - 0.38) / 449551 \cdot (0 + 1383346 + \\ 1926428 \cdot \ln(1.18)) \\ = 2.3452$$

Compute parameter [ϕ]:

$$= (1 + \nu^*)^F \\ = (1 + 0.38)^{2.35} \\ = 3.2446$$

Compute parameter [Q1]:



$$\begin{aligned} &= (ps - 1 - \phi \cdot Zv) / (1 + \phi \cdot Zm) \\ &= (1.05 - 1 - 3.24 \cdot 0.17) / (1 + 3.24 \cdot 0.69) \\ &= -0.155655 \end{aligned}$$

ASME Section VIII-2 Part 4.18.9.4 Step 5:

Determine factor [yb]:
= 0 (config a)

Calculate parameter [ws]:

$$\begin{aligned} &= ps \cdot ks \cdot \beta s \cdot \delta S (1 + h \cdot \beta s) \\ &= 1.05 \cdot 0 \cdot 0 \cdot 0.000 (1 + 280 \cdot 0) \\ &= 0.0000 \text{ mm}^2 \end{aligned}$$

Calculate parameter [ws*]:

$$\begin{aligned} &= ao^2 (ps^2 - 1) \cdot (ps - 1) / 4 - ws \\ &= 585^2 (1.05^2 - 1) \cdot (1.05 - 1) / 4 - 0 \\ &= 405.3960 \text{ mm}^2 \end{aligned}$$

Parameter [wc]:

$$\begin{aligned} &= pc \cdot kc [\delta C (CmTheta \cdot \beta C + Cq\Delta \cdot h \cdot \beta C^2) - \\ &\quad \zeta c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta C / 2)] \\ &= 1.05 \cdot 1015942912 [0.002094915 (1 + 0.0056 \cdot 1 \cdot 280) - \\ &\quad 0.000 (1/280 + 1 \cdot 0.0056/2)] \\ &= 32300.4121 \text{ mm}^2 \end{aligned}$$

Parameter [wc*]:

$$\begin{aligned} &= ao^2 [(pc^2 + 1) \cdot (pc - 1) / 4 - (ps - 1) / 2] - wc \\ &= 585^2 [(1.05^2 + 1) \cdot (1.05 - 1) / 4 - (1.05 - 1) / 2] - 32300 \\ &= -31895.0176 \text{ mm}^2 \end{aligned}$$

Pressure [P*S]:
= 0 For Pressure only cases or Configurations d,e,f,A,B,C,D

Pressure [P*C]:
= 0 For Pressure only cases or Configurations b,c,d,B,C,D

ASME Section VIII-2 Part 4.18.9.4 Step 6:

$$\begin{aligned} Pe &= Ps - Pt \\ &= 163 - 163 = 0 \text{ kgf/cm}^2 \end{aligned}$$

ASME Section VIII-2 Part 4.18.9.4 Step 7:

Determine factor [Q2]:

$$\begin{aligned} &= [((ws* \cdot Ps - ws* \cdot c \cdot Pt) - (ws \cdot P*s - wc \cdot P*c)) + \\ &\quad W* \cdot yb / (2 \cdot \pi)] / (1 + \phi \cdot Zm) \\ &= [((405 \cdot 163 - -31895 \cdot 163) - \\ &\quad (0 \cdot 0 - 32300 \cdot 0)) \cdot 0.01 + \\ &\quad 0 \cdot 0 / (2 \cdot \pi)] / (1 + 3.24 \cdot 0.69) \\ &= 16223.8184 \text{ kgf} \end{aligned}$$

Tubesheet Bending Stress - original thickness [σ]:

$$\begin{aligned} &= 6 \cdot Q2 / (\mu* \cdot (H - h'g)^2) \\ &= 6 \cdot 16224 / (0.26 \cdot (280 - 5)^2) \\ &= 489.1287 \text{ kgf/cm}^2 \end{aligned}$$

Allowable Tubesheet Bending Stress [σ allowed]:
= 1.5 · S = 1.5 · 1521 = 2282 kgf/cm²

Tubesheet Bending Stress - final thickness [σ_f]:

$$\begin{aligned} &= 6 \cdot Q2 / (\mu* \cdot (H - h'g)^2) \\ &= 6 \cdot 14.8 / (0.26 \cdot (8.85 - 5)^2) \\ &= 2281.8557 \text{ kgf/cm}^2 \end{aligned}$$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:

$$\begin{aligned} &= h + Cats + Catc \\ &= 8.85 + 0 + 0 \\ &= 8.8456 \text{ mm} \end{aligned}$$



H reqd. = HReqB = 8.8456 mm

ASME Section VIII-2 Part 4.18.9.4 Step 10:

Note:

For a given Channel thickness of 80 mm, the minimum Channel length adjacent to the tubesheet should be 563.719 mm

Channel Membrane Stress due to Joint Interaction [σ_c, m]:

$$= ac^2 \cdot Pt / ((Dc + tc) tc \cdot \cos(\theta))$$

$$= 613^2 \cdot 163 / ((1226 + 80) 80 \cdot \cos(0))$$

$$= 586.809 \text{ kgf/cm}^2$$

Channel Bending Stress due to Joint Interaction [σ_c, b]:

$$= 6 \cdot kc / tc^2 \cdot \{ (\delta_c \cdot Cm\Delta \cdot \beta_c - \zeta_c \cdot Cm\Theta) \cdot Pt$$

$$- 6 ((1 - \nu^2) / E) \cdot (ao/h)^3 \cdot (Cm\Theta +$$

$$h \beta_c / 2 \cdot Cm\Delta) \cdot [Pe (Zv + Zm \cdot Q1) + 2 / ao^2 \cdot Zm \cdot Q2] \}$$

$$= 6 \cdot 1015942912 / 80^2 \cdot \{ (0.002094915 \cdot 1 \cdot 0.0056 - 0.000 \cdot 1) \cdot 163$$

$$- 6 ((1 - 0.38^2) / 449551) \cdot$$

$$(585 / 280)^3 \cdot (1 + 280 \cdot 0.0056 / 2 \cdot 1) \cdot [0 (0.17 + 0.69$$

$$\cdot -0.16) + 2 / 585^2 \cdot 0.69 \cdot 16224] \}$$

$$= 669.9756 \text{ kgf/cm}^2$$

Channel Stress Summation vs. Allowable

$$\therefore | \sigma_{c,m} | + | \sigma_{c,b} | \leq 1.5 \cdot Sc$$

$$\therefore | 587 | + | 670 | \leq 2567 \text{ kgf/cm}^2$$

$$\therefore 1257 \text{ must be } \leq 2567 \text{ kgf/cm}^2$$

Computations completed for ASME tubesheet configuration: a

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	489.1 <=	2282.0 kgf/cm ²	Pass
Channel Stress (jt. inter.)	1256.8 <=	2566.5 kgf/cm ²	Pass

Thickness Results for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	8.846	280.000	Ok
Tube-Tubesheet Fillet Weld Leg :	0.000	2.000	Ok
Tube-Tubesheet Groove Weld Leg :	0.000	2.000	Ok

Floating Tubesheet results per ASME Div. 2 Part 4.18.9 2025

Results for 8 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend Allwd	Stresses Shear Allwd	Case Type	Pass/ Fail
D1uc	248.332	...	1793	2282	Ps+Pt-Th	D1 Ok
D2uc	264.824	...	2098	2282	Ps+Pt-Th	D2 Ok
D3uc	8.846	...	489	2282	Ps+Pt-Th	D3 Ok
D4uc	3.810	...	...	2282	Ps+Pt-Th	D4 Ok
D1c	248.669	...	1794	2282	Ps+Pt-Th-c	D1 Ok
D2c	267.544	...	2130	2282	Ps+Pt-Th-c	D2 Ok
D3c	9.157	...	493	2282	Ps+Pt-Th-c	D3 Ok
D4c	6.810	...	...	2282	Ps+Pt-Th-c	D4 Ok
Max:	267.5439	... mm	0.934	...	(Str. Ratio)	

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the



Load Case Tables

[(+)-Th]:

With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:

Operating Load Cases using the Maximum and Minimum Operating Pressures and Operating Temperatures

Tube, Shell and Channel Stress Summary:

Case#	---- Tube Stresses				Tube Loads		Shell Stress		Channel Stress		Stress Pass	
	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	Fail	
D1uc	...	...	...	...	...	...	...	...	3081	5133	Ok	
D2uc	...	...	...	...	...	...	...	2566	1824	2566	Ok	
D3uc	...	...	...	...	...	...	...	2566	1257	2566	Ok	
D4uc	...	...	...	...	...	...	...	2566	...	2566	Ok	
D1c	...	...	...	...	...	...	...	2566	3190	5133	Ok	
D2c	...	...	...	...	...	...	...	2566	1897	2566	Ok	
D3c	...	...	...	...	...	...	...	2566	1293	2566	Ok	
D4c	...	...	...	...	...	...	...	2566	...	2566	Ok	
Max RATIO	...	...	...	...	...	...	...	...	...	0.739		

Summary of Thickness Comparisons for 8 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	267.544	280.000	Ok
Tubesheet thickness flanged extension :	0.000	284.000	Ok
Tube thickness :	1.282	2.000	Ok
Tube-tubesheet fillet weld leg :	0.000	2.000	Ok
Tube-tubesheet groove weld leg :	0.000	2.000	Ok

Min Channel length of thk, (80.000) adj. to tubesheet: 563.719 mm

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	209.653	1.000	174.779	1.000
Tubesheet-Channel Junction	209.653	1.000	174.779	0.396
Tube Pressure Stress	384.950	1.000	310.724	1.000
Minimum MAWP	209.653		174.779	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	258.435	1.000	218.899	1.000
Tubesheet-Channel Junction	258.435	1.000	218.899	0.387
Tube Pressure Stress	473.460	1.000	363.930	1.000
Minimum MAPnc	258.435		218.899	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.



Tubesheet MDMT Calculations:

Determine the governing MDMT considering the governing condition:

Governing thickness on the shell side per figure 3.10 (a):

= tubesheet thickness/4

= 280/4

= 70.000 mm

Note: This Element/Detail was specified as being Impact Tested.

where the MDMT reduction ratio is:

= max(pt/Tubeside MAPnc, ps/Shellside MAPnc), must be ≤ 1

= max(163/258, 163/219)

= 0.745

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